

Diseases of Infancy and Childhood

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Children are not just little adults, and their diseases are not merely variants of adult diseases. Many childhood conditions are unique to, or at least take distinctive forms in, this stage of life and so are discussed separately in this chapter. Diseases of the perinatal period are sources of significant morbidity and mortality. Although the infant mortality rate in the United States declined from 20 deaths per 1000 live births in 1970 to 5.4 in 2021, it began to rise again during the COVID-19 pandemic. Moreover, African Americans continue to have an infant mortality rate more than twice (11 deaths per 1000 live births) that of White Americans (4.9 deaths). Worldwide, infant mortality rates vary widely, from as low as 1.8 deaths per 1000 live births in Slovenia to as high as 110.6 deaths per 1000 in Afghanistan. Rather dismayingly, the United States ranks thirtieth in infant mortality rate among developed nations in the Western hemisphere.

Early development can be divided into four stages (each associated with a somewhat different group of disorders: (1) the neonatal period (the first 4 weeks of life), (2) infancy (the first year of life), (3) 1 to 4 years of age, and (4) 5 to 14 years of age. Congenital anomalies, prematurity and low birth weight, sudden infant death syndrome, and maternal complications and injuries represent the leading causes of death in the first 12 months of life. Once the infant survives the first year of life, the outlook brightens considerably. In the next two age groups—1 to 4 years and 5 to 9 years—unintentional injuries resulting from accidents are the leading cause of death. Among the natural diseases, in order of importance, congenital anomalies and malignant neoplasms assume major significance. In the 10- to 14-year age group, accidents, malignancies, suicide, homicide, and congenital malformations are the leading causes of death.

The following discussion looks at specific conditions encountered during the various stages of infant and child development.

CONGENITAL ANOMALIES

Congenital anomalies are anatomic defects that are present at birth, but some, such as cardiac defects and renal anomalies, may not become clinically apparent until years later. The term *congenital* means “born with,” but it does not imply or exclude a genetic basis for the birth defect. It is estimated that about 120,000 (1 in 33) infants are born with a birth defect each year in the United States. In a sense, anomalies found in live-born infants represent the less serious developmental failures in embryogenesis as they are compatible with live birth. Perhaps 20% of fertilized ova are so anomalous that they are blighted at early stages. Others may be compatible with early fetal development but are later spontaneously aborted, mainly early in the first trimester. Less severe anomalies allow more prolonged intrauterine survival, with some disorders terminating in stillbirth and others permitting live birth despite the disabilities imposed.

Definitions

The process of morphogenesis (organ and tissue development) can be impaired by a variety of different errors.

- **Malformations represent primary errors of morphogenesis, in which there is an intrinsically abnormal developmental process**



Fig. 10.1 Examples of malformations. (A) Polydactyly (one or more extra digits) and syndactyly (fusion of digits) have little functional consequence when they occur in isolation. Similarly, cleft lip (B), with or without associated cleft palate, is compatible with life when it occurs as an isolated anomaly; in the present case, however, this neonate had an underlying malformation syndrome (trisomy 13) and died of severe cardiac defects. (C) The stillbirth illustrated represents a severe and essentially lethal malformation, wherein the midface structures are fused or ill-formed; in almost all cases, this degree of external dysmorphogenesis is associated with severe internal anomalies such as maldevelopment of the brain and cardiac defects. (A and C, Courtesy Dr. Reade Quinton. B, Courtesy Dr. Beverly Rogers, Department of Pathology, University of Texas Southwestern Medical Center, Dallas.)

(Fig. 10.1). Malformations can result from a single gene or chromosomal defect but are more commonly multifactorial in origin. Developmental anomalies may present in several patterns. Some, such as congenital heart defects and anencephaly (absence of part or all of the brain), involve single body systems, whereas in other cases multiple malformations involving many organs may coexist.

- **Disruptions result from secondary destruction of an organ or body region that was previously normal in development;** thus, in contrast with malformations, disruptions arise from an extrinsic disturbance in morphogenesis. *Amniotic bands*, denoting rupture of amnion with resultant formation of “bands” that encircle, compress, or attach to parts of the developing fetus, are the classic example of a disruption (Fig. 10.2). A variety of environmental agents may cause disruptions (see later). Disruptions are not heritable, hence they are not associated with increased risk of recurrence in subsequent pregnancies.
- **Deformations, like disruptions, represent an extrinsic disturbance of development rather than an intrinsic error of morphogenesis.** Deformations are common problems, affecting approximately 2% of newborn infants to varying degrees. Fundamental to the pathogenesis of deformations is localized or generalized compression of the growing fetus by abnormal biomechanical forces, leading eventually to a variety of structural abnormalities. The most common underlying factor responsible for deformations is *uterine constraint*. Between 35 and 38 weeks of gestation, rapid increase in the size of the fetus outpaces the growth of the uterus, and the relative amount of amniotic fluid (which normally acts as a cushion) decreases. Thus, even the normal fetus is subjected to some degree of uterine constraint. Several factors increase the likelihood of excessive compression of the fetus resulting in deformations. Maternal factors include first pregnancy, small uterus, malformed (e.g., bicornuate) uterus, and leiomyomas. Fetal or placental factors include oligohydramnios,

multiple fetuses, and abnormal fetal presentation. For example, club-foot can occur as a component of Potter sequence, described later.

- **A sequence is a cascade of anomalies triggered by one initiating aberration.** Approximately half the time, congenital anomalies occur singly; in the remaining cases, multiple congenital anomalies are recognized. In some instances, the constellation of anomalies may be explained by a single localized aberration in organogenesis (malformation, disruption, or deformation) that sets into motion secondary effects in other organs. A good example is the *oligohydramnios*

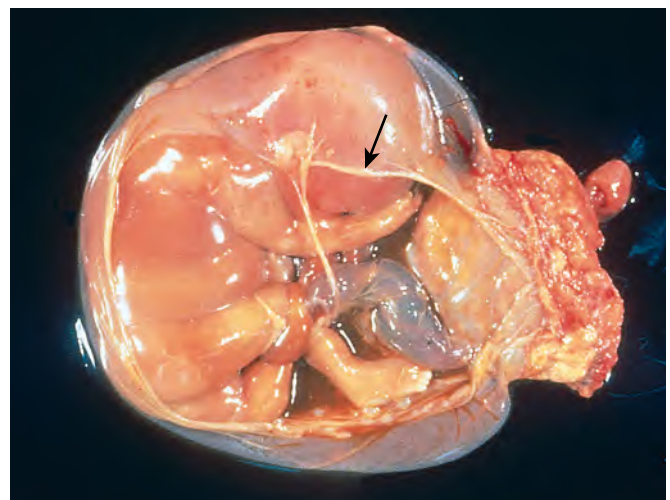


Fig. 10.2 Disruption of morphogenesis by an amniotic band. Note the placenta at the right of the diagram and the band of amnion extending from the top portion of the amniotic sac to encircle the leg of the fetus (arrow). (Courtesy Dr. Theonia Boyd, Boston Children's Hospital.)

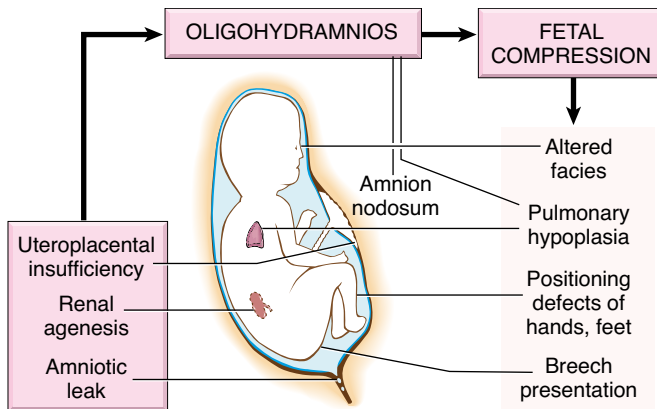


Fig. 10.3 Pathogenesis of oligohydramnios sequence.

(or *Potter*) sequence (Fig. 10.3). Oligohydramnios (decreased amniotic fluid) may be caused by a variety of maternal, placental, or fetal abnormalities. The most common cause of oligohydramnios is chronic leakage of amniotic fluid due to rupture of fetal membranes. Other causes include renal agenesis and urinary tract obstruction in the fetus (because fetal urine is a major constituent of amniotic fluid), and uteroplacental insufficiency resulting from maternal hypertension or severe preeclampsia. The fetal compression associated with significant oligohydramnios, in turn, results in a classic phenotype in the newborn infant, including flattened facies, positional abnormalities of the hands, and clubfeet (Fig. 10.4). The hips may be dislocated. Growth of the chest wall and the contained lungs is also compromised so that the lungs are frequently hypoplastic, sometimes leading to fetal demise. Nodules in the amnion (*amnion nodosum*) are frequently present.

- A **malformation syndrome** is a constellation of congenital anomalies, believed to be pathologically related, that, in contrast to a sequence, cannot be explained based on a single, localized, initiating defect. Syndromes are most often caused by a single etiologic agent, such as a viral infection or specific chromosomal abnormality, which simultaneously affects several tissues.

In addition to the general definitions, a few organ-specific terms should be defined. *Agenesis* refers to the complete absence of an organ and its associated primordium. A closely related term, *aplasia*, means an organ, tissue, or body part failed to develop normally. In some cases, aplasia means that a body part is missing. *Atresia* describes the absence of an opening, usually of a hollow visceral organ, such as the trachea or intestine. *Hypoplasia* refers to incomplete development or decreased size of an organ due to decreased numbers of cells, whereas *hyperplasia* refers to the converse, that is, the enlargement of an organ due to increased numbers of cells. An abnormality in an organ or a tissue because of an increase or a decrease in the size (rather than the number) of individual cells defines hypertrophy or hypotrophy, respectively. Finally, *dysplasia* in the context of malformations (versus *neoplasia*) describes an abnormal organization of cells.

Causes of Anomalies

Although we are learning a great deal about the molecular bases of some congenital anomalies, the exact cause remains unknown in 40% to 60% of cases. The era of molecular medicine promises to bring additional insights into the mechanisms by which malformations occur. The common known causes of congenital anomalies can be grouped into three major categories: genetic, environmental, and multifactorial (Table 10.1).



Fig. 10.4 Infant with oligohydramnios sequence. Note the flattened facial features and deformed right foot (talipes equinovarus).

TABLE 10.1 Causes of Congenital Anomalies in Live-Born Infants

Cause	Frequency (%)
Genetic	
Chromosomal aberrations	10–15
Mendelian inheritance	2–10
Environmental	
Maternal/placental infections	2–3
Rubella	
Toxoplasmosis	
Syphilis	
Cytomegalovirus	
Human immunodeficiency virus	
Maternal disease states	6–8
Diabetes	
Phenylketonuria	
Endocrinopathies, including severe obesity	
Drugs and chemicals	1
Alcohol, smoking	
Folic acid antagonists	
Androgens	
Phenytoin	
Thalidomide	
Warfarin	
13- <i>cis</i> -retinoic acid	
Others	
Irradiation	1
Multifactorial	20–25
Unknown	40–60

Modified from Stevenson RE, Hall JG, Everman DB, Solomon B, eds. *Human Malformations and Related Anomalies*. 3rd ed. Oxford University Press; 2016:12.

Genetic causes of malformations include all the previously discussed mechanisms of genetic disease (Chapter 5). Virtually all chromosomal syndromes are associated with congenital malformations. Examples include Down syndrome and other trisomies, Turner syndrome, and Klinefelter syndrome. Most chromosomal disorders arise during gametogenesis and hence are not familial. Single-gene mutations, characterized by Mendelian inheritance, may underlie some major malformations. For example, holoprosencephaly is the most common developmental defect of the forebrain and midface in humans; the Hedgehog signaling pathway plays a critical role in the morphogenesis of these structures, and loss-of-function mutations of individual components within this pathway are reported in families with a history of recurrent holoprosencephaly.

Environmental influences, such as viral infections, drugs, and maternal irradiation, may cause fetal anomalies. Among the viral infections listed in Table 10.1, rubella was a major scourge of the nineteenth and early twentieth centuries. Maternal rubella and the resultant *rubella embryopathy* have been virtually eliminated in high-resource countries because of maternal rubella vaccination, but in 2019 it was estimated that there were 32,000 infants born with congenital rubella syndrome worldwide. A variety of drugs and chemicals have been suspected to be teratogenic, but perhaps less than 1% of congenital malformations are caused by these agents. The list includes thalidomide, alcohol, anticonvulsants, warfarin (oral anticoagulant), and 13-*cis*-retinoic acid, which is used in the treatment of severe acne. For example, thalidomide, once used as a tranquilizer in Europe, causes an extremely high incidence (50% to 80%) of limb malformations. Alcohol, when consumed even in modest amounts during pregnancy, is an important environmental teratogen. Affected infants show prenatal and postnatal growth restriction, facial anomalies (microcephaly, short palpebral fissures, maxillary hypoplasia), and psychomotor disturbances. These in combination are labeled the *fetal alcohol syndrome* (also discussed in Chapter 9). Although cigarette smoke-derived nicotine has not been convincingly demonstrated to be a teratogen, there is a high incidence of spontaneous abortion, premature labor, and placental abnormalities in pregnant females who smoke. Infants born to mothers who smoke often have a low birth weight and may be prone to sudden infant death syndrome (discussed later). Considering these findings, it is best to avoid nicotine exposure altogether during pregnancy. Among maternal conditions listed in Table 10.1, diabetes is a common entity, and despite advances in antenatal obstetric monitoring and glucose control, the incidence of major malformations in infants of diabetic mothers stands between 6% and 10% in most series. Maternal hyperglycemia-induced fetal hyperinsulinemia results in fetal macrosomia (organomegaly and increased body fat and muscle mass); cardiac anomalies, neural tube defects, and other central nervous system (CNS) malformations are some of the major anomalies seen in *diabetic embryopathy*.

Multifactorial inheritance, which implies the interaction of environmental influences with two or more genes of small effect, is the most common genetic etiology of congenital malformations. Examples include relatively common malformations such as cleft lip, cleft palate, and neural tube defects. The dramatic reduction in incidence of neural tube defects by periconceptional intake of folic acid is one case where identifying an environmental factor has prevented development of multifactorial malformations.

Pathogenesis

The pathogenesis of congenital anomalies is complex and still poorly understood, but two general principles of developmental pathology are relevant regardless of the etiologic agent.

1. The timing of the prenatal teratogenic insult has an important impact on the occurrence and the type of anomaly produced (Fig. 10.5). The intrauterine development of humans can be divided into two phases: (1) the early embryonic period occupying the first 9 weeks of pregnancy and (2) the fetal period terminating at birth.
 - In the *early embryonic period* (first 3 weeks after fertilization), an injurious agent may either damage enough cells to cause death and abortion or only a few cells, allowing the embryo to recover without developing defects. **Between the third and the ninth weeks, the embryo is extremely susceptible to teratogenesis;** peak sensitivity occurs between the fourth and the fifth weeks, a critical period of organogenesis.
 - The *fetal period* that follows organogenesis is marked by growth and maturation of the organs and greatly reduced susceptibility to teratogenic agents. Instead, the fetus is susceptible to growth restriction or injury to already formed organs. Thus, a given agent may produce different anomalies if exposure occurs at different times of gestation.
2. The interplay between environmental teratogens and intrinsic genetic factors is exemplified by the fact that features of dysmorphogenesis caused by environmental insults can often be recapitulated by genetic defects in the pathways targeted by these teratogens. This is illustrated by the following representative examples.
 - *Cyclopamine* is a plant teratogen contained in the corn lily; pregnant sheep who feed on this plant give birth to lambs with severe craniofacial abnormalities including holoprosencephaly and “cyclopia” (single fused eye, hence the origin of the moniker cyclopamine). This compound is an inhibitor of Hedgehog signaling in the embryo, and, as stated earlier, mutations of Hedgehog genes are present in a subset of patients with holoprosencephaly.
 - *Valproic acid* is an antiepileptic and a recognized teratogen during pregnancy. Valproic acid disrupts expression of a family of highly conserved developmentally critical transcription factors known as *homeobox* (HOX) proteins. In vertebrates, HOX proteins regulate the patterning of limbs, vertebrae, and craniofacial structures. Not surprisingly, mutations in the *HOX* family of genes are responsible for congenital anomalies that mimic features observed in valproic acid embryopathy.
 - The vitamin A (retinol) derivative *all-trans-retinoic acid* is essential for normal development and differentiation, and its absence during embryogenesis results in a constellation of malformations affecting multiple organ systems, including the eyes, genitourinary system, cardiovascular system, diaphragm, and lungs (see Chapter 9 for effects of vitamin A deficiency in the postnatal period). Conversely, excessive exposure to retinoic acid is also teratogenic. Infants born to mothers treated with retinoic acid for severe acne have a predictable phenotype (retinoic acid embryopathy) including CNS, cardiac, and craniofacial defects, such as cleft lip and cleft palate. The latter may stem from retinoic acid-mediated deregulation of components of the transforming growth factor- β (TGF- β) signaling pathway, which is involved in palatogenesis. Mice with knockout of certain components of the TGF- β signaling pathway uniformly develop cleft palate, once again illustrating the functional relationship between teratogenic exposure and signaling pathways in the causation of congenital anomalies.

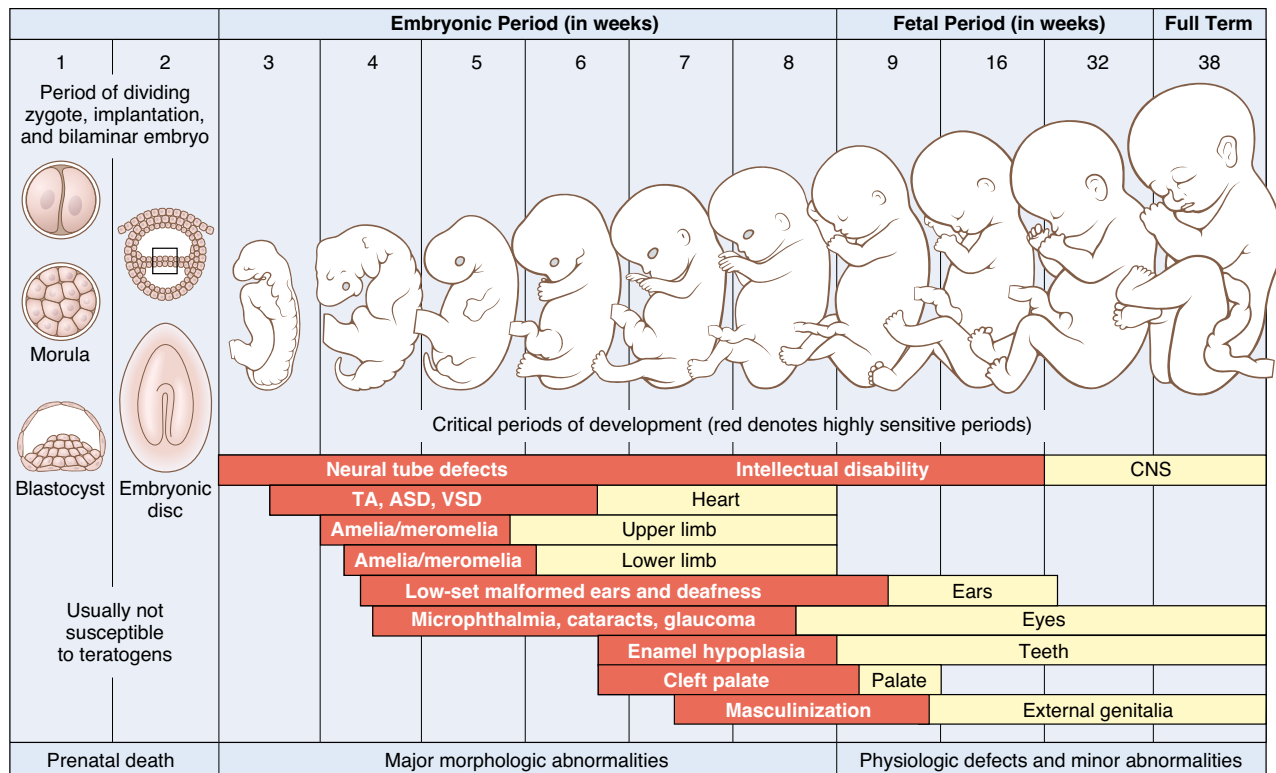


Fig. 10.5 Critical periods of development for various organ systems and associated malformations. ASD, Atrial septal defect; CNS, central nervous system; TA, tricuspid atresia; VSD, ventricular septal defect. (Modified from Moore KL: *The Developing Human*. 10th ed. Saunders; 2016:474.)

KEY CONCEPTS

Congenital Anomalies

- Congenital anomalies result from intrinsic abnormalities (malformations such as cleft palate) as well as extrinsic disturbances (deformations, disruptions such as oligohydramnios).
- Congenital anomalies can result from genetic (chromosomal abnormalities, gene mutations), environmental (infections, drugs, alcohol), and multifactorial causes.
- The timing of the in utero insult has profound influence on the extent of congenital anomalies, with earlier events usually having greater impact.
- The interplay between genetic and environmental causes of anomalies is emphasized by the fact that teratogens often target signaling pathways in which mutations have been reported as a cause for the same anomalies.

PREMATURITY AND FETAL GROWTH RESTRICTION

Prematurity, defined by a gestational age less than 37 weeks, is the second most common cause of neonatal mortality, behind only congenital anomalies. According to the Centers for Disease Control and Prevention (CDC), 10.4% of infants born in the United States in 2022 were preterm. This is a 1% decrease from 2021. However, preterm birth rates still vary by race and ethnicity. In 2022, the rate was 14.6% for Black females, 9.4% for White females, and 10.1% for Hispanic females. The major risk factors for prematurity include:

- **Preterm premature rupture of membranes (PPROM):** PPRM complicates about 3% of pregnancies and is responsible for as many as one-third of preterm deliveries. Rupture of membranes (ROM)

before the onset of labor can be spontaneous or induced. PPRM refers to spontaneous ROM occurring *before* 37 weeks of gestation (hence the annotation “preterm”). In contrast, PROM refers to spontaneous ROM occurring *after* 37 weeks of gestation. This distinction is important because after 37 weeks the associated risk to the fetus is considerably decreased. Several clinical risk factors have been identified for PPRM, including a prior history of preterm delivery, preterm labor and/or vaginal bleeding during the current pregnancy, maternal smoking, low socioeconomic status, and poor maternal nutrition. The fetal and maternal outcome after PPRM depends on the gestational age of the fetus (second-trimester PPRM has a dismal prognosis) and the effective prophylaxis of infections in the exposed amniotic cavity.

- **Intrauterine infection:** This is a major cause of preterm labor, being a causative factor in approximately 25% of all preterm births, and the earlier the gestational age at delivery, the higher the frequency of intraamniotic infection. The histologic correlates of intrauterine infection are inflammation of the placental membranes (chorioamnionitis) and inflammation of the fetal umbilical cord (funisitis). The most common microorganisms implicated in intrauterine infections leading to preterm labor are *Ureaplasma urealyticum*, *Mycoplasma hominis*, *Gardnerella vaginalis* (the dominant organism found in “bacterial vaginosis,” a polymicrobial infection), *Trichomonas*, gonorrhea, and *Chlamydia*. In developing countries, malaria and HIV are significant contributors to the burden of preterm labor and prematurity. Recent studies have begun to elucidate the molecular mechanisms of inflammation-induced preterm labor. Toll-like receptors (TLRs), which bind bacterial components as natural ligands (Chapter 6), have emerged as key players in this

process. It is postulated that signals produced by TLR engagement alter prostaglandin production, which, in turn, induces uterine smooth muscle contractions.

- *Uterine, cervical, and placental structural abnormalities:* Uterine distortion (e.g., uterine fibroids), compromised structural support of the cervix (“cervical incompetence”), placenta previa, and abruptio placentae (Chapter 22) are associated with an increased risk of prematurity.
- *Multiple gestation* (twin pregnancy)

The hazards of prematurity are manifold for the newborn and may give rise to one or more of the following:

- Neonatal respiratory distress syndrome, also known as hyaline membrane disease
- Necrotizing enterocolitis (NEC)
- Sepsis
- Intraventricular and germinal matrix hemorrhage

Fetal Growth Restriction

Although preterm infants have low birth weights, their weight is usually appropriate once adjusted for gestational age. In contrast, as many as one-third of infants who weigh less than 2500 g are born at term and are therefore undergrown rather than immature. These small-for-gestational-age (SGA) infants have fetal growth restriction (FGR), which may result from maternal, fetal, or placental abnormalities, although in many cases the specific cause is unknown.

Maternal Abnormalities

By far the most common factors associated with SGA infants are maternal conditions that result in decreased placental blood flow. Vascular diseases such as preeclampsia (toxemia of pregnancy) and chronic hypertension are often the underlying cause. The list of other maternal conditions associated with SGA infants is long, but some of the avoidable factors worth mentioning are maternal opiate abuse, alcohol intake, and heavy cigarette smoking. Drugs causing FGR include both classic teratogens, such as chemotherapeutic agents, and some commonly administered therapeutic agents, such as phenytoin (Dilantin). Maternal malnutrition (in particular, prolonged hypoglycemia) may also affect fetal growth.

Fetal Abnormalities

Fetal influences are those that intrinsically reduce growth potential of the fetus despite an adequate supply of nutrients from the mother. Prominent among such fetal conditions are chromosomal disorders, congenital anomalies, and congenital infections. Chromosomal abnormalities may be detected in up to 17% of fetuses sampled for FGR and in up to 66% of fetuses with documented ultrasonographic malformations. Among the first group, the abnormalities include triploidy (7%), trisomy 18 (6%), trisomy 21 (1%), trisomy 13 (1%), and a variety of deletions and translocations (2%). Fetal infection should be considered in all infants with FGR. Those most commonly responsible for FGR are the TORCH group of infections (*toxoplasmosis, other viruses and bacteria* such as syphilis, rubella, cytomegalovirus, and *herpesvirus*). Infants who are SGA because of fetal factors usually have symmetric growth restriction (also referred to as proportionate FGR), meaning that all organ systems are similarly affected.

Placental Abnormalities

During the third trimester of pregnancy, vigorous fetal growth places particularly heavy demands on the uteroplacental blood supply. Therefore, the adequacy of placental growth in the preceding mid-trimester is extremely important, and uteroplacental insufficiency is an important cause of growth restriction. This insufficiency may result

from umbilical-placental vascular anomalies (such as single umbilical artery and abnormal cord insertion), placental abruption, placenta previa, placental thrombosis and infarction, chronic villitis of unknown etiology, placental infection, or multiple gestations (Chapter 22). In some cases, the placenta (and the infant) may be small without any detectable underlying cause. Placental causes of FGR tend to result in asymmetric (or disproportionate) growth restriction of the fetus with relative sparing of the brain. Physiologically, this general type of FGR is viewed as a down-regulation of growth in the latter half of gestation because of limited availability of nutrients or oxygen.

The SGA infant faces a difficult course, not only during the struggle for survival in the perinatal period, but also in childhood and adult life. Depending on the underlying cause of FGR and, to a lesser extent, the degree of prematurity, there is a significant risk of morbidity in the form of a major disability such as cerebral dysfunction, learning disability, or hearing and visual impairment.

Neonatal Respiratory Distress Syndrome

There are many causes of respiratory distress in the newborn. The most common cause is neonatal respiratory distress syndrome (RDS), also known as hyaline membrane disease because of the deposition of a layer of hyaline proteinaceous material in the peripheral airspaces of infants who succumb to this condition. Others include excessive sedation of the mother, fetal head injury during delivery, aspiration of blood or amniotic fluid, and intrauterine hypoxia brought about by coiling of the umbilical cord about the neck. The incidence of RDS increases with decreasing gestational age, being 1% at 37 weeks, 10.5% at 34 weeks, and 93% in extremely preterm infants (28 weeks or less) without preventive treatment.

Pathogenesis

The fundamental defect in RDS is pulmonary immaturity and deficiency of surfactant. As described in Chapter 15, surfactant consists predominantly of *dipalmitoyl phosphatidylcholine* (lecithin), smaller amounts of phosphatidylglycerol, and two groups of surfactant-associated proteins. The first group is composed of the hydrophilic glycoproteins SP-A and SP-D, which play a role in pulmonary host defense (innate immunity). The second group consists of the hydrophobic surfactant proteins SP-B and SP-C, which, in concert with the surfactant lipids, are involved in the reduction of surface tension at the air-liquid barrier in the alveoli of the lung. With reduced surface tension in the alveoli, less pressure is required to keep them patent and hence aerated. The importance of surfactant proteins in normal lung function can be gauged by the occurrence of severe respiratory failure in neonates with congenital deficiency of surfactant caused by mutations in the *SFTPB* or *SFTPC* genes.

Surfactant production by type II alveolar cells is accelerated after the 35th week of gestation in the fetus. At birth, the first breath of life requires high inspiratory pressures to expand the lungs. With normal levels of surfactant, the lungs retain up to 40% of the residual air volume after the first breath; thus, subsequent breaths require far lower inspiratory pressures. With a deficiency of surfactant, the lungs collapse with each successive breath, so infants must work as hard with each successive breath as they did with the first. The problem of stiff atelectatic lungs is compounded by the soft thoracic wall that is pulled in as the diaphragm descends. Progressive atelectasis and reduced lung compliance lead to a chain of events as depicted in Fig. 10.6, resulting in protein-rich, fibrin-rich exudation into the alveolar spaces and the formation of hyaline membranes. Hyaline membranes are barriers to gas exchange, leading to carbon dioxide retention and hypoxemia. The hypoxemia itself further impairs surfactant synthesis, and a vicious cycle ensues.

Surfactant synthesis is modulated by a variety of hormones and growth factors, including cortisol, insulin, prolactin, thyroxine, and

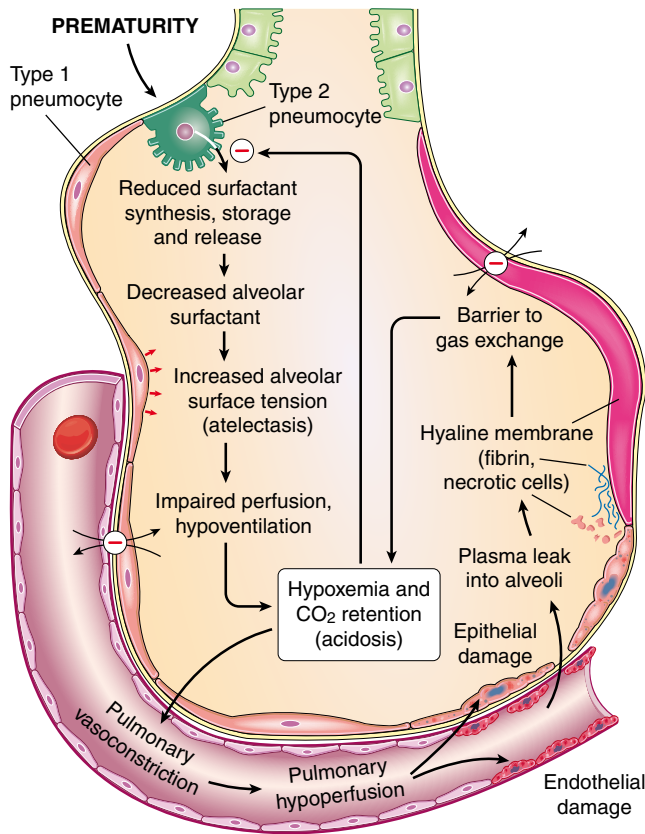


Fig. 10.6 Pathophysiology of respiratory distress syndrome.

TGF- β . The role of glucocorticoids is particularly important. Conditions associated with intrauterine stress and FGR that increase corticosteroid release lower the risk of developing RDS. Surfactant synthesis can be suppressed by the compensatory high blood levels of insulin in infants of diabetic mothers, which counteracts the effects of steroids. This may explain, in part, why infants of diabetic mothers have a higher risk of developing RDS. Labor is known to increase surfactant synthesis; hence, cesarean delivery before the onset of labor may increase the risk of RDS.

MORPHOLOGY

The lungs are distinctive on gross examination. Though of normal size, they are solid, airless, and reddish purple, similar to the color of the liver, and they usually sink in water, indicating the relative absence of entrapped air. Microscopically, alveoli are poorly developed, and those that are present are collapsed (**Fig. 10.7**). When the infant dies early in the course of the disease, necrotic cellular debris can be seen in the terminal bronchioles and alveolar ducts. The necrotic material becomes incorporated within **eosinophilic hyaline membranes** lining the respiratory bronchioles, alveolar ducts, and alveoli. The membranes are largely made up of fibrin admixed with cell debris derived chiefly from necrotic type II pneumocytes. The sequence of events that leads to the formation of hyaline membranes is depicted in **Fig. 10.6**. There is a remarkable paucity of neutrophilic inflammatory reaction associated with these membranes. The lesions of hyaline membrane disease are never seen in stillborn infants.

In infants who survive more than 48 hours, progressive reparative changes occur in the lungs. The alveolar epithelium proliferates under the surface of the membrane, which may detach into the airspace where it undergoes partial digestion or phagocytosis by macrophages. If it does not detach, fibroblasts grow into the membrane, and it becomes incorporated into the alveolar wall.

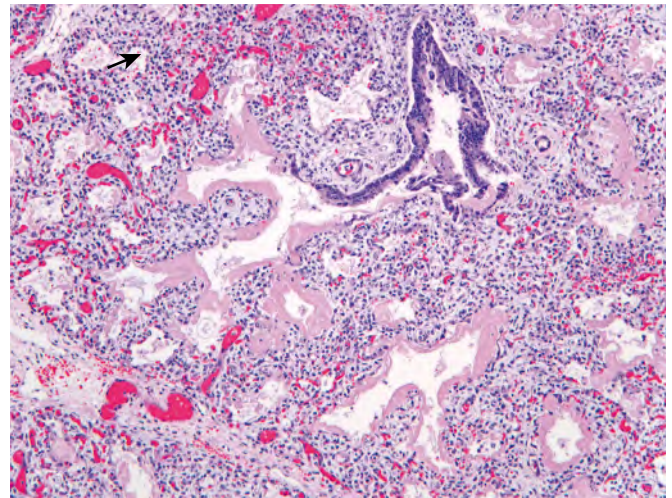


Fig. 10.7 Hyaline membrane disease. There are eosinophilic thick hyaline membranes lining the dilated alveoli. Note that the remaining airspaces are lined by cuboidal epithelium (arrow), indicating immaturity of the lung.

Clinical Features

Affected infants are almost always preterm and have weights appropriate for gestational age. There are strong associations with male sex, maternal diabetes, and cesarean delivery. Resuscitation may be necessary at birth, but usually within a few minutes rhythmic breathing and normal color are reestablished. Soon afterward, often within 30 minutes, breathing becomes more difficult, and within a few hours cyanosis becomes evident in the untreated infant. Fine rales can be heard over both lung fields. A chest radiograph usually reveals uniform minute reticulogranular densities, producing a so-called *ground-glass* picture. In the full-blown condition, the respiratory distress persists, cyanosis increases, and even the administration of 80% oxygen by a variety of ventilatory methods fails to improve the situation. If therapy staves off death for the first 3 or 4 days, however, the infant has an excellent chance of recovery.

The clinical course and prognosis for neonatal RDS vary depending on the maturity and birth weight of the infant and the promptness of surfactant therapy. A major thrust in the control of RDS focuses on prevention, either by delaying labor until the fetal lung reaches maturity or by inducing maturation of the lung in the at-risk fetus by use of antenatal steroids. Critical to these objectives is the ability to assess fetal lung maturity accurately. Because pulmonary secretions are discharged into the amniotic fluid, analysis of amniotic fluid phospholipids provides a good estimate of the level of surfactant in the alveolar lining. Prophylactic administration of exogenous surfactant at birth to extremely premature infants (gestational age <28 weeks) has been shown to be very beneficial. Together with newer assisted ventilation techniques, this has resulted in dramatic improvement in pulmonary function, a shortened course, and markedly reduced mortality. In uncomplicated cases, recovery begins to occur within 3 or 4 days.

Two complications of RDS, including retrolental fibroplasia (also called retinopathy of prematurity) in the eyes and bronchopulmonary dysplasia, are now less frequent and less severe when they do occur.

- *Retinopathy of prematurity* has a two-phase pathogenesis. During the hyperoxic phase of RDS therapy (phase I), expression of the proangiogenic vascular endothelial growth factor (VEGF) is markedly decreased, causing endothelial cell apoptosis; VEGF levels rebound after return to relatively hypoxic room air ventilation

(phase II), inducing retinal vessel proliferation (neovascularization) characteristic of the lesions in the retina (Chapter 29).

- **Bronchopulmonary dysplasia** is characterized by a striking decrease in alveolar septation (manifested as large, simplified alveolar structures) and a dysmorphic capillary configuration. The current view is that bronchopulmonary dysplasia is caused by a potentially reversible impairment in the development of alveolar septation at the so-called “saccular” stage. Multiple factors—hyperoxemia, hyperventilation, prematurity, inflammatory cytokines, and vascular maldevelopment—may contribute to bronchopulmonary dysplasia and probably act additively or synergistically to promote injury. The levels of a variety of proinflammatory cytokines (TNF, IL-1 β , IL-6, and IL-8) are increased in the alveoli of affected infants, suggesting a possible role for these cytokines in arresting pulmonary development. Treatment consists of respiratory support, fluid restriction, diuretics, and, in extremely preterm infants (age <28 weeks), glucocorticoids.

Infants who recover from RDS are also at increased risk for developing a variety of other complications associated with preterm birth; most important among these are patent ductus arteriosus, intraventricular hemorrhage, and necrotizing enterocolitis.

KEY CONCEPTS

Neonatal Respiratory Distress Syndrome

- Neonatal RDS (hyaline membrane disease) is a disease of prematurity; most cases occur in neonates born before 28 weeks' gestational age.
- The fundamental abnormality in RDS is insufficient pulmonary surfactant, which results in failure of lungs to inflate after birth.
- The characteristic morphologic feature of RDS is the presence of hyaline membranes (consisting of necrotic epithelial cells and plasma proteins) lining the airways.
- RDS can be ameliorated by prophylactic administration of steroids, surfactant therapy, and improved ventilation techniques.
- Long-term sequelae associated with RDS therapy include retinopathy of prematurity and bronchopulmonary dysplasia; the incidence of both complications has decreased with improvements in management of RDS.

Necrotizing Enterocolitis

Necrotizing enterocolitis (NEC) is most common in premature infants, with the incidence of the disease being inversely proportional to the gestational age. It occurs in approximately 1 in 10 very-low-birth-weight infants (<1500 g). Approximately 2500 cases occur annually in the United States.

The pathogenesis of NEC is unknown and suspected to be multifactorial. In addition to prematurity, most cases are associated with enteral feeding, suggesting that some postnatal insult (such as introduction of bacteria in the gut) sets in motion a cascade of events culminating in tissue destruction. Although infectious agents may play a role in the pathogenesis of NEC, no single bacterial pathogen has been linked to the disease. Many inflammatory mediators have been associated with NEC and it is suspected that they act by increasing mucosal permeability resulting from enterocyte apoptosis and compromising intercellular tight junctions, thus adding “fuel to the fire.” Ultimately, breakdown of mucosal barrier functions permits transmural migration of gut bacteria, leading to a vicious cycle of inflammation, mucosal necrosis, and further bacterial entry, eventually culminating in sepsis and shock (Chapter 4).

The clinical course is typical, with the onset of bloody stools, abdominal distention, and development of circulatory collapse. Abdominal radiographs often demonstrate gas within the intestinal wall (*pneumatosis intestinalis*). When detected early, NEC often can be managed conservatively, but many cases (20% to 60%) require resection of the necrotic segments of bowel. NEC is associated with high perinatal mortality; those who survive often develop post-NEC strictures from fibrosis caused by the healing process. Probiotic therapies are being evaluated for prevention of NEC.

MORPHOLOGY

NEC typically involves the terminal ileum, cecum, and right colon, although any part of the small or large intestine may be involved. The involved segment is distended, friable, and congested, or it can be frankly gangrenous; intestinal perforation with accompanying peritonitis may occur. Microscopically, mucosal, or transmural coagulative necrosis, ulceration, bacterial colonization, and submucosal gas bubbles may be seen (Fig. 10.8). Reparative changes, such as the formation of granulation tissue and fibrosis, may begin shortly after the acute episode.



Fig. 10.8 Necrotizing enterocolitis (NEC). (A) Postmortem examination in a severe case of NEC shows the entire small bowel is markedly distended with a perilously thin wall (usually this implies impending perforation). (B) The congested portion of the ileum corresponds to areas of hemorrhagic infarction and transmural necrosis microscopically. Submucosal gas bubbles (pneumatosis intestinalis) can be seen in several areas (arrows).

PERINATAL INFECTIONS

Fetal and perinatal infections are most often acquired through one of two routes—transcervically (also referred to as ascending) or transplacentally (hematologic). Occasionally, infections occur by a combination of the two routes when an ascending microorganism infects the endometrium and then invades the fetal bloodstream via the chorionic villi.

Transcervical (Ascending) Infections

Most bacterial and a few viral (e.g., herpes simplex II, SARS-CoV-2) infections can be acquired by the cervicovaginal route. Such infections may be acquired in utero or around the time of birth. In general, the fetus acquires the infection either by inhaling infected amniotic fluid into the lungs shortly before birth or by passing through an infected birth canal during delivery. As stated earlier, preterm birth is a common and unfortunate consequence of infection. Preterm birth due to infection may be related either to damage and rupture of the amniotic sac as a direct consequence of the inflammation, or to the induction of labor by prostaglandins released from infiltrating neutrophils. Inflammation of the placental membranes and cord are usually seen, but the presence of chorioamnionitis does not necessarily correlate with the severity of the fetal infection. In the fetus infected by inhalation of amniotic fluid, pneumonia, sepsis, and meningitis are the most common sequelae.

Transplacental (Hematologic) Infections

Most parasitic (e.g., *Toxoplasma*, malaria) and viral infections and a few bacterial infections (i.e., *Listeria*, *Treponema*) gain access to the fetal bloodstream transplacentally via the chorionic villi. Hematogenous transmission may occur at any time during gestation or occasionally, as may be the case with hepatitis B and HIV, at the time of delivery via maternal-to-fetal transfusion. The clinical manifestations of these infections are highly variable, depending largely on the gestational timing and microorganism involved.

Parvovirus B19, which causes erythema infectiosum or “fifth disease” of childhood in immunocompetent older children, can infect 1% to 5% of seronegative (nonimmune) pregnant females, and the vast majority have a normal pregnancy outcome. Adverse pregnancy outcomes in remaining cases include spontaneous abortion (particularly in the second trimester), stillbirth, hydrops fetalis (see later), and congenital anemia. Parvovirus B19 has a strong tropism for erythroid cells, and diagnostic viral inclusions can be seen in early erythroid progenitors in infected infants (Fig. 10.9).

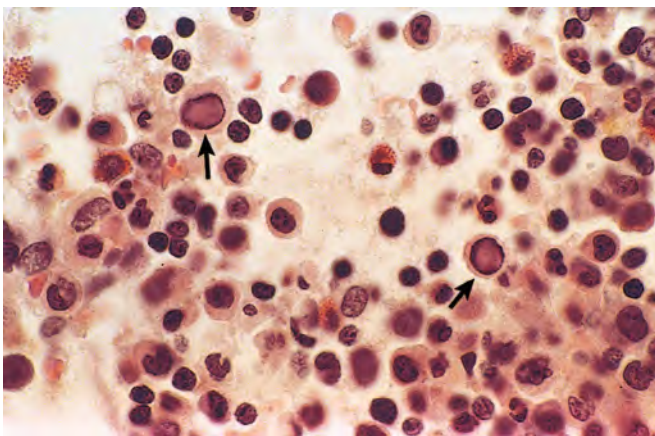


Fig. 10.9 Bone marrow from an infant infected with parvovirus B19. The arrows indicate two erythroid precursors with large homogeneous intranuclear inclusions and a surrounding peripheral rim of residual chromatin.

The **TORCH group of infections** (see earlier) are grouped together because they may evoke similar clinical and pathologic manifestations, including fever, encephalitis, chorioretinitis, hepatosplenomegaly, pneumonitis, myocarditis, hemolytic anemia, and vesicular or hemorrhagic skin lesions. Infections occurring early in gestation may cause chronic sequelae in the child, including growth retardation and intellectual disability, cataracts, congenital cardiac anomalies, and bone defects.

Sepsis

Perinatal sepsis can be grouped clinically into early onset (within the first 7 days of life) and late onset (from 7 days to 3 months). Most cases of early-onset sepsis are acquired at or shortly before birth and tend to result in clinical signs and symptoms of pneumonia, sepsis, and occasionally meningitis within 4 or 5 days of life. Group B *Streptococcus* is the most common cause of early-onset sepsis as well as early-onset bacterial meningitis. Infections with *Listeria* and *Candida* have longer latent periods between the time of microorganism inoculation and the appearance of clinical symptoms, and present as late-onset sepsis.

FETAL HYDROPS

Fetal hydrops refers to the accumulation of edema fluid in the fetus during intrauterine growth. In the past, hemolytic anemia caused by Rh blood group incompatibility between mother and fetus (immune hydrops) was the most common cause, but with successful prophylaxis of this disorder during pregnancy, nonimmune hydrops is more common. The intrauterine fluid accumulation is quite variable, from progressive, generalized edema of the fetus (*hydrops fetalis*), a usually lethal condition, to more localized edema, such as isolated pleural and peritoneal effusions, or postnuchal fluid accumulation (*cystic hygroma*, see later) that are compatible with life.

Immune Hydrops

Immune hydrops is a hemolytic disease caused by blood group antigen incompatibility between mother and fetus. When the fetus inherits red cell antigenic determinants from the father that are foreign to the mother, a maternal immune reaction may occur. The major antigens known to induce clinically significant immunologic reactions are certain Rh antigens and the ABO blood group antigens. The reaction occurs in second and subsequent pregnancies in an Rh-negative mother with an Rh-positive father.

Etiology and Pathogenesis

Immune hydrops is caused by an immune response in the mother to blood group antigens on fetal red cells, leading to the production of maternal antibodies that pass through the placenta and lyse fetal red cells (Fig. 10.10). Fetal red cells may reach the maternal circulation during the last trimester of pregnancy, when the cytotrophoblast is no longer present as a barrier, or during childbirth itself. The mother thus becomes sensitized to the foreign antigen. The initial exposure to Rh antigen evokes the formation of IgM antibodies that, unlike IgG antibodies, do not cross the placenta. Thus, Rh disease is uncommon with the first pregnancy. Exposure during a subsequent pregnancy generally leads to a brisk IgG antibody response and the risk of immune hydrops.

Of the numerous antigens included in the Rh system, only the D antigen is a major cause of Rh incompatibility. Several factors influence the immune response to RhD-positive fetal red cells that reach the maternal circulation.

- **Concurrent ABO incompatibility protects the mother against Rh immunization** because the fetal red cells are promptly coated and removed

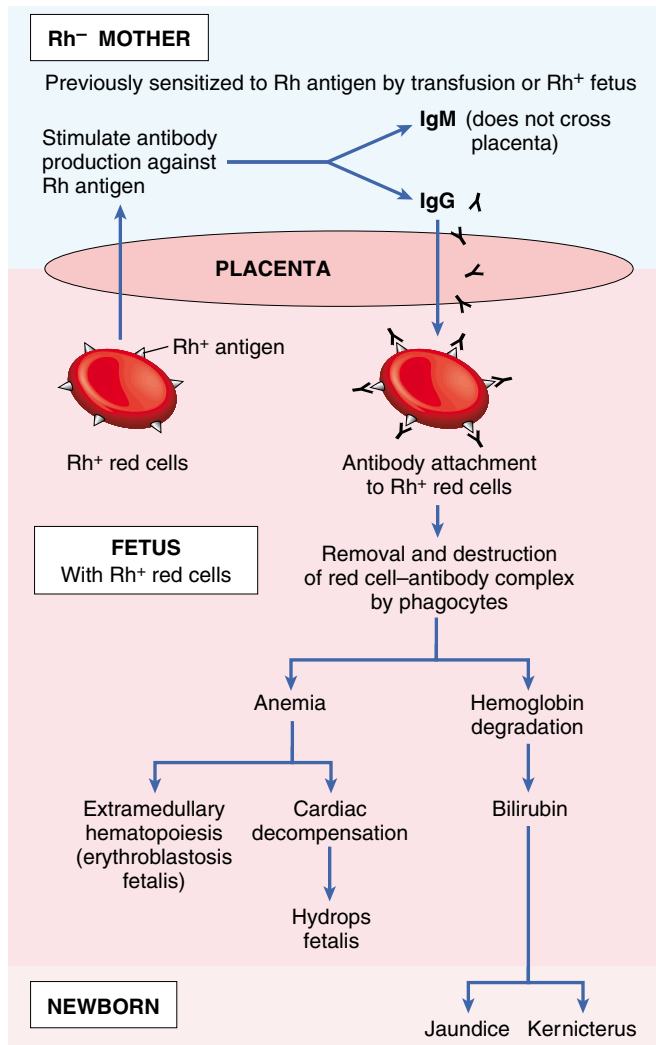


Fig. 10.10 Pathogenesis of immune hydrops fetalis.

from the maternal circulation by anti-A or anti-B IgM antibodies that do not cross the placenta.

- The antibody response depends on the dose of immunizing antigen; hence, hemolytic disease develops only when the mother has experienced a significant transplacental bleed (>1 mL of Rh-positive fetal red cells).

The incidence of maternal Rh isoimmunization has decreased significantly since the use of Rhesus immune globulin (RhIg) containing anti-D antibodies. Administration of RhIg to Rh-negative mothers at 28 weeks of gestation and within 72 hours of delivery significantly decreases the risk for hemolytic disease in Rh-positive neonates and in subsequent pregnancies; RhIg is also administered following abortions, because these too can lead to immunization. Antenatal identification and management of the at-risk fetus have been greatly facilitated by amniocentesis and chorionic villus and fetal blood sampling. In addition, cloning of the *RHD* gene has resulted in efforts to determine fetal Rh status using maternal serum because it contains fetal DNA. When identified, cases of severe intrauterine hemolysis may be treated by fetal intravascular transfusions via the umbilical cord and early delivery.

The pathogenesis of fetal hemolysis caused by maternal-fetal ABO incompatibility is slightly different from that caused by Rh

antigen incompatibility. ABO incompatibility occurs in approximately 20% to 25% of pregnancies, but hemolysis occurs in only 1 in 10 of such infants and is severe enough to require treatment in only 1 in 200 cases. Several factors account for this. First, most anti-A and anti-B antibodies are of the IgM type and hence do not cross the placenta. Second, neonatal red cells express A and B antigens at low levels. Third, many cells other than red cells express A and B antigens and thus absorb some of the transferred antibody. ABO hemolytic disease occurs almost exclusively in infants of group A or B who are born of group O mothers. For unknown reasons, certain group O females possess IgG antibodies directed against group A or B antigens (or both) even without prior sensitization. Therefore, the firstborn may be affected. There is no effective protection against ABO reactions, but severe immune hydrops due to ABO incompatibility is rare.

There are two consequences of excessive destruction of red cells in the neonate (see Fig. 10.10). The severity of these changes varies considerably, depending on the degree of hemolysis and the maturity of the infant.

- **Anemia** is a direct result of red cell loss. If hemolysis is mild, increased red cell production may suffice to maintain near-normal levels of red cells. However, with more severe hemolysis, progressive anemia develops and may result in hypoxic injury to the heart and liver. Because of liver injury, plasma protein synthesis decreases, and levels of these proteins may drop to as low as 2 to 2.5 mg/dL. Cardiac hypoxia may lead to cardiac decompensation and failure. The combination of reduced plasma oncotic pressure and increased hydrostatic pressure in the circulation (secondary to cardiac failure) results in generalized edema and anasarca, culminating in hydrops fetalis.
- **Jaundice** develops because hemolysis produces unconjugated bilirubin (Chapter 18), which passes through the infant's poorly developed blood-brain barrier. Being water-insoluble, bilirubin binds to lipids in the brain and can damage the CNS, causing kernicterus (see Fig. 10.13).

Nonimmune Hydrops

The three major causes of nonimmune hydrops include cardiovascular defects, chromosomal anomalies, and fetal anemia. Both structural and functional cardiovascular defects, such as congenital malformations and arrhythmias, may result in intrauterine cardiac failure and hydrops. Among the chromosomal anomalies, 45,X karyotype (Turner syndrome) and trisomies 21 and 18 are associated with fetal hydrops because of the accompanying structural cardiac anomalies. In Turner syndrome, abnormalities of lymphatic drainage from the neck may lead to postnuchal fluid accumulation (cystic hygromas) as well. Fetal anemia, not caused by Rh- or ABO-associated antibodies, can also result in hydrops. In fact, in some parts of the world (e.g., Southeast Asia), severe fetal anemia due to homozygous α -thalassemia, resulting from deletion of all four α -globin genes, is probably the most common cause of nonimmune hydrops (Chapter 14). Transplacental infection by parvovirus B19 is rapidly emerging as an important cause of hydrops (see earlier). The virus gains preferential entry into erythroid precursors (normoblasts), where it replicates, leading to apoptosis of red cell progenitors and red cell aplasia. Parvoviral intranuclear inclusions can be seen within circulating and marrow erythroid precursors (see Fig. 10.9). Approximately 10% of cases of nonimmune hydrops are related to monozygous twin pregnancies and twin-to-twin transfusion occurring through anastomoses between the two circulations. In up to 20% of cases, the cause is unknown.



Fig. 10.11 Hydrops fetalis. (A) There is generalized accumulation of fluid in the fetus. (B) Fluid accumulation is particularly prominent in the soft tissues of the neck, and this condition has been termed cystic hygroma. Cystic hygromas are characteristically seen, but not limited to constitutional chromosomal anomalies such as 45,X karyotypes. (Courtesy Dr. Beverly Rogers, Department of Pathology, University of Texas Southwestern Medical Center, Dallas.)

MORPHOLOGY

The anatomic findings in fetuses with intrauterine fluid accumulation vary with the severity of the disease and the underlying etiology. As previously noted, **hydrops fetalis** is the most severe and generalized manifestation (Fig. 10.11), and lesser degrees of edema such as isolated pleural, peritoneal, or post-nuchal fluid collections can occur. Accordingly, infants may be stillborn, die within the first few days, or recover completely. The presence of dysmorphic features suggests a chromosomal abnormality; postmortem examination may reveal an underlying cardiac anomaly.

In hydrops associated with fetal anemia, the fetus and placenta are characteristically pale; in most cases the liver and spleen are enlarged from cardiac failure and congestion. Additionally, the bone marrow demonstrates compensatory hyperplasia of erythroid precursors (parvovirus-associated red cell aplasia being a notable exception), and extramedullary hematopoiesis is present in the liver, spleen, lymph nodes, and possibly other tissues such as the kidneys, lungs, and even the heart (Fig. 10.12). The increased hematopoietic activity accounts for the presence in the peripheral circulation of large numbers of immature red cells, including reticulocytes and erythroblasts (**erythroblastosis fetalis**).

The most serious threat in immune fetal hydrops is CNS damage, known as **kernicterus** (Fig. 10.13). The affected brain is enlarged and edematous and, when sectioned, has a bright yellow color because of deposition of bilirubin, particularly the basal ganglia, thalamus, cerebellum, cerebral gray matter, and spinal cord. The precise level of bilirubin that induces kernicterus is unpredictable, but neural damage usually requires a blood bilirubin level greater than 20 mg/dL in term infants; in premature infants this threshold may be considerably lower.

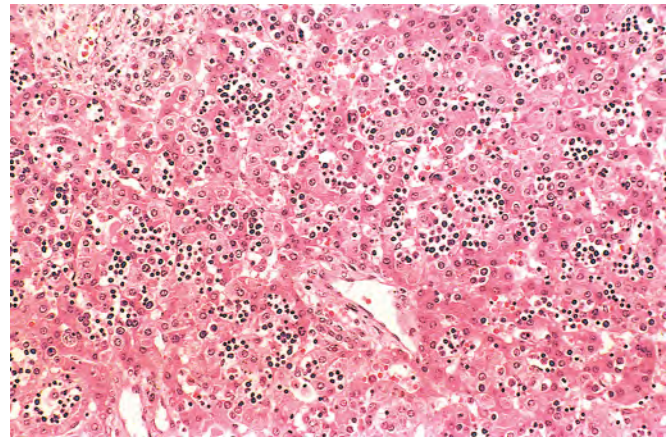


Fig. 10.12 Nonimmune hydrops fetalis. Numerous islands of extramedullary hematopoiesis (small blue cells) are scattered among mature hepatocytes in the liver of an infant with anemia and nonimmune hydrops fetalis.

hemolytic reactions), whereas the most gravely ill neonates present with intense jaundice, generalized edema, and signs of neurologic injury. These infants may be supported by a variety of measures, including *phototherapy* (visual light oxidizes toxic unconjugated bilirubin to harmless, readily excreted, water-soluble dipyrroles) and, in severe cases, total exchange transfusion of the infant.

Clinical Features

The clinical manifestations of fetal hydrops vary with the severity of the disease and can be inferred from the preceding discussion. Minimally affected infants display pallor, possibly accompanied by hepatosplenomegaly (to which may be added jaundice with more severe

KEY CONCEPTS

Fetal Hydrops

- Fetal hydrops refers to the accumulation of edema fluid in the fetus during intrauterine growth.



Fig. 10.13 Kernicterus. Note the yellow discoloration of the brain parenchyma due to bilirubin accumulation, which is most prominent in the basal ganglia deep to the ventricles.

- The degree of fluid accumulation is variable, from generalized hydrops fetalis to localized cystic hygromas.
- The most common causes of fetal hydrops are nonimmune (chromosomal abnormalities, cardiovascular defects, and fetal anemia), since immune hydrops has become less frequent as a result of Rh antibody prophylaxis.
- Erythroblastosis fetalis (circulating immature erythroid precursors) is a characteristic finding of fetal anemia-associated hydrops.
- Hemolysis-induced hyperbilirubinemia can result in bilirubin toxicity (kernicterus) in the basal ganglia and brainstem, particularly in premature infants.

INHERITED METABOLIC DISORDERS AND OTHER GENETIC DISORDERS

Inherited metabolic disorders, also commonly called **inborn errors of metabolism**, are genetic disorders that give rise to defects in the metabolism of carbohydrates, proteins, fatty acids, and other substances within the body. Since Sir Archibald Garrod coined the term in 1908, the number of recognized metabolic diseases has increased exponentially. Most inherited metabolic disorders are rare diseases that are generally autosomal recessive or X-linked traits (Chapter 5). Mitochondrial disorders (Chapter 5) form a distinct entity by themselves. Although each individual disease is rare, inherited metabolic disorders collectively affect 1 in 800 to 1 in 2500 live births. Some of the clinical features that suggest an underlying metabolic disorder in a neonate are listed in Table 10.2. The clinical manifestations of these diseases are generally the result of either abnormal metabolite accumulation or deficiency of the required product. Three genetic disorders, phenylketonuria (PKU), galactosemia, and cystic fibrosis, are selected for discussion here. PKU and galactosemia are reviewed because their early diagnosis (via neonatal screening programs) is particularly important, and death or intellectual disability can be prevented with appropriate dietary regimens. Cystic fibrosis is included because it is one of the most common, potentially lethal, diseases occurring in individuals of European descent.

TABLE 10.2 Abnormalities Suggesting Inborn Errors of Metabolism

General
Dysmorphic features
Deafness
Self-mutilation
Abnormal hair
Abnormal body or urine odor (“sweaty feet”; “mousy or musty”; “maple syrup”)
Hepatosplenomegaly; cardiomegaly
Hydrops
Neurologic
Hypotonia or hypertonia
Coma
Persistent lethargy
Seizures
Gastrointestinal
Poor feeding
Recurrent vomiting
Jaundice
Eyes
Cataracts
Cherry red macula
Dislocated lens
Glaucoma
Muscle, Joints
Myopathy
Abnormal mobility

Modified from Barness LA, Gilbert-Barness E. Metabolic diseases. In: Gilbert-Barness E, Barness LA, eds. *Potter's Pathology of the Fetus, Infant, and Child*. Mosby; 2007.

Phenylketonuria

There are several variants of this inherited metabolic disorder, which affects 1 in 10,000 live-born infants of Northern European descent. The most common form, referred to as *classic PKU*, is more common in persons of Scandinavian descent and less common in African American and Jewish populations.

PKU is an autosomal recessive disorder caused by a severe deficiency of the enzyme phenylalanine hydroxylase (PAH) leading to hyperphenylalaninemia. Affected infants are normal at birth but within a few weeks develop an elevated plasma phenylalanine level, which impairs brain development. Usually by 6 months of age, severe intellectual disability becomes evident; fewer than 4% of untreated phenylketonuric children have IQs (intelligence quotients) greater than 50 or 60, and the damage is irreversible. About one-third of these children never walk and two-thirds cannot talk. Seizures, other neurologic abnormalities, decreased pigmentation of hair and skin, a characteristic musty odor, and eczema often accompany the intellectual disability in untreated children. Hyperphenylalaninemia and the resultant intellectual disability can be avoided by restricting phenylalanine intake early in life. Hence, all states in the United States have adopted screening procedures to detect PKU in the immediate postnatal period. In adults, additional therapies for PKU include supplementation with the PAH cofactor tetrahydrobiopterin (BH₄), or enzyme substitution therapy with phenylalanine ammonia-lyase (PAL), an enzyme that metabolizes excess phenylalanine.

Many female patients with PKU, if treated with dietary restriction early in life, reach childbearing age and are clinically asymptomatic. Most of them have marked hyperphenylalaninemia, because dietary

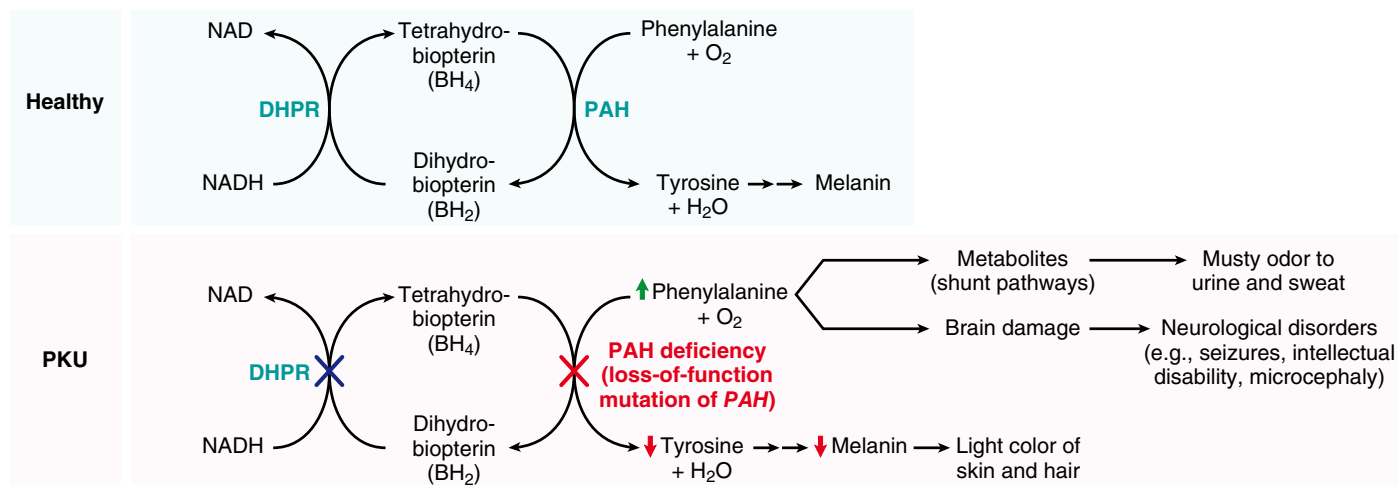


Fig. 10.14 Phenylalanine hydroxylase (PAH) system. *Top*, PAH system in a healthy person. *Bottom*, Phenylketonuria (PKU) can arise from deficiency of PAH (classic PKU, red X) or dihydropteridine reductase (DHPR; variant PKU, blue X). *NAD*, Nicotinamide adenine dinucleotide; *NADH*, reduced NADH.

treatment is discontinued after they reach adulthood. Between 75% and 90% of children born to such females have intellectual disability and are microcephalic, and 15% have congenital heart disease, even though the infants themselves are heterozygotes. This syndrome, termed *maternal PKU*, results from the teratogenic effects of phenylalanine or its metabolites that cross the placenta and affect specific fetal organs during development. The presence and severity of the fetal anomalies directly correlate with the maternal phenylalanine level, so it is imperative that maternal dietary restriction of phenylalanine be initiated before conception and continued throughout pregnancy.

The biochemical abnormality in PKU is an inability to convert phenylalanine into tyrosine. In normal children, less than 50% of the dietary intake of phenylalanine is used for protein synthesis. The remainder is converted to tyrosine by the phenylalanine hydroxylase system (Fig. 10.14). When phenylalanine metabolism is blocked because of a lack of functional PAH, increase flux through normally minor shunt pathways yield metabolites that are excreted in large amounts in the urine and sweat. These impart a strong musty or mousy odor to affected infants. It is believed that excess phenylalanine or its metabolites contribute to the brain damage in PKU. Concomitant lack of tyrosine (see Fig. 10.14), a precursor of melanin, is responsible for lightening of skin and hair color.

At the molecular level, almost 1000 mutant alleles of the *PAH* gene have been identified. Mutations in both *PAH* alleles are required to develop the disease. Infants with mutations resulting in severely reduced PAH activity present with markedly elevated blood phenylalanine levels and the classic features of PKU, and those with less severe deficiencies present with milder disease. Some *PAH* mutations result in only modest elevations of blood phenylalanine levels without associated neurologic damage. This condition, referred to as *benign hyperphenylalaninemia*, is important to recognize because these individuals may have positive screening tests but do not develop the stigmata of classic PKU. Because of the numerous possible disease-causing mutations in the *PAH* gene, molecular diagnosis is difficult, and measurement of serum phenylalanine levels is necessary to differentiate benign hyperphenylalaninemia from PKU; the levels in the latter are typically five-fold or more above normal. Once a biochemical diagnosis is established, the specific mutation causing PKU can be determined. With the identification

of the mutation, carrier testing of at-risk family members can be performed.

Although 98% of PKU is attributable to mutations in *PAH*, approximately 2% occur due to abnormalities in synthesis or recycling of the cofactor tetrahydrobiopterin (BH_4 ; see Fig. 10.14). Sapropterin, a synthetic BH_4 , is found to be effective in such patients. It is clinically important to recognize these variant forms of PKU because they cannot be treated by dietary restriction of phenylalanine.

KEY CONCEPTS

Phenylketonuria

- PKU is an autosomal recessive disorder caused by a lack of the enzyme phenylalanine hydroxylase and a consequent inability to metabolize phenylalanine.
- Clinical features of untreated PKU may include severe mental disability, seizures, and decreased pigmentation of skin, all of which can be avoided by restricting the intake of phenylalanine in the diet.
- Females with PKU who discontinue dietary treatment have high phenylalanine levels and can give birth to children with neurologic deficits resulting from transplacental passage of phenylalanine metabolites.

Galactosemia

Galactosemia is an autosomal recessive disorder of galactose metabolism resulting in the accumulation of galactose-1-phosphate in tissues. It affects 1 in 60,000 live-born infants. Normally, lactose, the major carbohydrate of mammalian milk, is split into glucose and galactose in the intestinal microvilli by lactase. Galactose is then converted to glucose in several steps catalyzed by distinct enzymes. Two variants of galactosemia have been identified. In the more common variant, there is a total lack of galactose-1-phosphate uridyl transferase (GALT). The rare variant arises from a deficiency of galactokinase. Because galactokinase deficiency leads to a milder form of the disease that is not associated with intellectual disability, it is not considered in this discussion.

As a result of GALT deficiency, galactose-1-phosphate accumulates in many tissues, including the liver, spleen, lens of the eye,

kidneys, heart muscle, cerebral cortex, and red cells. Alternative metabolic pathways are activated, leading to the production of galactitol (a polyol metabolite of galactose) and galactonate, an oxidized by-product of excess galactose, both of which accumulate in the tissues. Long-term toxicity in galactosemia has been ascribed to these metabolic intermediates. Heterozygotes may have a mild enzyme deficiency and are spared the clinical and pathologic consequences of the homozygous state.

The clinical picture is variable, probably reflecting the heterogeneity of mutations in the *GALT* gene. The liver, eyes, and brain bear the brunt of the damage. The early-to-develop hepatomegaly is due largely to fatty change, but in time, widespread scarring that closely resembles the cirrhosis of alcohol use disorder may supervene. Cataracts develop, probably because the lens absorbs water and swells as galactitol, produced by alternative metabolic pathways, accumulates, increasing osmotic pressure. Nonspecific alterations appear in the CNS, including loss of nerve cells, gliosis, and edema, particularly in the dentate nuclei of the cerebellum and the olivary nuclei of the medulla. Similar changes may occur in the cerebral cortex and white matter.

Affected infants fail to thrive almost from birth. Vomiting and diarrhea appear within a few days of milk ingestion. Jaundice and hepatomegaly usually become evident during the first week of life and may seem to be a continuation of the physiologic jaundice of the newborn. Cataracts develop within a few weeks, and within the first 6 to 12 months of life, intellectual disability appears. Even in untreated infants, however, the disability is usually not as severe as that seen in PKU. Accumulation of galactose and galactose-1-phosphate in the kidney impairs amino acid transport, resulting in aminoaciduria. There is an increased frequency of fulminant *E. coli* septicemia, possibly because of depressed neutrophil bactericidal activity. Hemolysis and coagulopathy in the newborn period can occur as well.

Newborn screening tests are widely utilized in the United States. They most commonly depend on fluorometric assay of GALT enzyme activity on a dried blood spot. A positive screening test must be confirmed by assay of GALT levels in red cells. Antenatal diagnosis is possible by assay of GALT activity in cultured amniotic fluid cells or measuring galactitol levels in amniotic fluid.

Many of the clinical and morphologic changes of galactosemia can be prevented or ameliorated by early removal of galactose from the diet for at least the first 2 years of life. Control instituted soon after birth prevents cataracts and liver damage and permits almost normal development. Even with dietary restrictions, however, older patients are frequently affected by speech defects and gonadal failure (especially premature ovarian failure) and, less commonly, ataxia.

KEY CONCEPTS

Galactosemia

- Galactosemia is an autosomal recessive disorder caused by lack of the galactose-1-phosphate uridylyltransferase, an enzyme required for conversion of galactose to glucose. Its absence leads to the accumulation of galactose-1-phosphate and its metabolites in tissues.
- Clinical features may include jaundice, liver damage, cataracts, neural damage, vomiting and diarrhea, and *E. coli* sepsis. Dietary restriction of galactose can prevent at least some of the more severe complications.

Cystic Fibrosis (Mucoviscidosis)

Cystic fibrosis is an inherited disorder of ion transport that affects fluid secretion in exocrine glands and in the epithelial lining of the respiratory, gastrointestinal, and reproductive tracts. In many individuals this disorder leads to abnormally viscous secretions that obstruct organ passages, resulting in most of the clinical features of this disorder,

such as chronic lung disease secondary to recurrent infections, pancreatic insufficiency, steatorrhea, malnutrition, hepatic cirrhosis, intestinal obstruction, and male infertility. These manifestations may appear at any point in life, from before birth to much later in childhood or even in adolescence.

Cystic fibrosis is the most common lethal genetic disease that affects individuals of Northern European descent, with an incidence of approximately 1 in 2500 live births. The carrier frequency in the United States is 1 in 20 among people of Northern European descent, but carriers are also present at lower frequency in other populations worldwide. Although cystic fibrosis follows an autosomal recessive transmission pattern, heterozygote carriers have a higher incidence of respiratory and pancreatic disease compared with the general population. In addition, there is a wide degree of phenotypic variation that results from three factors: pathogenic mutations, tissue-specific effects of the encoded gene product, and the influence of modifier genes.

Cystic Fibrosis Gene: Normal Structure and Function

The primary defect in cystic fibrosis is abnormal transport of chloride and bicarbonate ions mediated by an anion channel encoded by the **cystic fibrosis transmembrane conductance regulator (CFTR) gene on chromosome 7q31.2**. The 1480-amino acid polypeptide encoded by *CFTR* has two transmembrane domains (each containing six α -helices), two cytoplasmic nucleotide-binding domains (NBDs), and a regulatory domain (R domain) that contains protein kinase A phosphorylation sites (Fig. 10.15). The two transmembrane domains

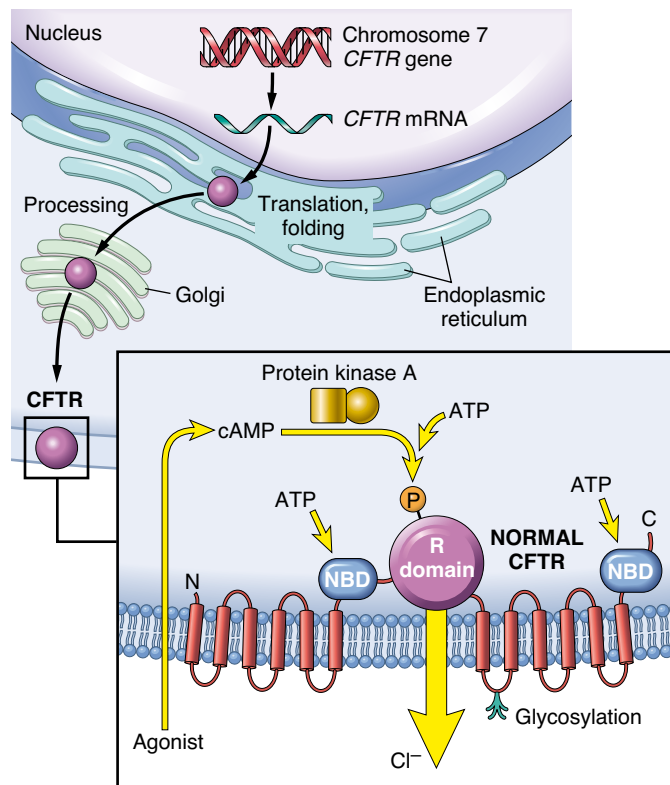


Fig. 10.15 Function of CFTR. CFTR consists of two transmembrane domains, two nucleotide-binding domains (NBDs), and a regulatory R domain. Agonists (e.g., acetylcholine) bind to epithelial cells and increase levels of cyclic adenosine monophosphate (cAMP), which activates protein kinase A; the latter phosphorylates R domain of the CFTR using adenosine triphosphate (ATP). This results in the opening of the chloride ion channel.

form a channel through which chloride passes. Activation of the CFTR channel is mediated by agonist-induced increases in cyclic adenosine monophosphate (cAMP), leading to activation of a protein kinase A, which phosphorylates the R domain. Adenosine triphosphate (ATP) binding and hydrolysis occurs at the NBD and is essential for opening and closing of the channel pore, which is enhanced by phosphorylation of the R domain. Several important facets of CFTR function have emerged in recent years:

- *CFTR regulates multiple ion channels and cellular processes.* Although initially characterized as a chloride-conductance channel, it is now recognized that CFTR regulates additional ion channels and cellular processes. These include so-called outwardly rectifying chloride channels, *inwardly rectifying potassium channels* (Kir6.1), the *epithelial sodium channel* (ENaC), gap junction channels, and cellular processes involved in ATP transport and mucus secretion. Of these, the interaction of CFTR with ENaC has possibly the most pathophysiologic relevance in cystic fibrosis. ENaC is situated on the apical surface of epithelial cells and is responsible for sodium uptake from the luminal fluid, rendering the luminal fluid hypotonic. ENaC is normally inhibited by CFTR; hence, in cystic fibrosis, ENaC activity increases, markedly augmenting sodium uptake across the apical membrane. The importance of this phenomenon is discussed later in the context of pulmonary and gastrointestinal pathology in cystic fibrosis. The one exception to this is human sweat ducts, where ENaC activity *decreases* because of *CFTR* mutations; therefore, hypertonic fluid with high sodium chloride (the

sine qua non of classic cystic fibrosis) is formed. This is the basis for the “salty” sweat that parents can often detect in their affected infants.

- *The functions of CFTR are tissue-specific; therefore, the impact of a mutation in CFTR is also tissue-specific.* The major function of CFTR in sweat gland ducts is to reabsorb luminal chloride ions and augment sodium reabsorption via ENaC (see earlier). Therefore, in the sweat ducts, loss of CFTR function leads to decreased reabsorption of sodium chloride and the production of hypertonic sweat (Fig. 10.16). However, in the respiratory and intestinal epithelium, CFTR is one of the most important channels for luminal secretion of chloride. At these sites, *CFTR* mutations result in loss or reduction of chloride secretion into the lumen (see Fig. 10.16), and luminal sodium absorption is increased due to loss of inhibition of ENaC activity. These changes in ion distribution result in increased passive water reabsorption from the lumen, lowering the water content of the surface fluid layer coating mucosal cells. Thus, unlike in the sweat ducts, there is no difference in the salt concentration of the surface fluid layer coating respiratory and intestinal mucosal cells in those with cystic fibrosis when compared with normal individuals. Instead, the pathogenesis of respiratory and intestinal complications in cystic fibrosis seems to stem from an isotonic but low-volume surface fluid layer. In the lungs, this dehydration leads to defective mucociliary action and the accumulation of hyperconcentrated, viscid secretions that obstruct the air passages and predispose to recurrent pulmonary infections.

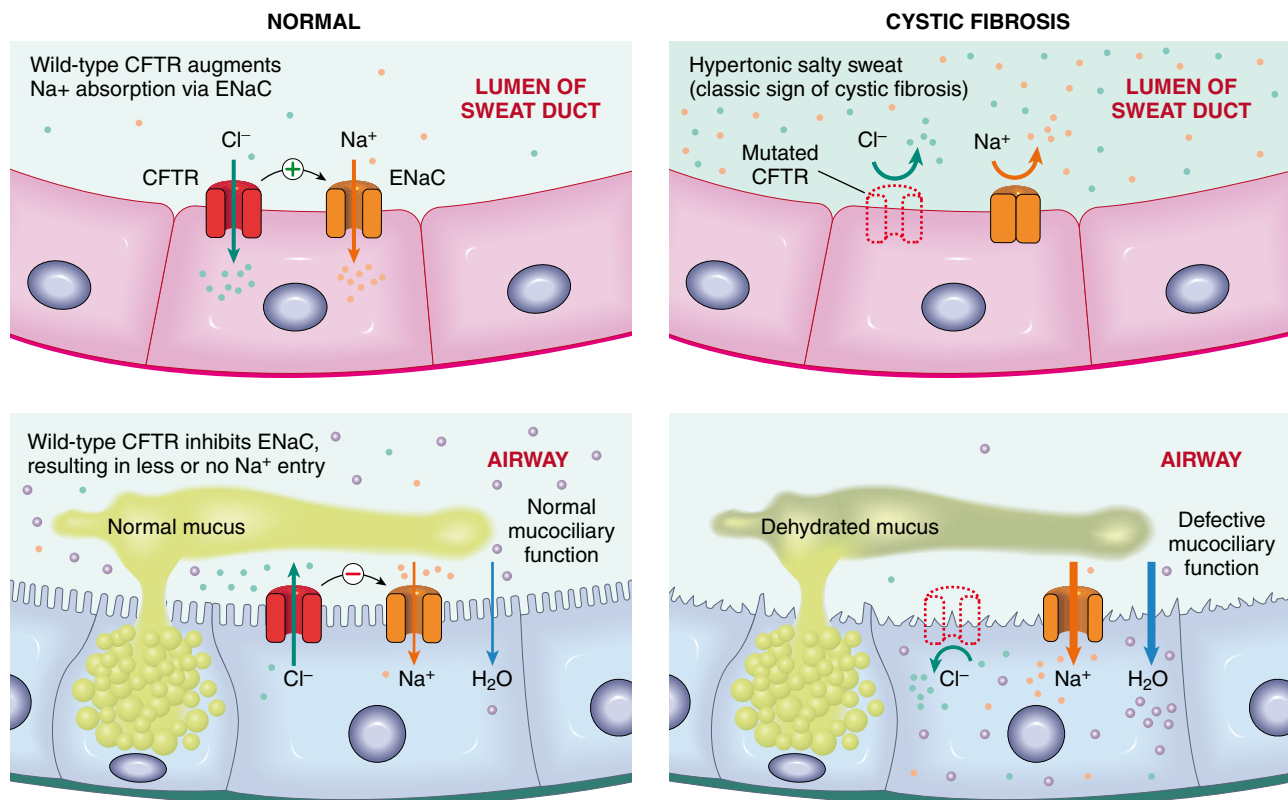


Fig. 10.16 Pathogenesis of cystic fibrosis. CFTR chloride channel defect in the sweat ducts (*top*) causes increased sodium chloride concentration in sweat. In the airway (*bottom*), patients with cystic fibrosis have decreased chloride (Cl^-) secretion and increased sodium (Na^+) and water (H_2O) reabsorption, leading to dehydration of the mucus layer coating epithelial cells, defective mucociliary action, and mucus plugging of airways. CFTR, Cystic fibrosis transmembrane conductance regulator; ENaC, epithelial sodium channel.

- CFTR *regulates transport of bicarbonate ions*. CFTR plays both a direct and indirect role in regulating bicarbonate transport across the apical epithelial membrane. The anion channel of CFTR is not specific for chloride and transports bicarbonate as well. In addition, CFTR has reciprocal interactions with the SLC26 family of anion exchangers, which are co-expressed on the apical surface with CFTR. In some CFTR mutants, chloride transport is completely or substantially preserved, and bicarbonate transport is markedly abnormal. Normal tissue secretes alkaline fluids; in contrast, fluids that are acidic (due to absence of bicarbonate ions) are secreted by epithelia harboring these mutant CFTR alleles. The acidity of secretions results in decreased luminal pH that can lead to a variety of adverse effects, such as increased mucin precipitation, plugging of ducts, and increased binding of bacteria to mucins. Pancreatic insufficiency, a feature of classic cystic fibrosis, is virtually always present when there are CFTR mutations with abnormal bicarbonate conductance.

Cystic Fibrosis Gene: Mutational Spectrum and Genotype-Phenotype Correlation

Since the CFTR gene was cloned in 1989, more than 2000 disease-associated mutations have been identified; only five disease-causing mutations have a relative frequency of 1% or more. Most of the mutations are missense alterations, but frameshift, splicing, and nonsense mutations also occur. The mutations can be grouped into six classes based on their effect on the CFTR protein (Fig. 10.17). Class I to III mutations result in less than 10% residual CFTR activity and are termed “severe” mutations, and class IV to VI mutations result in less than 20% residual CFTR activity and are termed “mild” mutations.

- Class I: *Defective protein synthesis (null mutations)*. These mutations result in premature termination of protein translation and are associated with complete lack of CFTR protein at the apical surface of epithelial cells.
- Class II: *Abnormal protein folding, processing, and trafficking (processing mutations)*. These mutations result in defective processing of the protein from the endoplasmic reticulum to the Golgi apparatus; the protein does not become fully folded and glycosylated and is instead degraded before it reaches the cell surface. The most common CFTR mutation is a class II mutation caused by deletion of three nucleotides coding for phenylalanine at amino acid position 508 ($\Delta F508$). Worldwide, this mutation is found in approximately 70% of patients of Northern European descent with cystic fibrosis. Class II mutations are associated with near-complete lack of CFTR protein at the apical surface of epithelial cells.
- Class III: *Defective regulation (gating mutations)*. Mutations in this class prevent activation of CFTR by abrogating ATP binding and hydrolysis, an essential prerequisite for ion transport (see earlier). Thus, there is a normal amount of CFTR on the apical surface, but it is nonfunctional.
- Class IV: *Decreased conductance (conduction mutations)*. These mutations typically occur in the transmembrane domain of CFTR, which forms the ionic pore for chloride transport. There is a normal amount of CFTR at the apical membrane, but with reduced chloride conduction.
- Class V: *Reduced abundance (production mutations)*. These mutations typically affect intronic splice sites or the CFTR promoter, such that there is a reduced amount of normal protein.
- Class VI: *Decreased membrane CFTR stability (instability mutations)*. These mutations give rise to fully processed and functional proteins but with greatly reduced membrane stability.

Because cystic fibrosis is an autosomal recessive disease, affected individuals harbor mutations on both alleles. However, the nature of mutations on each of the two alleles can have a remarkable effect on the overall phenotype, as well as on organ-specific manifestations (Fig. 10.18). Thus, two “severe” mutations that produce virtual absence of membrane CFTR function are associated with the classic cystic fibrosis phenotype (pancreatic insufficiency, sinopulmonary infections, and gastrointestinal symptoms), and the presence of a “mild” mutation on one or both alleles results in a less severe phenotype. This general dictum of genotype-phenotype correlation is most consistent for pancreatic disease, wherein the presence of one allele with a “mild” mutation associated with some CFTR activity can prevent the pancreatic insufficiency that is virtually always seen with homozygosity for “severe” mutations. In contrast, genotype-phenotype correlations are far less consistent in pulmonary disease, because of secondary modifiers (see later). As genetic testing for CFTR mutations has expanded, it has become evident that some patients who present with clinical features apparently unrelated to cystic fibrosis may also harbor CFTR mutations. These include individuals with idiopathic chronic pancreatitis, late-onset chronic pulmonary disease, idiopathic bronchiectasis, and obstructive azoospermia caused by bilateral absence of the vas deferens (see detailed discussion of individual phenotypes later). Most of these patients do not demonstrate other features of cystic fibrosis, despite the presence of bi-allelic CFTR mutations; these patients are classified as having *nonclassic* or *atypical cystic fibrosis*. Identifying these individuals is important not only for subsequent management, but also for genetic counseling.

Genetic and Environmental Modifiers

Although cystic fibrosis remains one of the best-known examples of the “one gene, one disease” axiom, there is considerable evidence that genes other than CFTR modify the frequency and severity of certain organ-specific manifestations, especially pulmonary manifestations and neonatal meconium ileus. Not surprisingly, polymorphisms in genes whose products modulate neutrophil function in response to bacterial infections act as modifier loci for the severity of pulmonary disease in cystic fibrosis. Examples of such modifier genes include mannose binding lectin 2 (MBL2), transforming growth factor $\beta 1$ (TGFB1), and interferon-related developmental regulator 1 (IFRD1). It is postulated that polymorphisms in these genes regulate the resistance of the lungs to exogenous infections with virulent microbes (see later), thus modifying the natural history of cystic fibrosis.

Environmental modifiers may also explain some of the significant phenotypic differences between individuals who share the same CFTR genotype. This is best exemplified in pulmonary disease, where CFTR genotype-phenotype correlations can be perplexing. As stated earlier, defective mucociliary action because of deficient hydration of the mucus results in an inability to clear bacteria from the airways. *Pseudomonas aeruginosa* species, in particular, colonize the lower respiratory tract, first intermittently and then chronically. Concurrent viral infections predispose to such colonization. The static mucus creates a hypoxic microenvironment in the airway surface fluid, which in turn favors the production of alginate, a mucoid polysaccharide capsule, by colonizing bacteria. Alginate production permits the formation of a biofilm that protects the bacteria from antibodies and antibiotics, allowing them to evade host defenses and produce a chronic destructive lung disease. Antibody- and cell-mediated immune reactions induced by the organisms result in further pulmonary destruction but are ineffective against the organism. Approximately 80% of cystic fibrosis patients are colonized by *Pseudomonas aeruginosa* by 20 years of age. Chronic colonization by these bacteria is a significant contributor to the morbidity and mortality of cystic fibrosis.

Normal	I	II	III	IV	V	VI
CFTR defect	No functional CFTR protein	CFTR trafficking/processing defect	Defective channel regulation	Decreased channel conductance	Reduced synthesis of CFTR	Decreased CFTR stability
Type of mutations	Nonsense; frameshift; canonical splice	Missense; amino acid deletion	Missense; amino acid change	Missense; amino acid change	Splicing defect; missense	Missense; amino acid change
Specific mutation examples	Gly542X Trp1282X Arg553X 621+1G→T	Phe508del Asn1303Lys Ile507del Arg560Thr	Gly551Asp Gly178Arg Gly551Ser Ser549Asn	Arg117His Arg347Pro Arg117Gys Arg334Ttp	3849+10kC→T 2789+5G→A 3120+1G→A 5T variant	4326delTC Gln412X 4279insA

Fig. 10.17 Classes of *CFTR* mutations. Mutations in the cystic fibrosis transmembrane conductance regulator (*CFTR*) gene can be divided into six classes. Class I mutations result in no protein production. Class II mutations (including the most prevalent, Phe508del or ΔF508) cause retention of a misfolded protein at the endoplasmic reticulum, and subsequent degradation in the proteasome. Class III mutations affect channel regulation, impairing channel opening (e.g., Gly551Asp). Class IV mutants show reduced conduction—i.e., decreased flow of ions (e.g., Arg117His). Class V mutations cause substantial reduction in mRNA or protein, or both. Class VI mutations cause substantial plasma membrane instability. (From Boyle MP, De Boeck K. A new era in the treatment of cystic fibrosis: correction of the underlying CFTR defect. *Lancet Respir Med*. 2013;1:158-163.)

Level of CFTR function	<5%	<10%	<20%	50%
Sinuses	Severe chronic sinusitis	Moderate chronic sinusitis		Increased rate of chronic sinusitis
Lungs	Severe lung disease	Variable lung disease		Increased rate of lung disease
Sweat glands	High sweat chloride	Intermediate sweat chloride		
Pancreas	Pancreatic insufficiency	Pancreatic sufficiency		
Intestines	Meconium ileus	Distal intestinal obstruction syndrome		
Vas deferens	Absent vas deferens			Increased risk of absent vas deferens

Fig. 10.18 Clinical manifestations of mutations in the cystic fibrosis gene. The clinical consequences vary from most severe to asymptomatic depending on the level of CFTR protein. (Modified from Chang EH, Zabner J. Precision genomic medicine in cystic fibrosis. *Clin Transl Sci*. 2015;8:606-610.)

It is evident, therefore, that in addition to genetic factors (e.g., class of mutation), a plethora of environmental modifiers (e.g., intercurrent and concurrent infections by microorganisms) can influence the severity and progression of lung disease in cystic fibrosis.

MORPHOLOGY

The anatomic changes are highly variable in distribution and severity. In individuals with nonclassic cystic fibrosis, the disease is quite mild and does not seriously disturb their growth and development. In others, the pancreatic involvement is severe and impairs intestinal absorption because of pancreatic insufficiency (Chapter 19), so malabsorption stunts development and postnatal growth. In others, the mucus secretion defect leads to defective mucociliary action, obstruction of bronchi and bronchioles, and disabling fatal pulmonary infections. In all variants, the **sweat glands are morphologically unaffected**.

Pancreatic abnormalities are present in approximately 85% to 90% of patients with cystic fibrosis. In milder cases, there may be only accumulation of mucus in the small ducts with some dilation of the exocrine glands. In more severe cases, usually seen in older children or adolescents, the ducts are completely plugged, causing atrophy of the exocrine glands and progressive fibrosis, leaving only islets within a fibrofatty stroma (Fig 10.19). The loss of pancreatic exocrine secretion impairs fat absorption, and the associated avitaminosis A may contribute to squamous metaplasia of the pancreatic duct lining epithelium, which is already injured by the inspissated mucus secretions. Thick viscid plugs of mucus may also be found in the small intestine of infants. Sometimes these plugs cause small bowel obstruction, known as **meconium ileus**.

Liver involvement follows the same basic pattern. Bile canaliculi are plugged by mucus material, accompanied by ductular proliferation and portal inflammation. Hepatic **steatosis** is not uncommon. Over time, **focal biliary cirrhosis** develops in approximately one-third of patients (Chapter 18), which can eventually involve the entire liver, resulting in diffuse hepatic nodularity. Such severe hepatic involvement is encountered in less than 10% of patients.

The **salivary glands** frequently show histologic changes similar to those described in the pancreas: progressive dilation of ducts, squamous metaplasia of the lining epithelium, and glandular atrophy followed by fibrosis.

The **pulmonary changes** are the most serious complications of this disease (Fig. 10.20). These stem from the viscous mucus secretions of the submucosal glands of the respiratory tree, leading to secondary obstruction and infection of air passages. The bronchioles are often distended with thick mucus associated

with marked hyperplasia and hypertrophy of the mucus-secreting cells. Superimposed infections give rise to severe chronic bronchitis and bronchiectasis (Chapter 15). In many instances, lung abscesses develop. *Staphylococcus aureus*, *Haemophilus influenzae*, and *Pseudomonas aeruginosa* are the three most common organisms responsible for lung infections. As mentioned earlier, a mucoid form of *P. aeruginosa* (alginate-producing) is particularly frequent and causes chronic inflammation. Even more sinister is the increasing frequency of infection with another group of pseudomonads, the *Burkholderia cepacia* complex, which includes at least nine different species; of these, infections with *B. cenocepacia* are the most common in cystic fibrosis patients. This opportunistic bacterium is particularly hardy, and infection with this organism has been associated with fulminant illness (“**cepacia syndrome**”), longer hospital stays, and increased mortality. Other opportunistic bacterial pathogens include *Stenotrophomonas maltophilia* and nontuberculous mycobacteria; allergic bronchopulmonary aspergillosis also occurs with increased frequency in cystic fibrosis.

Azoospermia and infertility are found in 95% of males who survive to adulthood; **congenital bilateral absence of the vas deferens** is a frequent finding in these patients. In some males, bilateral absence of the vas deferens may be the only feature suggesting an underlying *CFTR* mutation.

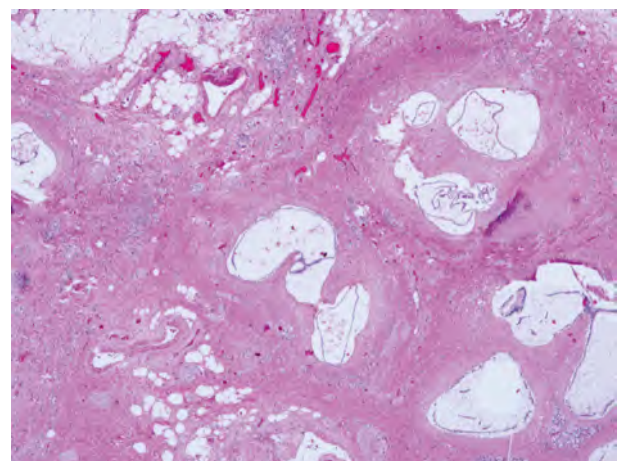


Fig. 10.19 End-stage pancreatic disease in cystic fibrosis. The ducts are markedly dilated, and exocrine glands are destroyed and replaced by fibrous tissue.

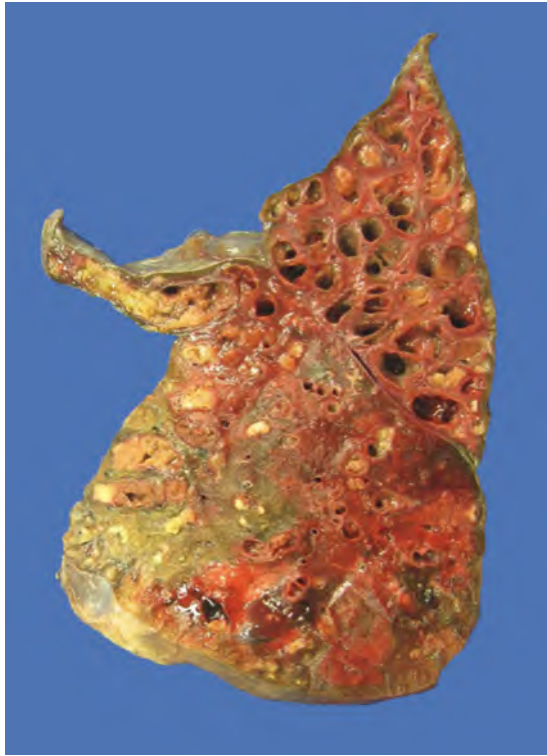


Fig. 10.20 Lung with cystic fibrosis. There is extensive mucus plugging and dilation of the tracheobronchial tree. The pulmonary parenchyma is consolidated by a combination of secretions and pneumonia, with the green color being associated with *Pseudomonas* infection.

Clinical Features

Few childhood diseases are as protean as cystic fibrosis in clinical manifestations (Table 10.3; Fig. 10.18). The symptoms are extremely varied and may appear at birth or years later, and involve one organ system or many. Approximately 5% to 10% of cases come to clinical attention at birth or soon after because of meconium ileus. Distal intestinal obstruction can also occur in older individuals, manifesting as recurrent episodes of right lower quadrant pain sometimes associated with a palpable mass of meconium, with or without associated intussusception, in the right iliac fossa.

Exocrine pancreatic insufficiency occurs in the majority (85% to 90%) of patients with cystic fibrosis and is associated with “severe” *CFTR* mutations on both alleles (e.g., $\Delta F508/\Delta F508$), whereas 10% to 15% of patients with one “severe” and one “mild” *CFTR* mutation (e.g., $\Delta F508/R117H$) or two “mild” *CFTR* mutations retain sufficient pancreatic exocrine function to not require enzyme supplementation (pancreas-sufficient phenotype). Pancreatic insufficiency is associated with protein and fat malabsorption and increased fecal loss. Manifestations of malabsorption (e.g., large foul-smelling stools, abdominal distention, and poor weight gain) may appear during the first year of life. The faulty fat absorption may induce deficiency of the fat-soluble vitamins, resulting in manifestations of avitaminosis A, D, E, or K. Hypoproteinemia may be severe enough to cause generalized edema. Persistent diarrhea may result in rectal prolapse in up to 10% of children with this disease. The pancreas-sufficient phenotype is usually not associated with gastrointestinal complications, and, in general, these individuals demonstrate excellent growth and development. A subset of patients with pancreas-sufficient cystic fibrosis have recurrent bouts of pancreatitis associated with acute abdominal pain and

TABLE 10.3 Clinical Features and Diagnostic Criteria for Cystic Fibrosis

Clinical Features of Cystic Fibrosis

1. Chronic sinopulmonary disease manifested by:
 - a. Persistent colonization/infection with typical cystic fibrosis pathogens, including *Staphylococcus aureus*, nontypeable *Haemophilus influenzae*, mucoid and nonmucoid *Pseudomonas aeruginosa*, *Burkholderia cepacia*
 - b. Chronic cough and sputum production
 - c. Persistent chest radiograph abnormalities (e.g., bronchiectasis, atelectasis, infiltrates, hyperinflation)
 - d. Airway obstruction manifested by wheezing and air trapping
 - e. Nasal polyps; radiographic or computed tomographic abnormalities of paranasal sinuses
 - f. Digital clubbing
2. Gastrointestinal and nutritional abnormalities, including:
 - a. Intestinal: meconium ileus, distal intestinal obstruction syndrome, rectal prolapse
 - b. Pancreatic: pancreatic insufficiency, recurrent acute pancreatitis, chronic pancreatitis
 - c. Hepatic: chronic hepatic disease manifested by clinical or histologic evidence of focal biliary cirrhosis or multilobular cirrhosis, prolonged neonatal jaundice
 - d. Nutritional: failure to thrive (severe acute malnutrition), hypoproteinemia, edema, complications secondary to fat-soluble vitamin deficiency
3. Salt-loss syndromes: acute salt depletion, chronic metabolic alkalosis
4. Male urogenital abnormalities resulting in obstructive azoospermia (congenital bilateral absence of vas deferens)

Criteria for Diagnosis of Cystic Fibrosis

One or more characteristic phenotypic features, or a history of cystic fibrosis in a sibling, parent, or child, or a positive newborn screening test result
and
An increased sweat chloride concentration, or a sweat chloride concentration in the intermediate range and identification of two disease-causing *CFTR* mutations, or a sweat chloride concentration in the intermediate range and demonstration of abnormal epithelial nasal ion transport (nasal potential difference or intestinal current measurement)

Modified from Farrell PM, White TB, Ren CL, et al. Diagnosis of cystic fibrosis: consensus guidelines from the Cystic Fibrosis Foundation. *J Pediatr*. 2017; 181S:S4-S15.

occasionally, life-threatening complications. These patients have other features of classic cystic fibrosis, such as pulmonary disease. “Idiopathic” chronic pancreatitis can also occur as an isolated late-onset finding in the absence of other stigmata of cystic fibrosis (Chapter 19); biallelic *CFTR* mutations (usually one “mild,” one “severe”) are demonstrable in the majority of these individuals who have nonclassic or atypical cystic fibrosis. Endocrine pancreatic insufficiency (i.e., diabetes mellitus) occurs in up to 50% of adults with cystic fibrosis and is thought to be caused by destruction of pancreatic parenchyma including the islets.

Cardiorespiratory complications, such as persistent lung infections, obstructive pulmonary disease, and cor pulmonale, are the most common cause of death in cystic fibrosis patients in the United States. By 18 years of age, 80% of patients with classic cystic fibrosis harbor *P. aeruginosa*, and 3.5% harbor *B. cepacia*. With the indiscriminate use of antibiotic prophylaxis against *Staphylococcus*, there has been an unfortunate resurgence of resistant strains of *Pseudomonas* in many patients. Individuals who carry one “severe” and one “mild” *CFTR* mutation may develop late-onset mild pulmonary disease, another example of nonclassic or atypical cystic fibrosis. Patients with mild pulmonary

disease usually have little or no pancreatic disease. Adult-onset “idiopathic” bronchiectasis has been linked to *CFTR* mutations in a subset of cases. Recurrent sinonasal polyps can occur in 10% to 25% of individuals with cystic fibrosis; hence, children who present with this finding should be tested for cystic fibrosis.

Significant *liver disease* occurs late in the natural history of cystic fibrosis and is gaining in clinical importance as life expectancies increase. In fact, after cardiopulmonary and transplantation-related complications, liver disease is the most common cause of death in cystic fibrosis. Most studies suggest that symptomatic or biochemical liver disease has its onset at or around puberty, with a prevalence of approximately 15%. However, asymptomatic hepatomegaly may be present in up to one-third of individuals. Obstruction of the common bile duct may occur due to stones or sludge; it presents with abdominal pain and the acute onset of jaundice. As previously noted, diffuse biliary cirrhosis develops in less than 10% of individuals with cystic fibrosis.

Approximately 95% of males with cystic fibrosis are infertile because of *obstructive azoospermia*. As mentioned earlier, this is most commonly due to congenital bilateral absence of the vas deferens, which is caused in 80% of cases by biallelic *CFTR* mutations.

In most cases, the diagnosis of cystic fibrosis is suspected based on persistently elevated sweat electrolyte concentrations (often a parent makes the diagnosis by recognizing the infant’s abnormally salty sweat), characteristic clinical findings (sinopulmonary disease and gastrointestinal manifestations), family history, or an abnormal newborn screening test. The most common newborn screening test is based on measuring the blood level of immunoreactive trypsinogen, which is produced by the pancreas; elevated levels result from pancreatic injury. Confirmation of the diagnosis of cystic fibrosis requires either a significantly elevated sweat serum chloride level; nasal transepithelial potential difference measurement consistent with cystic fibrosis; or identification of two disease-causing *CFTR* mutations by sequencing. A minority of patients with cystic fibrosis, especially those with at least one “mild” *CFTR* mutation, may have a normal or near-normal sweat test. Measurement of nasal transepithelial potential difference *in vivo* can be a useful adjunct under these circumstances; individuals with cystic fibrosis demonstrate a significantly more negative baseline nasal potential difference than controls. Sequencing the *CFTR* gene is the diagnostic gold standard.

There have been major improvements in the management of acute and chronic complications for cystic fibrosis, including more potent antimicrobial therapies, pancreatic enzyme replacement, and bilateral lung transplantation. New treatment modalities for restoring mutant *CFTR* function are also now available or are being tested that fall into three mechanistic categories:

- **Potentiators.** These agents keep the “gate” of the *CFTR* channel open. Hence, they are most useful in gating (class III) and conduction (class IV) mutations. They are also helpful for patients with production (class V) mutations, by improving the function of reduced amounts of *CFTR*.
- **Correctors.** These agents assist in proper folding of the *CFTR* protein, thereby increasing its trafficking to the cell surface. Hence, they are useful in treating patients with processing (class II) mutations including $\Delta F508$, the most common *CFTR* mutation.
- **Amplifiers.** These agents increase the amount of *CFTR* protein the cell makes and thus can be useful for patients with production (class V) mutations.

Combination therapies consisting of both “potentiator” agents and small-molecule *CFTR* “correctors” have been approved for the treatment of patients with specific *CFTR* mutations, including those with two copies of $\Delta F508$. For treatment of patients with class I nonsense

mutations, molecules that allow “ribosomal read-through” of premature stop codons are in the pipeline. Such therapy is also being tested in muscular dystrophies, in which stop codons prevent synthesis of dystrophin (Chapter 27). Overall, improvements in the management of cystic fibrosis have extended the median life expectancy to 50 years.

KEY CONCEPTS

Cystic Fibrosis (CF)

- Cystic fibrosis is an autosomal recessive disease caused by mutations in the *CFTR* gene encoding a transmembrane regulator (*CFTR*).
- *CFTR* is an ion channel that regulates the transport of chloride, sodium, and bicarbonate ions. The defect in chloride ion transport results in high salt concentrations in sweat and in viscous luminal secretions in respiratory and gastrointestinal tracts. Defective bicarbonate transport in pancreatic ducts contributes to obstruction due to precipitation of mucin.
- *CFTR* mutations can be severe (e.g., $\Delta F508$), resulting in multi-system disease, or mild, with limited disease extent and severity.
- Cardiopulmonary complications constitute the most common cause of death; pulmonary infections, especially with resistant *Pseudomonas* or *Burkholderia* species, are frequent. Bronchiectasis and right-sided heart failure are long-term sequelae.
- Pancreatic insufficiency is extremely common; infertility caused by congenital bilateral absence of vas deferens is a characteristic finding in adult males with CF.
- Liver disease, including cirrhosis, is increasing in frequency as life expectancy increases.
- Molecular therapies that enhance the transport or stability of mutant *CFTR* protein are useful in patients who harbor certain *CFTR* alleles.

SUDDEN INFANT DEATH SYNDROME

According to the National Institute of Child Health and Human Development (NICHD), sudden infant death syndrome (SIDS) is defined as “the sudden death of an infant under 1 year of age which remains unexplained after a thorough case investigation, including performance of a complete autopsy, examination of the death scene, and review of the clinical history.” It is important to emphasize that many cases of sudden death in infancy may have an unexpected anatomic or biochemical basis discernible at autopsy (Table 10.4), and these should not be labeled as SIDS, but rather as sudden unexpected infant death (SUID). The Centers for Disease Control and Prevention estimates that SIDS accounts for approximately one-half of SUID cases in the United States. An aspect of SIDS that is not stressed in the definition is that the infant usually dies while asleep, mostly in the prone or side position, hence the pseudonyms of crib death or cot death.

Epidemiology

As infant deaths due to nutritional problems and infections have come under control in high-resource countries, including the United States, SIDS has assumed greater importance as a cause of infant death. SIDS is the leading cause of death between 1 month and 1 year of age in the United States and the fourth leading cause of death overall in infancy, after congenital anomalies, diseases of prematurity and low birth weight, and maternal complications. Mostly because of nationwide “Back to Sleep” (now “Safe to Sleep”) campaigns, there has been a significant drop in SIDS-related mortality, from an estimated 120 deaths per 100,000 live births in 1992 to 35 per 100,000 in 2017. This number translates to about 1360 deaths due to SIDS in the United States. Despite the decline in SIDS and SUID in all races and ethnicities, the rate of SUID among non-Hispanic Black (70 per 100,000 live births) and

TABLE 10.4 Risk Factors and Postmortem Findings Associated With Sudden Infant Death Syndrome

Parental
Young maternal age (<20 years) Maternal smoking during pregnancy Drug abuse in <i>either</i> parent, specifically paternal marijuana and maternal opiate, cocaine use Short intergestational intervals Late or no prenatal care Low socioeconomic group African American and American Indian ethnicity (? socioeconomic factors)
Infant
Brainstem abnormalities, associated with delayed development of arousal and cardiorespiratory control Prematurity and/or low birth weight Male sex Product of multiple birth SIDS in a prior sibling Antecedent respiratory infections Germline polymorphisms in autonomic nervous system genes
Environment
Prone or side sleep position Sleeping on a soft surface Hyperthermia Cosleeping in first 3 months of life
Postmortem Abnormalities Detected in Cases of Sudden Unexpected Infant Death ^a
Infections Viral myocarditis Bronchopneumonia Unsuspected congenital anomaly Congenital aortic stenosis Anomalous origin of the left coronary artery from the pulmonary artery Traumatic child abuse Intentional suffocation (filicide) Genetic and metabolic defects Long QT syndrome (<i>SCN5A</i> and <i>KCNQ1</i> mutations) Fatty acid oxidation disorders (<i>MCAD</i> , <i>LCHAD</i> , <i>SCHAD</i> mutations) Histiocytoid cardiomyopathy (<i>MTCYB</i> mutations) Abnormal inflammatory responsiveness (partial deletions in <i>C4a</i> and <i>C4b</i>)

^aSudden infant death syndrome (SIDS) is not the only cause of sudden unexplained infant death, but rather is a diagnosis of exclusion. Therefore, performance of an autopsy may often reveal findings that would explain the cause of sudden death in an infant. These cases should *not*, strictly speaking, be labeled as SIDS.

C4, Complement component 4; *KCNQ1*, potassium voltage-gated channel, KQT-like subfamily, member 1; *LCHAD*, long-chain 3-hydroxyacyl coenzyme A dehydrogenase; *MCAD*, medium-chain acyl coenzyme A dehydrogenase; *MTCYB*, mitochondrial cytochrome b; *SCHAD*, short-chain 3-hydroxyacyl coenzyme A dehydrogenase; *SCN5A*, sodium channel, voltage-gated, type V, alpha subunit.

American Indian/Alaskan Native (77 per 100,000 live births) infants was more than double that of non-Hispanic White infants (35 per 100,000 live births) in 2017. SIDS rates for Asian/Pacific Islander and Hispanic infants were much lower than the rate for non-Hispanic White infants. Differences in the prevalence of supine positioning and other sleep environment conditions between racial and ethnic populations may contribute to these disparities. Worldwide, in countries where unexpected infant deaths are diagnosed as SIDS only after postmortem examination, the death rates from SIDS range from 10 per 100,000 live births in the Netherlands to 80 per 100,000 in New Zealand.

Approximately 90% of all SIDS deaths occur during the first 6 months of life, most between 2 and 4 months of age. This narrow window of peak susceptibility is a unique characteristic that is independent of other risk factors (described later) and the geographic locale. Most infants who die of SIDS die at home, usually during a period of sleep at night.

Pathogenesis

The circumstances surrounding SIDS have been explored in great detail, and it is generally accepted that it is a multifactorial condition. A “triple-risk” model of SIDS has been proposed, which postulates the interaction of three factors: (1) a vulnerable infant, (2) a critical developmental period in homeostatic control, and (3) an exogenous stressor. According to this model, several factors make the infant vulnerable to sudden death during a critical developmental period (i.e., the first 6 months of life). These vulnerability factors may relate to the parents, infant, or exogenous environmental stressors (see Table 10.4).

Although numerous factors have been proposed to account for a vulnerable infant, the most compelling hypothesis is that SIDS reflects a delayed development of “arousal” and cardiorespiratory control. The brainstem, and in particular, the medulla oblongata, plays a critical role in the body’s arousal response to noxious stimuli such as episodic hypercarbia, hypoxia, and thermal stress encountered during sleep. The serotonergic (5-HT) system of the medulla is implicated in these arousal responses, as well as regulation of other critical homeostatic functions such as respiratory drive, blood pressure, and upper airway reflexes. Abnormalities in serotonin-dependent signaling in the brainstem may be the underlying basis for SIDS in some infants.

Epidemiologic and genetic studies have identified additional vulnerability factors for SIDS in the triple-risk model. Infants who are born before term or who are low birth weight are at increased risk, and risk increases with decreasing gestational age or birth weight. Male sex is associated with a slightly greater incidence of SIDS. SIDS in a prior sibling is associated with a five-fold relative risk of recurrence, highlighting the importance of a genetic predisposition; traumatic child abuse must be carefully excluded under these circumstances. Most SIDS babies have an immediate prior history of a mild respiratory tract infection, but no single causative organism has been isolated. These infections may predispose an already vulnerable infant to even greater impairment of cardiorespiratory control and delayed arousal. In this context, laryngeal chemoreceptors have emerged as a putative “missing link” between upper respiratory tract infections, the prone position, and SIDS. When stimulated, these laryngeal chemoreceptors typically elicit an inhibitory cardiorespiratory reflex. Stimulation of the chemoreceptors is augmented by respiratory tract infections, which increase the volume of secretions, and by the prone position, which impairs swallowing and clearing of the airways, even in healthy infants. In a previously vulnerable infant with impaired arousal, the resulting inhibitory cardiorespiratory reflex may prove fatal. Genetic vulnerability factors in the infant include polymorphisms of genes related to serotonergic signaling and autonomic innervation, pointing to the importance of these processes in the pathophysiology of SIDS.

Maternal smoking during pregnancy has consistently emerged as a risk factor in epidemiologic studies of SIDS, with children exposed in utero to nicotine having more than double the risk of SIDS as children born to nonsmokers. Young maternal age, frequent childbirths, and inadequate prenatal care are all risk factors associated with increased incidence of SIDS in the offspring.

Among the potential “environmental stressors,” prone or side sleeping positions, sleeping with parents in the first 3 months, sleeping on soft surfaces, and thermal stress are probably the most important modifiable risk factors for SIDS. The prone or side positions

predispose an infant to one or more recognized noxious stimuli (hypoxia, hypercarbia, and thermal stress) during sleep. The side position was considered a reliable alternative to the prone sleeping position, but the American Academy of Pediatrics now recognizes the supine sleeping position as the only position that reduces the risk of SIDS. The Safe to Sleep campaign has resulted in substantial reductions in SIDS-related deaths since its inception in 1994.

SIDS is a diagnosis of exclusion, requiring careful examination of the death scene and a complete postmortem examination. The latter can reveal an unsuspected cause of sudden death in as many as 20% or more of these infants. Infections (e.g., viral myocarditis or bronchopneumonia) are the most common causes of sudden “unexpected” death, followed by unsuspected congenital anomalies. In part because of advancements in molecular diagnostics and knowledge of the human genome, several genetic causes of sudden unexpected infant death have emerged (see Table 10.4). For example, fatty acid oxidation disorders, characterized by defects in mitochondrial fatty acid oxidative enzymes, may be responsible for as many as 5% of sudden unexplained infant deaths.

MORPHOLOGY

In infants who have died of suspected SIDS, a variety of findings have been reported at postmortem examination. They are usually subtle and of uncertain significance and are not present in all cases. Multiple petechiae are the most common finding (~80% of cases); these are usually present on the thymus, visceral and parietal pleura, and epicardium. Grossly, the lungs are usually congested, and vascular engorgement with or without pulmonary edema is demonstrable microscopically in most cases. These changes possibly represent agonal events, because they are found with comparable frequencies in explained sudden deaths in infancy. Within the upper respiratory system (larynx and trachea), there may be some histologic evidence of recent infection (correlating with the clinical symptoms), although the changes are not sufficiently severe to account for death and should not detract from the diagnosis of SIDS. The CNS demonstrates gliosis of the brainstem and cerebellum. Morphometric studies have revealed quantitative brainstem abnormalities such as hypoplasia of the arcuate nucleus or a decrease in brainstem neuronal populations in several cases; these observations are not uniform, however. Nonspecific findings include frequent persistence of hepatic extramedullary hematopoiesis and periadrenal brown fat; it is tempting to speculate that these latter findings relate to chronic hypoxemia, retardation of normal development, and chronic stress. Thus, autopsy usually fails to provide a clear cause of death, and this may well be related to the etiologic heterogeneity of SIDS. The importance of a postmortem examination rests in identifying other causes of SUID, such as unsuspected infection, congenital anomaly, or a genetic disorder (see Table 10.4), the presence of any of which would exclude a diagnosis of SIDS; and in ruling out the unfortunate possibility of traumatic child abuse.

KEY CONCEPTS

Sudden Infant Death Syndrome

- SIDS is a disorder of unknown cause, defined as the sudden death of an infant younger than 1 year of age that remains unexplained after a thorough case investigation including performance of an autopsy. Most SIDS deaths occur between the ages of 2 and 4 months.
- The most likely basis for SIDS is a delayed development of arousal reflexes and cardiorespiratory control.
- Numerous environmental risk factors have been proposed, of which the prone sleeping position is best recognized hence the success of the “Back to Sleep” program in reducing the incidence of SIDS.

TUMORS AND TUMOR-LIKE LESIONS OF INFANCY AND CHILDHOOD

Only 2% of all malignant tumors occur in infancy and childhood; nonetheless, cancer (including leukemia) accounts for about 16% of deaths in the United States in children between 5 and 14 years of age, and only accidents cause significantly more deaths. Benign tumors are even more common than cancers. Most benign tumors are of little concern, but on occasion they cause serious complications because of their location or size.

It is sometimes difficult to distinguish true tumors or neoplasms from tumor-like lesions in the infant and child. In this context, two special categories of tumor-like lesions should be distinguished from true tumors.

- The term *heterotopia* (or *choristoma*) is applied to microscopically normal cells or tissues that are present in abnormal locations. Examples of heterotopias include a rest of pancreatic tissue found in the wall of the stomach or small intestine, or a small mass of adrenal cells found in the kidney, lungs, or ovaries. These heterotopic rests are usually of little significance, but they can be confused clinically with neoplasms. Rarely, they are sites of origin of true neoplasms, producing paradoxes such as an adrenal cortical carcinoma arising in the ovary.
- The term *hamartoma* refers to an excessive, focal overgrowth of cells and tissues native to the organ in which it occurs. Although the cellular elements are mature and identical to those found in the remainder of the organ, they do not reproduce the normal architecture of the surrounding tissue. The line of demarcation between a hamartoma and a benign neoplasm is often unclear because both lesions can be clonal. Hemangiomas, lymphangiomas, rhabdomyomas of the heart, adenomas of the liver, and developmental cysts within the kidneys, lungs, or pancreas are interpreted by some as hamartomas and by others as true neoplasms. Their unequivocally benign histology, however, does not preclude bothersome and rarely life-threatening clinical problems in some cases.

Benign Tumors and Tumor-like Lesions

Virtually any tumor may be encountered in children, but within this wide array, hemangiomas, fibrous lesions, and teratomas deserve special mention. You will notice that the most common solid neoplasms of childhood are soft tissue tumors of mesenchymal derivation. This contrasts with adults, in whom the most common tumors, benign or malignant, have an epithelial origin. Benign tumors of various tissues are described in greater detail in appropriate chapters; a few comments are made here about their special features in childhood.

Hemangiomas and Vascular Malformations

Mass-forming lesions of vascular origin are conceptually divided into proliferative vascular tumors and nonproliferative vascular malformations. The distinction between these two groups is often unclear clinically and morphologically, and the nomenclature can be somewhat inconsistent, with the term “hemangioma” most strictly referring to only benign vascular tumors but often used to describe certain types of vascular malformations as well. Vascular lesions are the most common mass-forming lesions of infancy (Chapter 11). Architecturally, they do not differ from those occurring in adults. In children, most occur in the skin, particularly on the face and scalp, where they produce flat to elevated, irregular, red-blue masses. Some of the larger flat lesions are referred to as *port-wine stains*. Hemangiomas may enlarge along with the growth of the child, but in many instances, they spontaneously regress (Fig. 10.21). In addition to their cosmetic significance, they can represent one facet of a hereditary disorder, for example von Hippel-Lindau disease (Chapter 28). A subset of CNS cavernous vascular

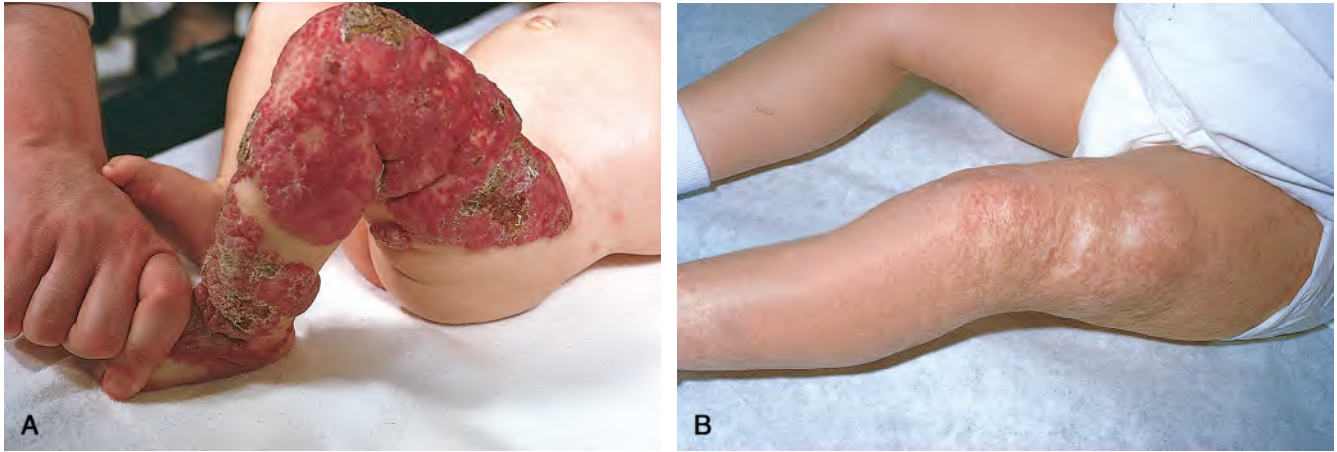


Fig. 10.21 Congenital hemangioma at birth (A) and at 2 years of age (B) after spontaneous regression. (Courtesy Dr. Eduardo Yunis, Children's Hospital of Pittsburgh.)

malformations can occur in the familial setting; these families harbor mutations in one of three genes (*KRIT1*, *CCM2*, or *PDCD10*).

Fibrous Tumors

A diverse assortment of fibrous tumors, thought to derive from fibroblasts or myofibroblasts, occur in infants and children. These include sparsely cellular proliferations of spindle-shaped cells (designated as fibromatosis) and richly cellular lesions histologically indistinguishable from fibrosarcomas occurring in adults (designated as infantile fibrosarcoma). Biologic behavior cannot be predicted based on histology alone; despite their histologic similarities with adult fibrosarcoma, infantile fibrosarcoma generally has an excellent prognosis. A characteristic chromosomal translocation, $t(12;15)(p13;q25)$, has been described in infantile fibrosarcomas, which results in generation of an *ETV6::NTRK3* fusion transcript. The normal *ETV6* gene product is a transcription factor, and the *NTRK3* gene product (also known as TRKC) is a tyrosine kinase. Like other tyrosine kinase fusion proteins found in human neoplasms, the *ETV6*-*NTRK3* fusion protein is constitutively active and stimulates signaling through the oncogenic RAS-MAPK and PI3K/AKT pathways (Chapter 7). Among soft-tissue tumors, the *ETV6::NTRK3* fusion transcript is most commonly found in infantile fibrosarcomas, making it a useful diagnostic marker. Infantile fibrosarcoma also provides a compelling example of molecular alterations driving targeted therapy; for infantile fibrosarcoma with *ETV6::NTRK3* fusion, selective NTRK inhibitors induce dramatic anti-tumor response in patients with previously unresectable tumors.

Teratomas

Teratomas illustrate the relationship of histologic maturity to biologic behavior. Teratomas may occur as benign, well-differentiated cystic lesions (mature teratomas), as lesions of indeterminate potential (immature teratomas), or as unequivocally malignant teratomas (usually admixed with another germ cell tumor component such as yolk sac tumor) (Chapter 21). They exhibit two peaks in incidence: the first at approximately 2 years of age and the second in late adolescence or early adulthood. The first peak is comprised of congenital neoplasms; the later-occurring lesions may also be of prenatal origin but are more slowly growing. Sacrococcygeal teratomas are the most common teratomas of childhood, accounting for 40% or more of cases (Fig. 10.22). They occur with a frequency of 1 in 20,000 to 40,000 live births, and they are four times more common in girls than in boys. Approximately 10% of sacrococcygeal teratomas are associated with congenital anomalies,



Fig. 10.22 Sacrococcygeal teratoma. Note the size of the lesion in relation to the fetus.

primarily defects of the hindgut and cloacal region and other midline defects (e.g., meningocele, spina bifida) not believed to result from local effects of the tumor. Approximately 75% of these tumors are mature teratomas, comprised of a disordered arrangement of mature tissue types of endodermal, mesodermal, and/or ectodermal origin; about 12% are unequivocally malignant and lethal. The remainder are immature teratomas; their malignant potential correlates with the extent of immature tissue, usually immature neuroepithelial elements present. Most benign teratomas are encountered in younger infants (<4 months), whereas children with malignant lesions tend to be somewhat older. Other sites for teratomas in childhood include the testis (Chapter 21), ovary (Chapter 22), and various midline locations, such as the mediastinum, retroperitoneum, and head and neck.

Malignant Tumors

Cancers of infancy and childhood differ biologically and histologically from their counterparts that occur later in life. The main differences, some of which have already been alluded to, include:

- Incidence and type of tumor
- Relatively frequent demonstration of a close relationship between abnormal development (teratogenesis) and tumor induction (oncogenesis)

- Prevalence of underlying familial genetic aberrations
- Tendency of fetal and neonatal malignancies to regress or differentiate spontaneously
- Improved survival or cure of many childhood tumors, so that more attention is now being devoted to minimizing the adverse delayed effects of chemotherapy and radiation therapy in survivors, including the development of second malignancies

Incidence and Types

The most frequent childhood cancers arise in the hematopoietic system, nervous tissue (including the central and sympathetic nervous system, adrenal medulla, and retina), soft tissue, bone, and kidney. This is in sharp contrast to adults, in whom the skin, lung, breast, prostate, and colon are the most common sites of tumors.

Neoplasms that exhibit sharp peaks in incidence in children younger than 15 years of age and their approximate age distribution are indicated in Table 10.5. Leukemia is the most common form of cancer in childhood, but because it is highly curable, brain tumors and leukemia are both responsible for about 25% of deaths.

Histologically, many of the malignant nonhematopoietic pediatric neoplasms are unique. In general, they tend to have a more primitive (embryonal) undifferentiated appearance, often characterized by sheets of cells with small round nuclei, and frequently show features of organogenesis specific to the site of tumor origin. Because of this latter characteristic, these tumors are frequently designated by the suffix *-blastoma*, for example, nephroblastoma (Wilms tumor), hepatoblastoma, and neuroblastoma. Because of their primitive histologic appearance, many childhood tumors have been collectively referred to as *small round blue cell tumors*. The differential diagnosis of such tumors includes neuroblastoma, Wilms tumor, lymphoma (Chapter 13), rhabdomyosarcoma (Chapter 26), Ewing sarcoma (Chapter 26), medulloblastoma (Chapter 28), and retinoblastoma (Chapter 29). If the anatomic site of origin is known, diagnosis is usually possible on histologic grounds alone. Occasionally, a combination of molecular analysis, immunoperoxidase stains, or electron microscopy is required. Two of these tumors are discussed here: the neuroblastic tumors, specifically neuroblastoma, and Wilms tumor. The remaining tumors are discussed in their respective organ-specific chapters.

Neuroblastic Tumors

The term “neuroblastic tumor” includes tumors of the sympathetic ganglia and adrenal medulla that are derived from primordial

neural crest cells populating these sites. Neuroblastoma is the most important member of this family. It is the most common extracranial solid tumor of childhood, and the most frequently diagnosed infant malignancy. The prevalence is about 1 in 7000 live births, and there are approximately 700 cases diagnosed each year in the United States. The median age at diagnosis is 18 months; approximately 40% of cases are diagnosed in infancy. Most neuroblastomas occur sporadically, but 1% to 2% are familial; in such cases the neoplasms may involve both adrenal glands or multiple primary autonomic sites. Germline mutations in the anaplastic lymphoma kinase (*ALK*) gene (Chapter 13) are a major cause of familial predisposition to neuroblastoma. Somatic gain-of-function *ALK* mutations are also observed in approximately 10% of sporadic neuroblastomas.

Despite the remarkable progress made in the therapy of this disease, long-term prognosis for high-risk subsets remains poor, with 5-year survival in the range of 40%. As will be evident later, age and stage have a remarkable effect on prognosis, and, in general, children younger than 18 months of age tend to have a significantly more favorable prognosis than older individuals with similar disease burdens.

MORPHOLOGY

In childhood, about 40% of neuroblastomas arise in the adrenal medulla. The remainder occur anywhere along the sympathetic chain, with the most common locations being the paravertebral region of the abdomen (25%) and posterior mediastinum (15%). Tumors may arise in numerous other sites, including the pelvis and neck.

Neuroblastomas range in size from minute nodules (so-called **in situ lesions**) to large masses more than 1 kg in weight. In situ neuroblastomas are reported to occur 40 times more frequently than clinically overt tumors. The great majority of these silent lesions spontaneously regress, leaving only a focus of fibrosis or calcification in the adult; this has led some to question the neoplastic connotation for the in situ lesions.

Some neuroblastomas are sharply demarcated by a fibrous pseudocapsule, but others are infiltrative and invade surrounding structures, including the kidneys, renal vein, and vena cava, and envelop the aorta. On transection, they are composed of soft, gray-tan tissue. Larger tumors have areas of necrosis, cystic softening, and hemorrhage. Occasionally, foci of punctate intratumoral calcification can be palpated.

Histologically, classic neuroblastomas are composed of small, primitive-appearing cells with dark nuclei, scant cytoplasm, and poorly defined cell borders growing in solid sheets. Such tumors may be difficult to differentiate morphologically from other small round blue cell tumors. Mitotic activity, nuclear breakdown (“karyorrhexis”), and pleomorphism may be prominent. The background often demonstrates a faintly eosinophilic fibrillary material (**neuropil**) that corresponds to neuritic processes of the primitive neuroblasts. Stereotypically, tumors have **Homer-Wright pseudorosettes**, in which tumor cells are concentrically arranged about a central space filled with neuropil (Fig. 10.23). Other helpful features include positive immunohistochemical reactions for neuron-specific enolase and the transcription factor PHOX2B. Some neoplasms show signs of maturation that can be spontaneous or therapy-induced. Larger cells having more abundant cytoplasm, large vesicular nuclei, and a prominent nucleolus, representing ganglion cells in various stages of maturation, may be found in tumors admixed with primitive neuroblasts (**ganglioneuroblastoma**). Even better differentiated lesions contain many more large cells resembling mature ganglion cells with no residual neuroblasts; such neoplasms merit the designation **ganglioneuroma** (Fig. 10.24). Maturation of neuroblasts into ganglion cells is usually accompanied by the appearance of **Schwann cells**. The origin of Schwann cells in neuroblastoma remains an issue of contention. Some investigators believe they represent a reactive population recruited by the tumor cells. However, studies using micro-

TABLE 10.5 Common Malignant Neoplasms of Infancy and Childhood

0–4 Years	5–9 Years	10–14 Years
Leukemia	Leukemia	Renal cell carcinoma
Retinoblastoma	Neuroblastoma	Hepatocellular carcinoma
Neuroblastoma	Hepatocellular neoplasm	Soft tissue sarcomas
Wilms tumor	Soft tissue sarcomas	Ewing sarcoma
Hepatoblastoma	Ewing sarcoma	Osteogenic sarcoma
Soft tissue sarcomas (especially rhabdomyosarcoma)	Central nervous system tumors	Lymphoma, including Hodgkin lymphoma
Teratoma	Lymphoma	Thyroid carcinoma
Central nervous system tumors		

dissection techniques have demonstrated that the Schwann cells harbor at least a subset of the same genetic alterations found in neuroblasts, and therefore are a component of the malignant clone. Irrespective of histogenesis, documenting the presence of schwannian stroma is essential, because its presence is associated with a favorable outcome (see later).

Metastases, when they develop, appear early and widely. In addition to local infiltration and lymph node spread, there is a pronounced tendency to spread through the bloodstream to the liver, lungs, bone marrow, and bones.

Staging. In the past several years, the International Neuroblastoma Staging System (INSS), a staging system partially based on the extent of surgical resection, has been largely supplanted by the International Neuroblastoma Risk Group Staging System (INRGSS; Table 10.6), which depends on tumor imaging characteristics prior to treatment and is detailed in Table 10.7. Tumor stage is of paramount importance in determining prognosis.

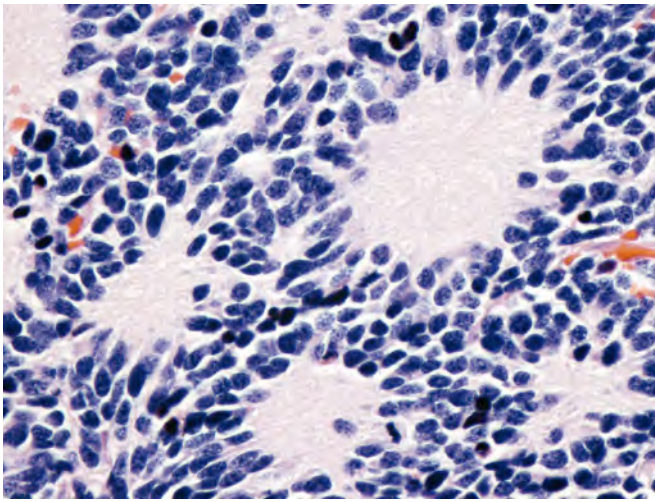


Fig. 10.23 Adrenal neuroblastoma. This tumor is composed of small cells embedded in a finely fibrillar matrix (neuropil). Several Homer-Wright pseudorosettes are seen in this image.

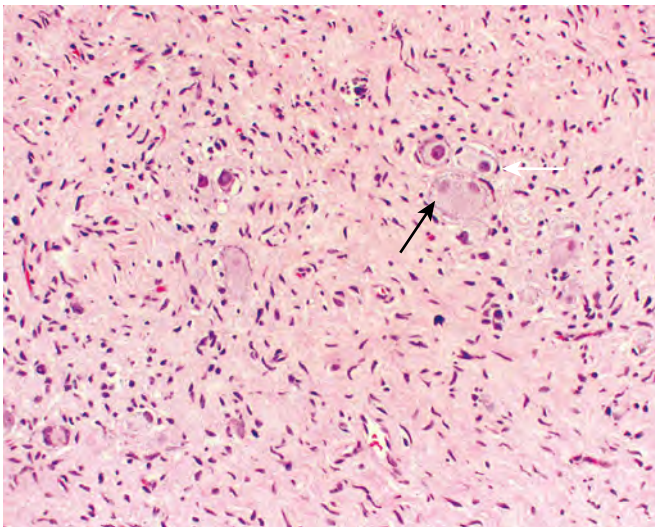


Fig. 10.24 Ganglioneuromas are characterized by clusters of large cells with eccentric nuclei, prominent nucleoli, and abundant eosinophilic cytoplasm, representing neoplastic ganglion cells (arrows). Spindle-shaped Schwann cells are present in the background stroma.

TABLE 10.6 International Neuroblastoma Risk Group Staging System

Stage	Description
L1	Localized tumor not involving vital structures as defined by the list of image-defined risk factors and confined to one body compartment
L2	Locoregional tumor with presence of one or more image-defined risk factors
M	Distant metastatic disease (except stage MS)
MS	Metastatic disease in children younger than 18 months with metastases confined to skin, liver, and/or bone marrow

From Monclair T, Brodeur GM, Ambros PF, et al: The International Neuroblastoma Risk Group (INRG) Staging System: An INRG Task Force Report. *J Clin Oncol*. 2009;27:298.

TABLE 10.7 Image-Defined Risk Factors in Neuroblastic Tumors

Ipsilateral Tumor Extension Within Two Body Compartments
Neck-chest, chest-abdomen, abdomen-pelvis
Neck
Tumor encasing carotid and/or vertebral artery and/or internal jugular vein
Tumor extending to base of skull
Tumor compressing the trachea
Cervicothoracic Junction
Tumor encasing brachial plexus roots
Tumor encasing subclavian vessels and/or vertebral and/or carotid artery
Tumor compressing the trachea
Thorax
Tumor encasing the aorta and/or major branches
Tumor compressing the trachea and/or principal bronchi
Lower mediastinal tumor infiltrating the costovertebral junction between T9 and T12
Thoracoabdominal
Tumor encasing the aorta and/or vena cava
Abdomen/Pelvis
Tumor infiltrating the porta hepatis and/or the hepatoduodenal ligament
Tumor encasing branches of the superior mesenteric artery at the mesenteric root
Tumor encasing the origin of the coeliac axis, and/or of the superior mesenteric artery
Tumor invading one or both renal pedicles
Tumor encasing the aorta and/or vena cava
Tumor encasing the iliac vessels
Pelvic tumor crossing the sciatic notch
Intraspinal Tumor Extension Whatever the Location, Provided That:
More than one-third of the spinal canal in the axial plane is invaded and/or the perimedullary leptomenigeal spaces are not visible and/or the spinal cord signal is abnormal
Infiltration of Adjacent Organs/Structures
Pericardium, diaphragm, kidney, liver, duodenopancreatic block, and mesentery
Conditions to Be Recorded, but <i>Not</i> Considered IDRFs
Multifocal primary tumors
Pleural effusion, with or without malignant cells
Ascites, with or without malignant cells

From Monclair T, Brodeur GM, Ambros PF, et al: The International Neuroblastoma Risk Group (INRG) Staging System: An INRG Task Force Report. *J Clin Oncol*. 2009;27:298.

Clinical Features

In young children (younger than 2 years), neuroblastomas generally present with large abdominal masses, fever, and possibly weight loss. In older children, they may not come to attention until metastases produce manifestations, such as bone pain, respiratory symptoms, or gastrointestinal complaints. Neuroblastomas may metastasize widely through the hematogenous and lymphatic systems, particularly to liver, lungs, bones, and bone marrow. Proptosis and ecchymosis may be present due to spread to the periorbital region, a common metastatic site. In neonates, disseminated neuroblastomas may present with multiple cutaneous metastases that cause deep blue discoloration of the skin. About 90% of neuroblastomas, regardless of location, produce catecholamines (similar to the catecholamines associated with pheochromocytomas), which are an important diagnostic feature (i.e., elevated blood levels of catecholamines and elevated urine levels of the metabolites vanillylmandelic acid [VMA] and homovanillic acid [HVA]). Despite the elaboration of catecholamines, hypertension is much less frequent with these neoplasms than with pheochromocytomas (Chapter 24). Ganglioneuromas, unlike their malignant counterparts, tend to produce either asymptomatic mass lesions or symptoms related to compression.

The clinical course of neuroblastomas is extremely variable. Several clinical, histopathologic, molecular, and biochemical factors have been identified that have a bearing on prognosis (Table 10.8); based on the collection of prognostic factors present in a given patient, they are classified as “low,” “intermediate,” or “high” risk. With improvements in therapy, long-term survival exceeds 90% of patients in the first two groups, but less than 50% of patients in the high-risk category are long-term survivors. The most pertinent prognostic factors include:

- **Tumor stage and patient age at time of diagnosis.** Neuroblastomas without extensive disease spread (INSS stage L1 or L2) tend to have an excellent prognosis, irrespective of age (“low” or “intermediate”

risk); tumors exhibiting amplification of the *MYCN* oncogene are a notable exception. Infants with localized primary tumors and metastases to the liver, bone marrow, and skin INRGSS stage MS) represent a special subtype; these tumors may regress spontaneously. The biologic basis of this welcome behavior is not clear. The age of 18 months has emerged as a critical point of dichotomy in terms of prognosis. Children younger than 18 months of age, and especially those in the first year of life, have an excellent prognosis regardless of the stage of the neoplasm. Children older than 18 months of age generally fall into at least the “intermediate” risk category, and those with higher-stage tumors or with unfavorable prognostic features like *MYCN* amplification in the neoplastic cells are considered “high” risk.

- **Tumor morphology.** An age-linked morphologic classification of neuroblastic tumors has been proposed that divides them into favorable and unfavorable histologic subtypes. The specific morphologic features that bear on prognosis are listed in Table 10.8.
- ***MYCN* amplification status.** Amplification of the *MYCN* proto-oncogene in neuroblastomas markedly impacts the prognosis, particularly in tumors that would otherwise be classified as having a good prognosis. The presence of *MYCN* amplification “bumps” the tumor to the high-risk category, irrespective of age, stage, or histology. *MYCN* is located on the distal short arm of chromosome 2 (2p24). Amplification of *MYCN* does not occur at 2p24, but as extrachromosomal double minute chromosomes or homogeneously staining regions on other chromosomes (Fig. 10.25). *MYCN* amplification is present in about 20% of primary tumors, most presenting as advanced-stage disease. *MYCN* amplification is currently the most important genetic abnormality used in risk stratification of neuroblastic tumors (see later).
- **Ploidy** of the tumor cells correlates with outcome in children younger than 2 years of age but loses its independent prognostic significance in older children. Broadly, neuroblastomas can be divided into two categories: near-diploid and hyperdiploid (whole-chromosome gains), with the latter being associated with a more favorable prognosis. It is postulated that neuroblastomas with hyperdiploidy have an underlying defect in the mitotic machinery leading to nondisjunction and whole-chromosome gains. The more aggressive near-diploid tumors harbor generalized genomic instability, with multiple segmental chromosomal aberrations that result in a complex karyotype with adverse prognostic implications. One unusual form of segmental aberration in aggressive neuroblastomas is called chromothripsis (Chapter 7), which involves localized fragmentation of a chromosome segment followed by random assembly of the fragments. In a subset of neuroblastomas, chromothripsis can result in amplification of *MYCN* or other oncogenes, or losses in tumor suppressor loci.

Although age, stage, histology, *MYCN* status, and DNA ploidy are the core criteria used for formal risk stratification and therapeutic decision-making, various segmental chromosomal losses can also have prognostic significance. Hemizygous deletion of the distal short arm of chromosome 1 in the region of band 1p36, which occurs in 25% to 35% of primary tumors, is strongly correlated with *MYCN* amplification, advanced disease stage, and an increased risk of disease relapse. Hemizygous loss of chromosome 11q is another common deletion event with adverse prognosis that is not correlated with *MYCN* amplification.

Whole-genome sequencing has uncovered alterations in a variety of genes. Some of these alterations have prognostic implications. These include those whose products are involved in *neuritogenesis* (a process in neuronal differentiation that includes the sprouting of

TABLE 10.8 Prognostic Factors in Neuroblastomas

Variable	Favorable	Unfavorable
Stage ^a	Stage L1, MS	Stage L2, M
Age ^a	<18 months	>18 months
Histology ^a		
Evidence of schwannian stroma and gangliocytic differentiation	Present, >50%	Absent, or <50%
Mitosis-karyorrhexis index ^b	<200/5000 cells	>200/5000 cells
DNA ploidy ^a	Hyperdiploid (whole-chromosome gains)	Near-diploid (segmental chromosome losses; chromothripsis)
<i>MYCN</i> status ^a	Not amplified	Amplified
Chromosome 1p loss	Absent	Present
Chromosome 11q loss	Absent	Present
TRKA expression	Present	Absent
TRKB expression	Absent	Present
Mutation of neuritogenesis genes	Absent	Present

^aIncluded in the most used parameters in clinical practice for assessment of prognosis and risk stratification.

^bMitosis-karyorrhexis index is defined as the number of mitotic or karyorrhectic cells per 5000 tumor cells in random foci.

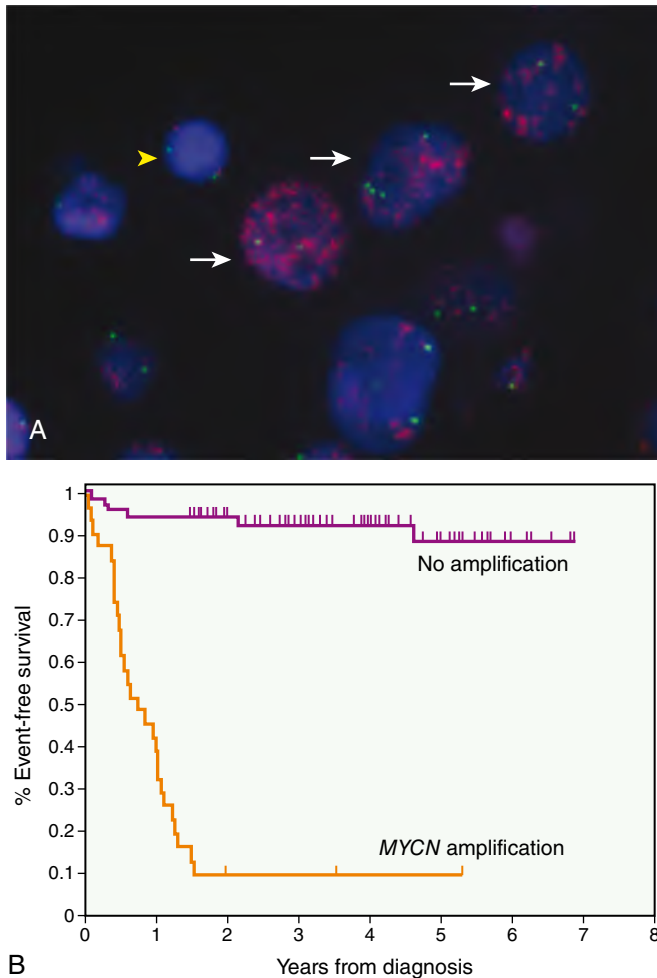


Fig. 10.25 MYCN amplification in neuroblastoma. (A) Fluorescence in situ hybridization using a fluorescent probe for *MYCN* on a tissue section. The neuroblastoma cells (white arrow) have large areas of staining (red); this corresponds to amplified *MYCN* in the form of homogeneously staining regions. A background normal cell (yellow arrowhead) shows only two (normal) *MYCN* (red) and chromosome 2 (green) signals. (B) A Kaplan-Meier survival curve of infants younger than 1 year of age with metastatic neuroblastoma. The 3-year event-free survival of infants whose tumors lacked *MYCN* amplification was 93%, whereas those with tumors with *MYCN* amplification had only a 10% event-free survival. (B, From Brodeur GM. Neuroblastoma: biological insights into a clinical enigma. *Nat Rev Cancer*. 2003;3:203-216.)

neurites, which later become dendrites and axons). Selected examples of mutated genes in this category include *alpha thalassemia/mental retardation, X linked (ATRX)*; genes involving chromatin remodeling, such as *AT-rich interaction domains 1A and 1B (ARID1A and ARID1B)*; and neurotrophin receptors (*NTRK1, NTRK2, and NTRK3*). The impact of these changes on prognosis is under investigation.

Although discussion of the treatment modalities for neuroblastoma is beyond the scope of this book, we mention in passing two promising experimental approaches. The first involves the use of retinoids as an adjunct therapy for inducing the differentiation of neuroblastoma. Recall that the retinoic acid pathway plays a critical role in cellular differentiation during embryogenesis. Second, clinical trials are evaluating the utility of small-molecule ALK inhibitors in treating tumors harboring activating *ALK* mutations.

KEY CONCEPTS

Neuroblastoma

- Neuroblastomas and related tumors arise from neural crest derived cells in the sympathetic ganglia and adrenal medulla.
- Neuroblastomas are undifferentiated, whereas ganglioneuroblastomas and ganglioneuromas demonstrate evidence of differentiation (Schwannian stroma and ganglion cells). Homer-Wright pseudorosettes, are characteristic of neuroblastoma.
- Age, stage, and *MYCN* amplification and ploidy are the most important prognostic features; children younger than 18 months usually have a better prognosis than older children, whereas children with higher-stage tumors or *MYCN* amplification fare worse.
- Neuroblastomas secrete catecholamines, whose metabolites (VMA/ HVA) can be used for screening patients.

Wilms Tumor

Wilms tumor (nephroblastoma) afflicts approximately 1 in 10,000 children in the United States, making it the most common primary renal tumor of childhood and the fourth most common pediatric malignancy in the United States. The peak incidence for Wilms tumor is between 2 and 5 years of age, and 95% of tumors occur before 10 years of age. Approximately 5% to 10% of Wilms tumors involve both kidneys, either simultaneously (synchronous) or one after the other (metachronous). Bilateral Wilms tumors have a median age of onset approximately 10 months earlier than tumors restricted to one kidney, and these patients are presumed to harbor a germline mutation in one of the Wilms tumor-predisposing genes (see later). The biology of this tumor illustrates several important aspects of childhood neoplasms, such as the relationship between malformations and neoplasia, the histologic similarities between organogenesis and oncogenesis, the two-hit theory of recessive tumor suppressor genes (Chapter 7), the role of premalignant lesions, and the potential for treatment to affect prognosis and outcome. Improvements in cure rates for Wilms tumor (from as low as 30% a few decades ago to approximately 90% currently) represent one of the greatest successes of pediatric oncology.

Pathogenesis and Genetics

The risk of Wilms tumor is increased with at least three recognizable groups of congenital malformations associated with distinct chromosomal loci. Although Wilms tumors arising in this setting account for no more than 10% of cases, these syndromic tumors have provided important insights into the biology of this neoplasm.

- The first group of patients has *WAGR syndrome*, characterized by Wilms tumor, aniridia, genitourinary anomalies, and intellectual disability. Their lifetime risk of developing Wilms tumor is approximately 33%. Individuals with WAGR syndrome carry constitutional (germline) deletions of 11p13. Studies on these patients led to the identification of the first Wilms tumor-associated gene, *WT1*, and a contiguous deleted autosomal dominant gene for aniridia, *PAX6*, both located on chromosome 11p13. Patients with deletions restricted to *PAX6*, with normal *WT1* function, develop aniridia but are not at increased risk for Wilms tumors. The presence of germline *WT1* deletions in WAGR syndrome represents the “first hit”; the development of Wilms tumor in these patients frequently correlates with the occurrence of a nonsense or frameshift mutation in the second *WT1* allele (“second hit”).
- A second group of patients at much higher risk for Wilms tumor (~90%) has *Denys-Drash syndrome*, which is characterized by gonadal dysgenesis (male pseudohermaphroditism) and early-onset

nephropathy leading to renal failure. The characteristic glomerular lesion in these patients is diffuse mesangial sclerosis (Chapter 20). As in patients with WAGR, these patients also demonstrate germline abnormalities in *WT1*. In patients with Denys-Drash syndrome, however, the genetic abnormality is a dominant-negative missense mutation in the zinc-finger region of the *WT1* protein that affects its DNA-binding properties. This mutation interferes with the function of the remaining wild-type allele, yet strangely, it is sufficient only in causing genitourinary abnormalities but not tumorigenesis. Wilms tumors arising in Denys-Drash syndrome demonstrate biallelic inactivation of *WT1*. In addition to Wilms tumors, these individuals are also at increased risk for developing germ cell tumors called gonadoblastomas (Chapter 21), almost certainly a consequence of disruption of normal gonadal development.

WT1 encodes a DNA-binding transcription factor that is expressed within several tissues, including the kidney and gonads, during embryogenesis. The *WT1* protein is critical for normal renal and gonadal development. It has multiple binding partners, and the identity of this partner can affect whether *WT1* functions as a transcriptional activator or repressor in a given cellular context. Numerous transcriptional targets of *WT1* have been identified, including genes encoding glomerular podocyte-specific proteins and proteins involved in induction of renal differentiation. Despite the importance of *WT1* in nephrogenesis and its unequivocal role as a tumor suppressor gene, only about 10% of patients with sporadic (nonsyndromic) Wilms tumors have *WT1* mutations, suggesting that most of these tumors are caused by mutations in other genes.

- A third group with an increased risk of developing Wilms tumor are children with *Beckwith-Wiedemann syndrome* (BWS), characterized by enlargement of body organs (organomegaly), macroglossia, hemihypertrophy, omphalocele, and abnormal large cells in the adrenal cortex (adrenal cytomegaly). BWS has served as a model for tumorigenesis associated with genomic imprinting (Chapter 5). The chromosomal region implicated in BWS localizes to band 11p15.5 (the so-called *WT2* locus), distal to the *WT1* locus. This region contains multiple genes that are normally expressed from only one of the two parental alleles, with transcriptional silencing (i.e., imprinting) of the other parental homologue by methylation of the promoter region. Unlike WAGR and Denys-Drash syndromes, the genetic basis for BWS is considerably more heterogeneous in that no single 11p15.5 gene is involved in all cases. Moreover, the phenotype of BWS, including the predisposition to tumorigenesis, is influenced by the specific imprinting abnormalities present. One of the genes in the 11p15.5 region—insulin-like growth factor-2 (*IGF2*)—is normally expressed solely from the paternal allele, and the maternal allele is silenced by imprinting. In some Wilms tumors, loss of imprinting (i.e., re-expression of the maternal *IGF2* allele) can be demonstrated, leading to overexpression of the *IGF2* protein. In other instances, there is selective deletion of the imprinted maternal allele combined with duplication of the transcriptionally active paternal allele in the tumor (uniparental paternal disomy), which has the identical functional effect of *IGF2* overexpression. Because the *IGF2* protein is an embryonal growth factor, it could conceivably explain the features of overgrowth associated with BWS as well as the increased risk for Wilms tumors in these patients.

In contrast with syndromic Wilms tumors, the molecular abnormalities underlying sporadic (i.e., nonsyndromic) tumors, which account for 90% of cases overall in children, are only beginning to be elucidated. Some of them are associated with specific histologic features. Gain-of-function mutations of the gene encoding β -catenin (Chapter 7) have been demonstrated in approximately 10% of sporadic Wilms tumors. Other recurrent mutations occur in genes encoding

proteins involved in micro-RNA (miRNA) processing (*DROSHA*, *DGCR8*, and *DICER1*); these are seen in 15% to 20% of Wilms tumors with predominantly blastemal histology (see later). It is postulated that aberrations in miRNA processing lead to reduced levels of many mature miRNAs, in particular those in the miR-200 family, which is involved in “*mesenchymal to epithelial transformation*” during renal morphogenesis. The lack of mesenchymal to epithelial transformation likely leads to persistent blastemal “rests” in the kidney (see later), which evolve into Wilms tumors. Finally, tumors with *TP53* mutations are associated with an especially poor prognosis and often have a distinctive anaplastic histologic appearance, described later.

MORPHOLOGY

Grossly, Wilms tumor tends to present as a large, solitary, well-circumscribed mass. Approximately 10% are either bilateral or multicentric at the time of diagnosis. On cut section, the tumor is soft, homogeneous, and tan-gray with occasional foci of hemorrhage, cyst formation, and necrosis (Fig. 10.26).

Microscopically, Wilms tumors are characterized by recognizable attempts to recapitulate different stages of nephrogenesis. **The classic triphasic combination of blastemal, stromal, and epithelial cell types is observed in most lesions**, although the percentage of each component is variable (Fig. 10.27A). Sheets of small blue cells with few distinctive features characterize the blastemal component. Epithelial differentiation is usually in the form of abortive tubules or glomeruli. Stromal cells are usually fibroblastic or myxoid in nature, although skeletal muscle differentiation is not uncommon. Rarely, other heterologous elements are identified, including squamous or mucinous epithelium, smooth muscle, adipose tissue, cartilage, osteoid, and neurogenic tissue. Approximately 5% of tumors are **anaplastic**, defined as the presence of cells with large, hyperchromatic, pleomorphic nuclei and abnormal mitoses (see Fig. 10.27B). The presence of anaplasia correlates with the presence of *TP53* mutations and the emergence of resistance to chemotherapy. Recall that p53 elicits pro-apoptotic signals in response to DNA damage (Chapter 7). The loss of p53 function might explain the relative unresponsiveness of anaplastic cells to cytotoxic chemotherapy.

Nephrogenic rests are putative precursor lesions of Wilms tumors and are seen in the renal parenchyma adjacent to approximately 25% to 40% of unilateral tumors; this frequency rises to nearly 100% in cases of bilateral Wilms tumors. In many instances the nephrogenic rests share genetic alterations with the adjacent Wilms tumor, pointing to their preneoplastic status. The appearance of nephrogenic rests varies from expansile masses that resemble Wilms tumors (hyperplastic rests) to sclerotic rests consisting predominantly of fibrous tissue and occasional admixed immature tubules or glomeruli. It is important to document the presence of nephrogenic rests in the resected specimen, because these patients are at an increased risk of developing Wilms tumors in the contralateral kidney and require frequent and regular surveillance for many years.

Clinical Features

Most children with Wilms tumors present with a large abdominal mass that may be unilateral or, when very large, extend across the midline and down into the pelvis. Hematuria, pain in the abdomen after a traumatic incident, intestinal obstruction, and hypertension are other patterns of presentation. In a considerable number of these patients, pulmonary metastases are present at the time of primary diagnosis.

As stated, most patients with Wilms tumor can expect to be cured of their malignancy. Anaplastic histology is perhaps the most critical determinant of adverse prognosis. Even anaplasia restricted to the kidney (i.e., without extrarenal spread) confers an increased risk of

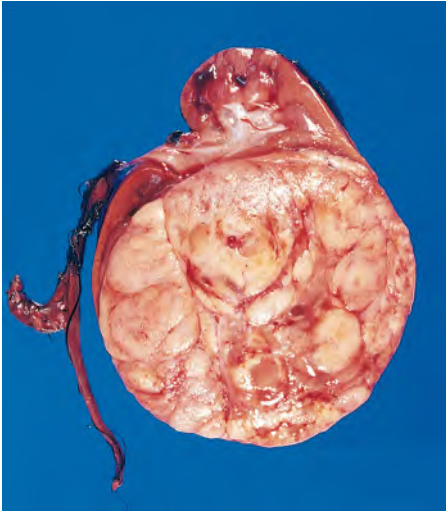


Fig. 10.26 Wilms tumor in the lower pole of the kidney with the characteristic tan color and well-circumscribed margins.

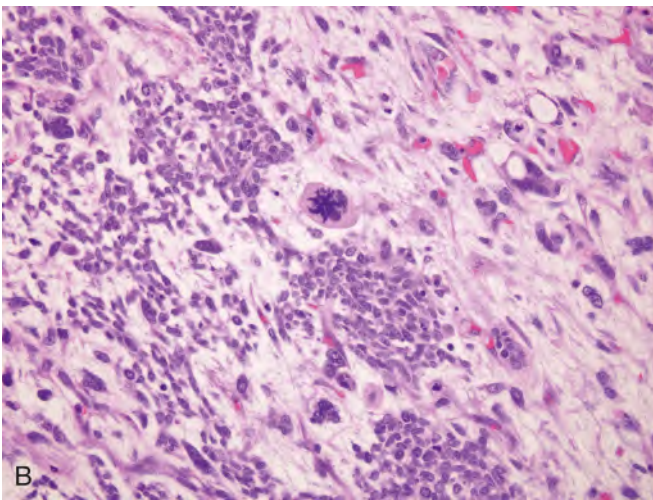
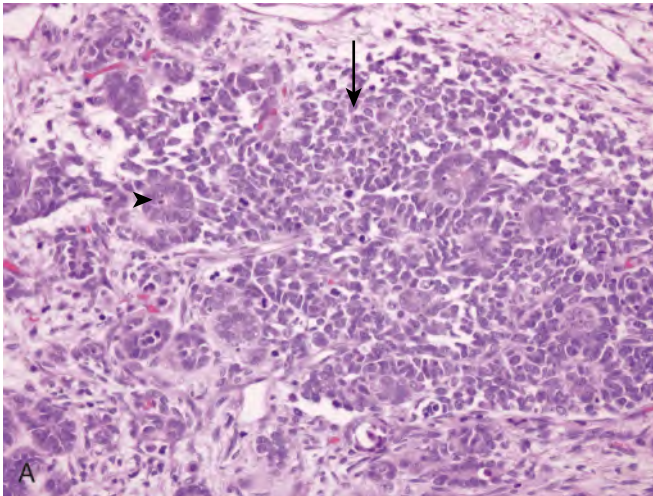


Fig. 10.27 Wilms tumor. (A) Tightly packed blue cells (blastemal component, arrow) and interspersed primitive tubules (epithelial component, arrowhead). Although multiple mitotic figures are seen, none are atypical in this field. (B) Anaplasia, characterized by cells with hyperchromatic, pleomorphic nuclei, and abnormal mitoses (center).

recurrence and death, emphasizing the need for accurate identification of this histologic feature. Molecular parameters that correlate with adverse prognosis include loss of heterozygosity of chromosomes 1p and 16q, and gain of chromosome 1q in tumor cells. Along with the increased survival of individuals with Wilms tumor have come reports of an increased risk of developing second primary tumors, including bone and soft tissue sarcomas, leukemia and lymphoma, and breast cancers. Although some of these neoplasms result from the presence of a germline mutation in a cancer predisposition gene, others are a consequence of therapy, most commonly radiation administered to the cancer field. This tragic, albeit uncommon, outcome has mandated that radiation therapy be used judiciously in the treatment of this and other childhood cancers.

KEY CONCEPTS

Wilms Tumor

- Wilms tumor is the most common renal neoplasm of childhood. Patients with three syndromes are at increased risk for Wilms tumors: Denys-Drash syndrome, Beckwith-Wiedemann syndrome, and WAGR syndrome (Wilms tumor, aniridia, genital abnormalities, and mental retardation).
- WAGR syndrome and DDS are associated with *WT1* gene inactivation, whereas Beckwith-Wiedemann arises through imprinting abnormalities principally involving the *IGF2* gene.
- The morphologic components of Wilms tumor include blastemal (small, round, blue cells) and epithelial and stromal elements.
- Nephrogenic rests are precursor lesions of Wilms tumors.

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Blood Vessels

Richard N. Mitchell and Marc K. Halushka

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Vascular disorders are responsible for more morbidity and mortality than any other category of human disease. Although arterial disease is most significant clinically, venous disorders also take a considerable toll. Two principal mechanisms underlie vascular disease:

- *Narrowing (stenosis) or complete obstruction* of vessel lumens, either progressively (e.g., by atherosclerosis) or precipitously (e.g., by thrombosis or embolism)
- *Weakening* of vessel walls, leading to dilation or rupture

To better appreciate the pathogenesis of vascular disorders, it is important to first understand the structure and functions of normal vessels.

VASCULAR STRUCTURE AND FUNCTION

The general architecture and cellular composition of blood vessels are similar throughout the cardiovascular system. However, structural specializations that reflect distinct functional roles characterize specific kinds of vessels (Fig. 11.1). For example, arterial walls are thicker than veins at the same level of branching to accommodate pulsatile flow and higher blood pressures. Arterial wall thickness gradually diminishes as the vessels become smaller, but the ratio of wall thickness to lumen diameter increases, allowing these muscular vessels to exert control over blood flow and pressure. Many disorders of the vasculature affect only particular

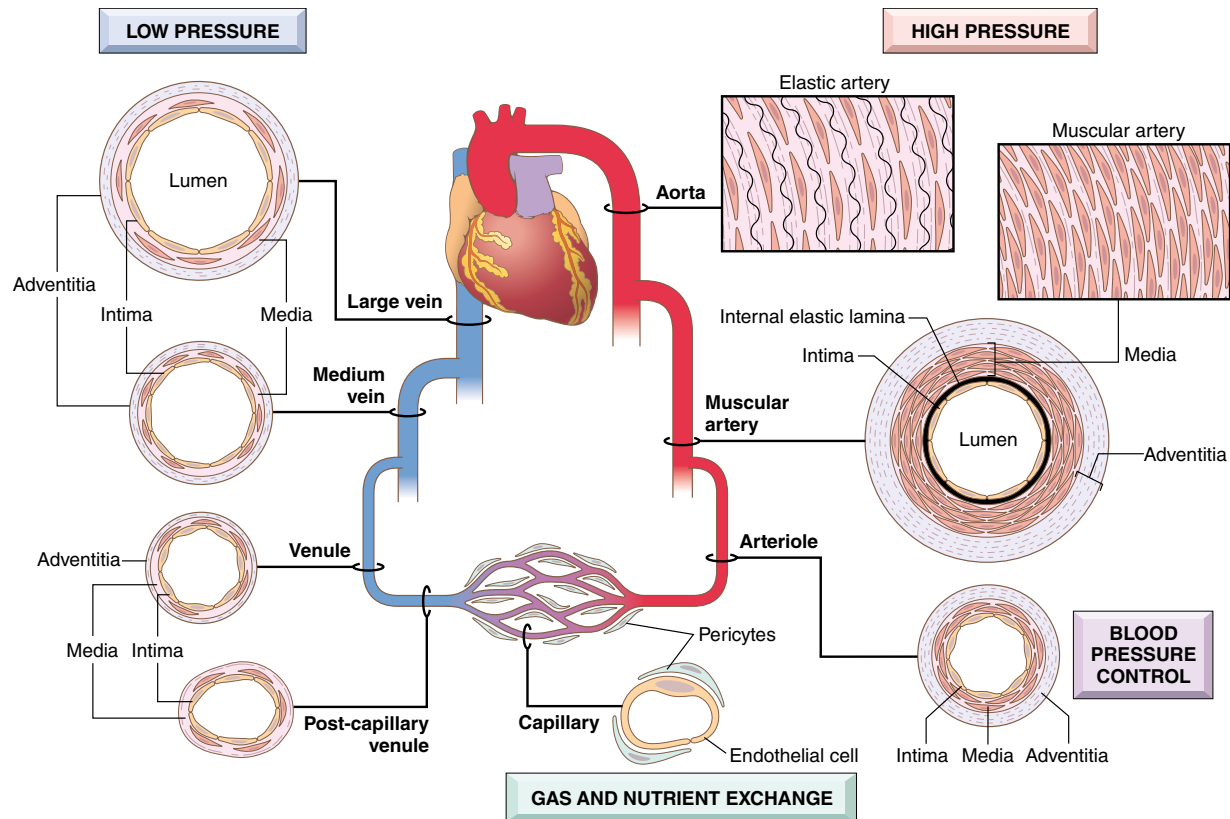


Fig. 11.1 Regional specializations of the vasculature. The aorta and other elastic arteries have substantial elastic tissue to accommodate high pulsatile forces, with the capacity to recoil and transmit energy into forward blood flow. These vessels have lamellar units composed of repetitions of a layer of elastic fibers, a smooth muscle cell, and intervening extracellular matrix. Purely muscular arteries have elastic fibers only at the intersection of the intima and media (internal elastic lamina) or media and adventitia (external elastic lamina, not shown). In comparison, the venous system has relatively thinner medial layers that permit greater capacitance, and the capillary wall permits ready diffusion of oxygen and nutrients because it is comprised only of an endothelial cell and sparse encircling pericytes. The different structural and functional attributes also influence the disorders that can affect the various parts of the vascular tree. Thus, loss of aortic elastic tissue will result in aneurysm, while stasis in a dilated venous bed can result in thrombosis.

types of vessels and have characteristic anatomic distributions. Thus, atherosclerosis affects mainly elastic and muscular arteries, hypertension affects small muscular arteries and arterioles, and different varieties of vasculitis characteristically involve only vessels of a certain caliber.

The principal constituents of the walls of blood vessels are endothelial cells (ECs) that line the vascular lumens, and smooth muscle cells (SMCs) that provide architectural strength and synthesize a variety of extracellular matrix (ECM) elements including elastin, collagen, and glycosaminoglycans. In arteries and veins, these constituents are organized into three concentric layers—*intima*, *media*, and *adventitia*, which are better defined in the arteries than in veins.

- The *intima* normally consists of a single layer of ECs attached to a basement membrane with a thin underlying layer of ECM; the intima is demarcated from the media by the internal elastic lamina.
- The *media* of *elastic arteries* (e.g., the aorta) are composed of concentric layers resembling tree rings. The high elastin content of the media allows these vessels to expand during systole and recoil during diastole. With aging and the loss of elasticity, the aorta and larger arteries become less compliant; besides transmitting higher pressures into distal tissues, the arteries of older individuals often become progressively tortuous and dilated (*ectatic*).
- The media of *muscular arteries* is composed predominantly of circumferentially oriented SMCs. Arteriolar SMC contraction

(vasoconstriction) or relaxation (vasodilation) is regulated by inputs from the autonomic nervous system and local metabolic factors.

- *Arterioles are the principal points of physiologic resistance to blood flow.* Since the resistance to fluid flow through a tube is inversely proportional to the fourth power of the diameter (i.e., halving the diameter increases resistance 16-fold), small decreases in the lumen size of arterioles caused by structural changes or vasoconstriction can have profound effects on blood pressure.
- The *adventitia* is external to the media and in many arteries is separated from the media by an external elastic lamina. The adventitia is typically composed of loose connective tissue containing nerve fibers. Diffusion of oxygen and nutrients from the lumen is sufficient to sustain thin-walled vessels and the innermost SMCs of all vessels. In large- and medium-sized vessels, however, small arterioles within the adventitia (called *vasa vasorum*—literally, “vessels of vessels”) perfuse the outer one-third to one-half of the media.

As already alluded to, arteries are divided into three types based on their size and structural features: (1) large or elastic arteries, which include the aorta, the major branches of the aorta (innominate, subclavian, common carotid, and iliac), and pulmonary arteries; (2) medium-sized or muscular arteries comprising smaller branches of the aorta (e.g., coronary and renal arteries); and (3) small arteries (≤ 2 mm

in diameter) and *arterioles* (20 to 100 μm in diameter) within tissues and organs.

Capillaries are slightly smaller (5 μm) than the diameter of a red blood cell (7 to 8 μm); they have an EC lining but no media, but variable numbers of pericytes, cells that resemble SMCs, lie beneath the endothelium. Pericytes are important for the maintenance of capillary integrity. Collectively, capillaries have a huge cross-sectional area and a relatively low flow rate. The combination of thin walls and slow flow makes capillaries ideally suited for the exchange of diffusible substances between blood and tissues. Because functionally useful oxygen diffusion is limited to a distance of approximately 100 μm , the capillary network of most tissues is very rich. Tissues with high metabolic rates such as myocardium and brain have the highest density of capillaries.

Blood from capillary beds flows into *postcapillary venules* and then sequentially through collecting venules into small, medium, and large veins. In most forms of inflammation, vascular leakage and leukocyte exudation occur preferentially from postcapillary venules (Chapter 3).

Relative to arteries at the same level of branching, veins have larger diameters, larger lumens, and thinner and less well-organized walls (see Fig. 11.1). These structural features augment the capacitance of the venous side of the circulation, which, on average, contains roughly two-thirds of total blood volume. Reverse flow in veins (e.g., due to gravity) is prevented in the extremities by venous valves.

Lymphatics are thin-walled channels lined by lymphatic-specific endothelial cells; these drain interstitial fluid, known as lymph, from the tissues, eventually returning this fluid to the blood stream via the thoracic duct, which empties into the left subclavian vein. Lymphatics also transport inflammatory cells and antigens from the periphery to lymph nodes, thereby facilitating antigen presentation and immune responses. This is an important protective mechanism against pathogens but can be a double-edged sword as these channels can also disseminate disease by transporting microbes or tumor cells to distant sites.

KEY CONCEPTS

Vascular Structure and Function

- All vessels except capillaries share a general three-layered architecture consisting of an endothelial-lined intima, surrounded by smooth muscle media, and supportive adventitia.
- The SMCs and ECM of arteries, veins, and capillaries vary according to hemodynamic demands (e.g., pressure, pulsatility) and functional requirements.
- The specific composition of the vessel wall at any given site within the vascular tree influences the nature and consequences of pathologic injuries.
- Capillaries are delicate blood vessels and ideal sites for the exchange of diffusible substances between blood and tissues.
- Postcapillary venules are important segment of microvasculature and play a critical role in inflammation.
- The lymphatic system maintains a healthy body fluid level and defends the body against infection by transporting inflammatory cells and helping antigen presentation.

VASCULAR ANOMALIES

Although rarely symptomatic, anatomic variants of the usual vascular supply are important to recognize, as failure to do so may lead to surgical complications or impede therapeutic interventions (e.g., placement of coronary artery stents). Among *congenital vascular anomalies*, four are of particular clinical significance:

- *Developmental* or *berry aneurysms* occur at branch points in cerebral vessels; when ruptured, these can cause fatal intracerebral hemorrhage (Chapter 28).
- *Arteriovenous fistulas* are direct connections (usually small) between arteries and veins that bypass capillaries. They occur most commonly as developmental defects but can also result from rupture of an arterial aneurysm into an adjacent vein, from penetrating injuries that pierce arteries and veins, or from inflammation and necrosis of adjacent vessels. Surgically generated arteriovenous fistulas provide vascular access for hemodialysis. Like berry aneurysms, arteriovenous fistulas can rupture. Arteriovenous fistulas that shunt large amounts of blood from the arterial to the venous side may put sufficient strain on the heart to cause high-output cardiac failure.
- *Fibromuscular dysplasia* is a focal irregular thickening in medium and large muscular arteries including renal, carotid, splanchnic, and vertebral vessels. The cause is unknown. Segments of the vessel wall are focally thickened by medial and intimal hyperplasia and fibrosis, resulting in luminal stenosis. In the renal arteries, it can cause of renovascular hypertension (Chapter 20). Immediately adjacent vessel segments can have markedly attenuated media (on angiography the vessels are said to have a “string of beads” appearance). These points of thinning are prone to develop aneurysms that may rupture.
- *Anomalous coronary artery origin* is a developmental anomaly in which both coronary arteries arise over the same coronary cusp of the aortic valve. Although usually benign, if the anomalous coronary vessel passes between the aorta and pulmonary artery it can be compressed, for example, during exercise, limiting blood flow and resulting in sudden death.

VASCULAR WALL RESPONSE TO INJURY

The integrated functioning of ECs and SMCs impacts physiologic and pathophysiologic responses to hemodynamic and biochemical stimuli. Their function (and dysfunction) are described briefly, followed by discussion of specific vascular disorders.

Endothelial Cells

Endothelial cells (ECs) form a specialized single cell layer of thin and elongated cells lining the blood vessels. Although ECs throughout the vascular tree share many attributes, populations that line different portions of the vascular tree (large vessels vs. capillaries, arteries vs. veins) have distinct gene expression profiles, behaviors, and morphologic appearances. Thus, ECs in liver sinusoids or in renal glomeruli are fenestrated (presumably to facilitate filtration), while central nervous system ECs (along with the associated perivascular cells) create an impermeable *blood-brain barrier*.

ECs are versatile multifunctional cells with a wealth of synthetic and metabolic properties. In the normal state they have several constitutive activities that are critical for vessel homeostasis and circulatory function (Table 11.1). Unless injured or otherwise activated, ECs have a non-thrombogenic surface that maintains blood in a fluid state (Chapter 4). They also modulate medial SMC tone (thereby influencing vascular resistance); metabolize hormones such as angiotensin; regulate inflammation; and affect the growth of other cell types, particularly SMCs. Although interendothelial junctions are largely impermeable in normal vessels, vasoactive agents (e.g., histamine) cause contraction of ECs and allow the rapid egress of fluids, electrolytes, and protein; in inflammation, even leukocytes slip between adjacent ECs (Chapter 3).

ECs can respond to various stimuli by adjusting their steady state (constitutive) functions and by expressing newly acquired (inducible) properties—a process termed *endothelial activation* (Fig. 11.2). There are numerous inducers of endothelial activation, including cytokines and bacterial products, which elicit inflammation and, in severe cases, can promote the systemic inflammatory response syndrome and shock (Chapter 4). Additional activators include hemodynamic stresses and

TABLE 11.1 Endothelial Cell Properties and Functions

Property/Function	Mediators/Products
Maintenance of permeability barrier	
Elaboration of anticoagulant, antithrombotic, and fibrinolytic regulators	Prostacyclin Thrombomodulin Heparin-like molecules Plasminogen activator
Elaboration of prothrombotic molecules	von Willebrand factor Tissue factor Plasminogen activator inhibitor
Production of extracellular matrix	Collagen, proteoglycans
Modulation of blood flow and vascular reactivity	Vasoconstrictors: endothelin, ACE Vasodilators: NO, prostacyclin
Regulation of inflammation and immunity	IL-1, IL-6, chemokines Adhesion molecules: VCAM-1, ICAM, E-selectin, P-selectin Histocompatibility antigens
Regulation of cell growth	Growth stimulators: PDGF, CSF, FGF Growth inhibitors: heparin, TGF- β

ACE, Angiotensin-converting enzyme; CSF, colony-stimulating factor; FGF, fibroblast growth factor; ICAM, intercellular adhesion molecule; IL, interleukin; LDL, low-density lipoprotein; NO, nitric oxide; PDGF, platelet-derived growth factor; TGF- β , transforming growth factor- β ; VCAM, vascular cell adhesion molecule.

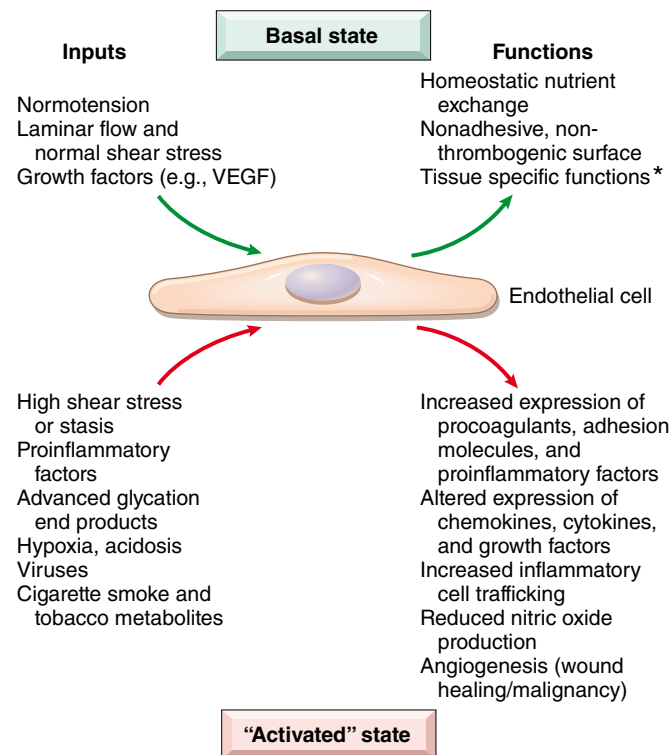


Fig. 11.2 Basal (top) and activated (bottom) endothelial cell states. Normal blood pressure, laminar flow/shear stress, and stable growth factor levels promote a basal endothelial cell state that maintains a nonthrombotic, nonadhesive surface with appropriate vascular wall smooth muscle tone. Injury or exposure to certain mediators such as turbulent flow results in endothelial activation, a state where endothelial cells develop a procoagulant surface that can be adhesive for inflammatory cells and express factors that cause smooth muscle contraction and/or proliferation and matrix synthesis. *Endothelium in different vascular sites (e.g., endothelium with tight junctions to maintain blood-brain barrier, fenestrated endothelium in glomeruli) behaves differently. VEGF, Vascular endothelial growth factor.

lipid products—both contributing to the pathogenesis of atherosclerosis (see later), and advanced glycation end-products (AGEs) that underlie the pathogenic sequelae of diabetes (Chapter 24). Viruses, complement components, and hypoxia also activate ECs.

Depending on the stimulus, activated ECs express adhesion molecules (Chapter 3) and major histocompatibility complex molecules as well as cytokines, chemokines, growth factors, vasoactive molecules, procoagulant or anticoagulant factors, and a variety of other biologically active products. ECs influence the vasoreactivity of the underlying SMCs through the production of both vasodilating (relaxing) factors (e.g., nitric oxide [NO]) and vasoconstrictive factors (e.g., endothelin). Normal EC function is characterized by a balance of these responses (see Fig. 11.2).

Endothelial dysfunction (see Fig. 11.2) refers to an alteration in endothelial phenotype—seen in many different conditions—that is often both proinflammatory and prothrombotic. It is responsible, at least in part, for the initiation of thrombus formation, atherosclerosis, and the vascular lesions of hypertension and other disorders. Some endothelial dysfunction can be rapid in onset (within minutes), reversible, and independent of new protein synthesis (e.g., EC contraction induced by histamine and other vasoactive mediators that cause gaps in venular endothelium) (Chapter 3). Other changes, such as the upregulation of adhesion molecules, involve alterations in gene expression and protein synthesis and require hours or even days to develop.

Vascular Smooth Muscle Cells

Vascular SMCs are the predominant cellular element of the vascular media, playing important roles in normal vascular repair and pathologic processes such as atherosclerosis. Normally, SMCs have a non-proliferative, contractile phenotype; however, they have the capacity to proliferate when appropriately stimulated. They can also synthesize collagen, elastin, and proteoglycans and elaborate growth factors and cytokines. SMCs are responsible for the vasoconstriction or dilation that occurs in response to physiologic or pharmacologic stimuli.

Intimal Thickening: A Stereotyped Injury Response to Vascular Injury

Vascular injury leading to EC dysfunction or loss stimulates SMC recruitment, proliferation, and ECM synthesis; the result is intimal thickening that can compromise vascular flow. Healing of injured vessels is analogous to the healing that occurs in other damaged tissue (Chapter 3) but can be counterproductive because the resulting vascular luminal narrowing reduces downstream blood flow. ECs involved in repair can migrate from adjacent uninjured areas into denuded areas or can derive from circulating precursors.

Medial SMCs or circulating smooth muscle precursor cells also migrate into the intima, proliferate, and synthesize ECM in much the same way that fibroblasts fill in a wound. Neointima is formed by accumulation of smooth muscle cells and extracellular matrix in the intimal layer. Neointimal SMCs have a phenotype that is distinct from that of the contractile medial SMCs; specifically, they are proliferative and have increased biosynthetic capabilities and reduced contractile function. This change in SMC phenotype is stimulated by cytokines and growth factors derived from platelets, ECs, and macrophages, as well as thrombin and activated complement factors. With time and restoration and/or normalization of the EC layer, neointimal SMCs can return to a nonproliferative state.

Thus, neointimal thickening is the *stereotypical response of the vessel wall to any insult*; while it is commonly seen as a core element of atherosclerotic plaque (see later), this response also underlies the vascular stenoses associated with stenting, surgical grafting, vascular inflammation (vasculitis), and vasculopathy associated with solid-organ rejection.

KEY CONCEPTS

Response of Vascular Wall Cells to Injury

- All vessels are lined with ECs and all ECs share certain homeostatic properties.
- ECs in specific vascular beds have special features that allow for tissue-specific functions (e.g., fenestrated ECs in renal glomeruli).
- EC function is tightly regulated in both the basal and activated states. Various physiologic and pathophysiologic stimuli induce endothelial activation and dysfunction that alter the EC phenotype (e.g., procoagulative vs. anticoagulative, proinflammatory vs. antiinflammatory, and nonadhesive vs. adhesive).
- Injury to the vessel wall results in a stereotyped healing response involving SMC proliferation, ECM deposition, and intimal expansion.
- The recruitment and activation of the SMCs involves signals from cells (e.g., ECs, platelets, and macrophages) as well as mediators derived from the coagulation and complement cascades.
- Excessive thickening of the intima can result in luminal stenosis and vascular obstruction.

HYPERTENSIVE VASCULAR DISEASE

Systemic and local tissue blood pressures must be maintained within a narrow range to prevent untoward consequences. Low pressures (hypotension) result in inadequate organ perfusion and can lead to dysfunction or tissue death. Conversely, high pressure (hypertension) can cause end-organ damage and is one of the major risk factors for atherosclerosis stated numerous times already.

Like height and weight, blood pressure is a continuously distributed variable in all populations. Detrimental effects of blood pressure increase continuously as the pressure rises—no rigidly defined threshold level of blood pressure identifies those who have an increased risk for cardiovascular disease. Both the systolic and the diastolic blood pressure are important in determining risk. Individuals with diastolic pressures below 80 mm Hg or systolic pressures below 120 mm Hg are considered to have normal blood pressure (normotensive). Blood pressure is considered borderline elevated if the systolic pressure is 120 to 130 mm Hg. Overt hypertension (high blood pressure) reflects a measurement of ≥ 90 mm Hg diastolic or ≥ 130 mm Hg systolic. Somewhere between one-third and one-half of all adults in the general population, across many nations, are hypertensive based on these criteria. However, such cutoffs do not reliably assess risk in all patients; for example, when other risk factors such as diabetes are present, lower thresholds are applicable.

Table 11.2 lists the major causes of hypertension. Although the molecular pathways that regulate normal blood pressure are reasonably well understood, the causes of hypertension in most individuals remain largely unknown. A small number of patients (approximately 5% to 10%) are said to have *secondary hypertension* resulting from an underlying renal disease (e.g., renal artery stenosis), adrenal disease (e.g., primary aldosteronism, Cushing syndrome, or pheochromocytoma), or another identifiable cause. The remaining 90% to 95% of hypertension is idiopathic—so-called *essential hypertension*. It seems likely that essential hypertension is a multifactorial disorder resulting from the cumulative effects of multiple genetic polymorphisms and interacting environmental factors.

The prevalence and vulnerability to complications of hypertension increase with age. Besides increasing the risk of atherosclerosis, hypertension can cause cardiac hypertrophy and heart failure (*hypertensive heart disease*) (Chapter 12), multi-infarct dementia (Chapter 28), aortic dissection (discussed later in this chapter), and renal failure (Chapter 20). Unfortunately, hypertension typically remains asymptomatic until late in its course, and even severely elevated

TABLE 11.2 Types and Causes of Hypertension (Systolic and Diastolic)

Essential Hypertension
Accounts for 90%–95% of all cases; contributions from genetics and diet
Secondary Hypertension
Renal
Acute glomerulonephritis
Chronic renal disease
Polycystic disease
Renal artery stenosis
Renal vasculitis
Renin-producing tumors
Endocrine
Adrenocortical hyperfunction (Cushing syndrome, primary aldosteronism, congenital adrenal hyperplasia, licorice ingestion)
Hormones (glucocorticoids, estrogen including oral contraceptives, sympathomimetics and tyramine-containing foods, monoamine oxidase inhibitors)
Pheochromocytoma
Acromegaly
Hyperthyroidism (thyrotoxicosis)
Pregnancy-induced (preeclampsia)
Cardiovascular
Coarctation of the aorta
Polyarteritis nodosa
Increased intravascular volume
Increased cardiac output
Rigidity of the aorta
Neurologic
Psychogenic
Increased intracranial pressure
Sleep apnea
Acute stress, including surgery

pressures can be clinically silent for years. Left untreated, roughly one-half of hypertensive patients die of ischemic heart disease or congestive heart failure, and another one-third die of stroke. Treatment with blood pressure-lowering drugs dramatically reduces the incidence and death rates attributable to all forms of hypertension-related pathology.

A small percentage of hypertensive persons (as much as 5%) show a rapidly rising blood pressure that, if untreated, leads to death within 1 to 2 years. This form of hypertension, called *malignant hypertension*, is characterized by severe pressure elevations (i.e., systolic pressure >180 mm Hg, diastolic pressure >120 mm Hg), renal failure, and retinal hemorrhages and exudates, with or without papilledema (swelling of the optic nerve due to increased intracranial pressures). It can develop in previously normotensive persons but more often is superimposed on preexisting “benign” hypertension.

In this section, we will first briefly outline normal blood pressure homeostasis, followed by a discussion of pathogenic mechanisms that underlie hypertension and a description of hypertension-associated pathologic changes in vessels.

Blood Pressure Regulation

Blood pressure is a function of cardiac output and peripheral vascular resistance, both of which are influenced by multiple genetic and environmental factors (Fig. 11.3). The integration of the various inputs ensures adequate systemic perfusion, despite regional demand differences.

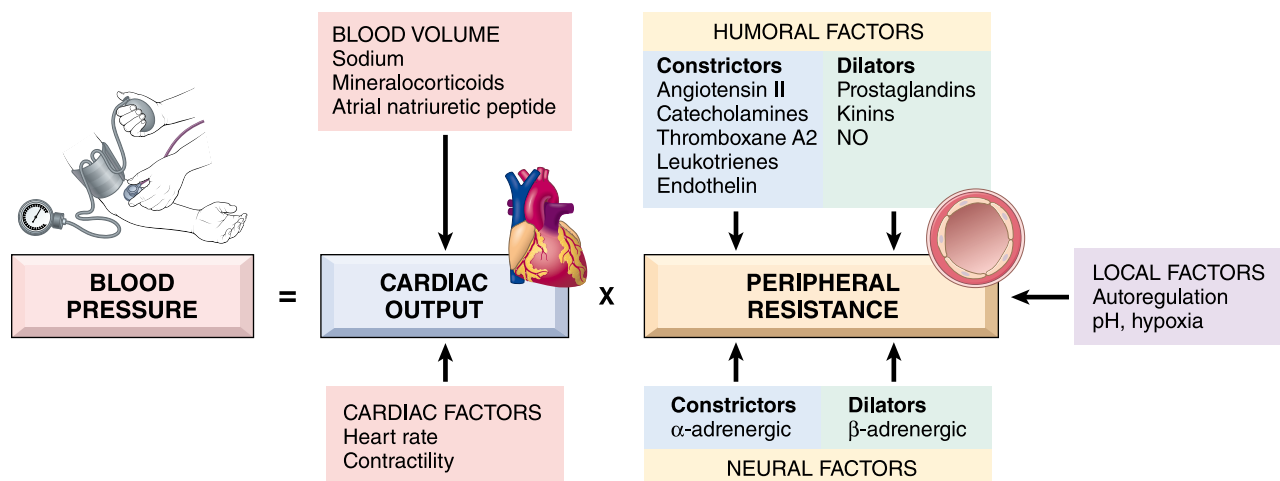


Fig. 11.3 Blood pressure regulation. Diverse influences on cardiac output (e.g., blood volume and myocardial contractility) and peripheral resistance (neural, humoral, and local effectors) impact the output blood pressure. NO, Nitrous oxide.

- **Cardiac output** is a function of stroke volume and heart rate. The most important determinant of stroke volume is the filling pressure, which is regulated through sodium homeostasis and its effect on blood volume. Heart rate and myocardial contractility (a second factor affecting stroke volume) are both regulated by the α- and β-adrenergic systems, which also have important effects on vascular tone.
- **Peripheral resistance** is regulated predominantly at the level of the arterioles by neural and hormonal inputs. Vascular tone reflects a balance between vasoconstrictors (including angiotensin II, catecholamines, and endothelin) and vasodilators (including kinins, prostaglandins, and NO). Resistance vessels also exhibit autoregulation, whereby increased blood flow induces vasoconstriction to protect tissues against hyperperfusion. Finally, blood pressure is fine-tuned by tissue pH and hypoxia to accommodate local metabolic demands.

Factors released from the kidneys, adrenals, and myocardium interact to influence vascular tone and to regulate blood volume by adjusting sodium balance (Fig. 11.4). The kidneys filter 170 L of plasma containing 23 moles of salt daily. Thus, with a typical diet containing 100 mEq of sodium, 99.5% of the filtered salt must be reabsorbed to maintain total body sodium levels. About 98% of the filtered sodium is reabsorbed by several constitutively active transporters. The small amount of remaining sodium is subject to resorption by the epithelial sodium channel (ENaC), which is tightly regulated by the renin-angiotensin-aldosterone system; it is this pathway that determines net sodium balance.

The kidneys and heart contain cells that sense changes in blood pressure or volume. In response, these cells release circulating effectors that act in concert to maintain normal blood pressure.

- **Renin** is a proteolytic enzyme produced by renal juxtaglomerular cells, which are myoepithelial cells adjacent to the glomerular afferent arterioles. Renin is released in response to low blood pressure in afferent arterioles, elevated levels of circulating catecholamines, or low sodium levels in the distal convoluted renal tubules. The latter occurs when the glomerular filtration rate falls (e.g., when the cardiac output is low), leading to increased sodium resorption by the proximal tubules.
- **Renin cleaves plasma angiotensinogen to angiotensin I, which in turn is converted to angiotensin II by angiotensin-converting enzyme (ACE)**, mainly a product of vascular endothelium. Angiotensin II raises blood pressure by (1) inducing vascular contraction,

- (2) stimulating aldosterone secretion by the adrenal gland, and
- (3) increasing tubular sodium resorption. Adrenal aldosterone increases blood pressure by its effect on blood volume; aldosterone increases sodium resorption (and thus water) in the distal convoluted tubules, which increases blood volume.

- **The kidney also produces a variety of vascular relaxing substances** (including prostaglandins and NO) that can counterbalance the vasopressor effects of angiotensin.
- **Myocardial natriuretic peptides** (atrial natriuretic peptide [ANP] and B-type natriuretic peptide [BNP]) are released from atrial myocardium (constitutively) and ventricular myocardium (in volume overload states). ANP and BNP reduce blood pressure by causing systemic vasodilation, and by inhibiting sodium resorption in the distal renal tubules, leading to sodium excretion and diuresis (see Fig. 11.4). Elevated circulating levels of ANP or BNP indicate a hypervolemic state (increased blood volume)—most typically associated with *congestive heart failure* (Chapter 12).

KEY CONCEPTS

Blood Pressure Regulation

- Blood pressure is determined by vascular resistance and cardiac output.
- Vascular resistance is regulated at the level of the arterioles, influenced by neural and hormonal inputs.
- Cardiac output is determined by heart rate and stroke volume, which is strongly influenced by blood volume. Blood volume in turn is regulated mainly by renal sodium excretion or resorption.
- Renin, a major regulator of blood pressure, is secreted by the kidneys in response to decreased blood pressure in afferent arterioles. In turn, renin cleaves angiotensinogen to angiotensin I; subsequent endothelial catabolism produces angiotensin II, which regulates blood pressure by increasing vascular SMC tone and by increasing adrenal aldosterone secretion, thereby increasing renal sodium resorption.

Mechanisms of Secondary Hypertension

For many of the secondary forms of hypertension, the underlying pathways are reasonably well understood.

- In **renovascular hypertension**, renal artery stenosis causes decreased glomerular flow and pressure in the afferent arteriole of the

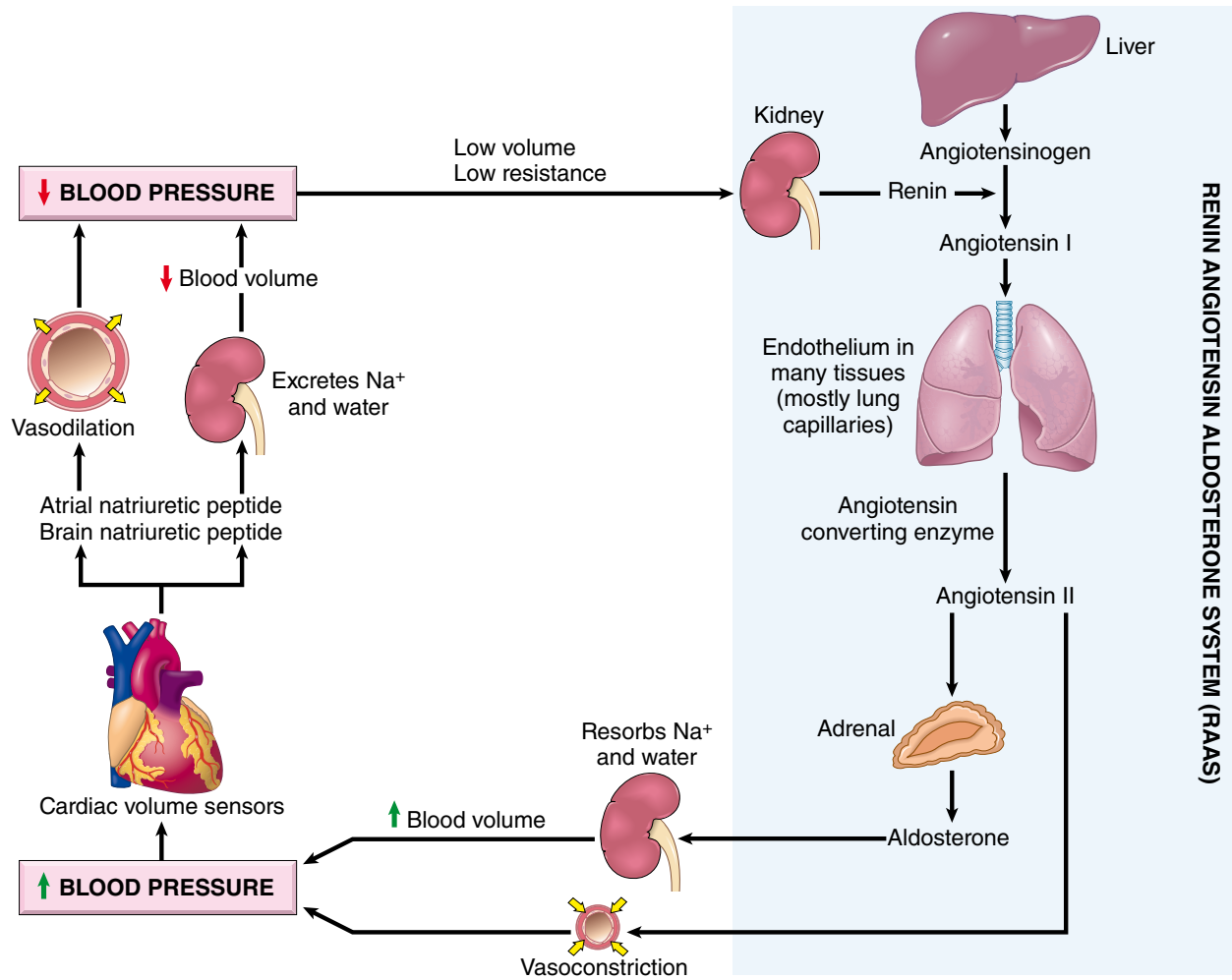


Fig. 11.4 Multiple organs help regulate blood pressure homeostasis. The interplay of the renin-angiotensin-aldosterone system and natriuretic peptides impact on blood pressure.

glomerulus. As already discussed, this stimulates renin secretion, leading to increased blood volume and vasoconstriction via the angiotensin and aldosterone pathways (see Fig. 11.4).

- **Primary hyperaldosteronism** is one of the most frequent causes of secondary hypertension (Chapter 24). It may be idiopathic or less commonly caused by aldosterone-secreting adrenal adenomas.
- **Single-gene disorders** cause rare, severe forms of hypertension.
 - *Gene defects affecting enzymes involved in aldosterone metabolism* (e.g., aldosterone synthase, 11 β -hydroxylase, 17 α -hydroxylase) cause increased aldosterone secretion, resulting in increases in salt and water resorption, plasma volume expansion, and, ultimately, hypertension.
 - *Mutations affecting proteins that influence sodium reabsorption.* For example, a moderately severe form of salt-sensitive hypertension, called *Liddle syndrome*, is caused by mutations in an epithelial Na⁺ channel (ENaC γ) that lead to exaggerated distal tubular reabsorption of sodium in response to aldosterone.

Mechanisms of Essential Hypertension

As mentioned earlier, in most cases hypertension results from complex interactions between genetic and environmental influences.

- **Genetic factors** definitely impact blood pressure regulation, as demonstrated through studies of twins and genetically related versus

adopted children. Moreover, as noted earlier, several single-gene disorders cause relatively rare forms of hypertension (and hypotension) by altering net sodium reabsorption in the kidney. Large genome-wide association studies point to more than 500 genetic variants that are linked to blood pressure levels.

- **Insufficient renal sodium excretion** in the presence of normal arterial pressure may be a key common feature in the development of hypertension. Insufficient sodium excretion may lead sequentially to an increase in fluid volume, increased cardiac output, and peripheral vasoconstriction, thereby elevating blood pressure. At the higher blood pressure, enough sodium is excreted by the kidneys to equal intake and prevent further fluid retention. Thus, a new steady state of sodium balance is achieved ("resetting of pressure natriuresis"), but at the expense of an increase in blood pressure.
- **Vasoconstrictive factors** or stimuli that cause structural changes in the vessel wall can lead to an increase in peripheral resistance and may also play a role in essential hypertension.
- **Environmental factors** such as stress, obesity, smoking, physical inactivity, and heavy salt consumption all are implicated in hypertension. Indeed, there is strong evidence linking dietary sodium intake with the prevalence of hypertension in different populations.

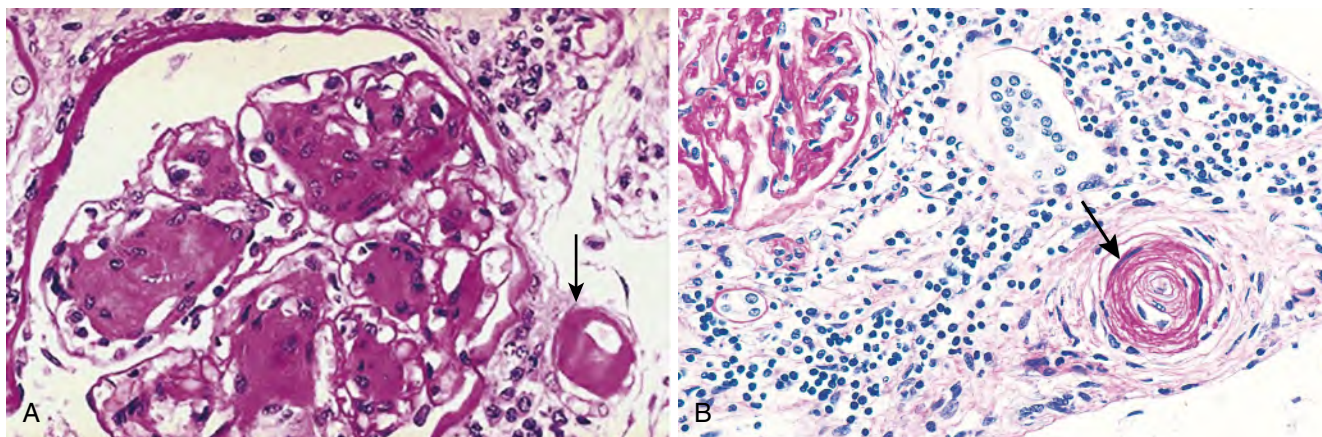


Fig. 11.5 Vascular pathology in hypertension. (A) Hyaline arteriosclerosis. Markedly thickened arteriole (arrow) with protein deposition (hyalinization) and a markedly narrowed lumen is seen to the right of the glomerulus. Note also hyalinized glomeruli in this patient, who also had diabetes (nodular glomerulosclerosis). (B) Hyperplastic arteriosclerosis (onion-skinning) causing luminal obliteration (arrow) (periodic acid-Schiff stain). (A, From Klatt EC. *Atlas of Pathology*. 3rd ed. Elsevier; 2014. B, Courtesy Helmut Rennke, MD, Brigham and Women's Hospital, Boston.)

Vascular Pathology in Hypertension

Hypertension not only accelerates atherogenesis (see later) but also causes degenerative changes in the walls of large and medium arteries that can lead to disorders such as aortic dissection and cerebrovascular hemorrhage. Three forms of small vessel disease are hypertension-related (Fig. 11.5).

MORPHOLOGY

1. **Hyaline arteriosclerosis.** Arterioles show homogeneous, pink hyaline thickening with associated luminal narrowing (see Fig. 11.5A). These changes reflect both plasma protein leakage across injured ECs and increased SMC matrix synthesis in response to the chronic hemodynamic pressures of hypertension. Although the vessels of older patients (either normotensive or hypertensive) also frequently exhibit hyaline arteriosclerosis, it is more generalized and severe in patients with hypertension and diabetes (Chapter 24). In **nephrosclerosis** due to chronic hypertension, the arteriolar narrowing causes diffuse impairment of renal blood supply and glomerular scarring (Chapter 20).

2. **Hyperplastic arteriosclerosis.** This lesion occurs in severe hypertension; vessels exhibit concentric, **laminated** (“onion-skin”) thickening of the walls with luminal narrowing (see Fig. 11.5B). The laminations consist of SMCs with thickened, reduplicated basement membrane; in malignant hypertension, they are accompanied by fibrinoid deposits and vessel wall necrosis.

3. **Necrotizing arteriolitis**, particularly in the kidney (Chapter 20).

Pulmonary hypertension can be caused by several entities including left heart failure, congenital heart disease, valve disorders, obstructive or interstitial lung disease, and recurrent thromboemboli. The arterioles in such affected lungs typically show histologic changes ranging from fibrotic intimal thickening to medial hyperplasia. These are described in greater detail in Chapter 15.

- Essential hypertension represents 90% to 95% of cases and is a complex, multifactorial disorder involving both environmental influences and genetic variants that influence sodium resorption, aldosterone pathways, and the renin-angiotensin-aldosterone system.
- Hypertension is occasionally caused by single-gene disorders or is secondary to diseases of the kidney, adrenal, or other endocrine organs.
- Sustained hypertension is marked by increased blood volume and increased peripheral resistance, both of which may stem from increased sodium resorption by the kidney.
- Histologically, hypertension is associated with thickening of arterial walls caused by hyaline deposits (hyaline arteriosclerosis) and, in severe cases, by proliferation of ECs or SMCs and replication of the basement membrane (hyperplastic arteriosclerosis).

ARTERIOSCLEROSIS

Arteriosclerosis literally means “hardening of the arteries”; it is a generic term for arterial wall thickening and loss of elasticity. There are four general patterns, with different clinical and pathologic consequences.

- **Arteriosclerosis** affects small arteries and arterioles and may lead to ischemic injury. The two anatomic variants associated with hypertension—hyaline and hyperplastic—are discussed above.
- **Mönckeberg medial sclerosis** is characterized by calcifications of the medial walls of muscular arteries, typically starting along the internal elastic membrane. Adults older than age 50 years are most commonly affected, particularly those with renal insufficiency and secondary hypercalcemia. The calcifications do not encroach on the vessel lumen and are usually not clinically significant.
- **Fibromuscular intimal hyperplasia** occurs in muscular arteries larger than arterioles. It is driven by inflammation (as in a healed arteritis or transplant-associated arteriopathy; see Chapter 12) or by mechanical injury (e.g., associated with stents or balloon angioplasty; see later) and can be considered a healing response. The affected vessels can become quite stenotic; indeed, such intimal hyperplasia underlies in-stent restenosis and is a major cause of loss-of-function in solid-organ transplants.

KEY CONCEPTS

Hypertension

- Hypertension is a common disorder affecting roughly one-half of adults in the United States; it is a major risk factor for atherosclerosis, congestive heart failure, and renal failure.

- *Atherosclerosis*, from Greek root words for “gruel” and “hardening,” is the most frequent and clinically important vasculopathy and is discussed here.

ATHEROSCLEROSIS

Atherosclerosis underlies the pathogenesis of coronary, cerebral, and peripheral vascular disease and causes more morbidity and mortality (roughly half of all deaths) in the Western world than any other disorder. Coronary artery disease is an important manifestation of atherosclerosis, which often leads to death from ischemic heart disease (Chapter 12); indeed, myocardial infarction is responsible for almost a quarter of all deaths in the United States. Significant morbidity and mortality are also caused by aortic and carotid atherosclerotic disease, resulting in aneurysm and stroke, respectively.

The likelihood of atherosclerosis is strongly influenced by acquired (e.g., cholesterol levels, smoking, hypertension), inherited (e.g., low-density lipoprotein [LDL] receptor gene mutations), and sex- and age-associated risk factors. Acting in concert, they cause intimal lesions called *atheromas* (also called *atheromatous* or *atherosclerotic plaques*) that, when advanced, protrude into vessel lumens. A typical atheromatous plaque is a raised lesion with a soft grumous core of lipid (mainly cholesterol and cholesterol esters) covered by a fibrous cap (see later). Besides mechanically obstructing blood flow, atherosclerotic plaques can rupture, leading to rapid and often catastrophic formation of an obstructive thrombus. Atherosclerotic plaques also increase the diffusion distance from the lumen to the media, leading to SMC ischemia with resultant wall weakening and aneurysm formation.

Epidemiology

Although atherosclerosis-associated ischemic heart disease is virtually ubiquitous in high-resource nations, risk reduction and improved therapies have combined to sharply decrease the associated morbidity and mortality over the past several decades. At the same time, reduced mortality from infectious diseases and the adoption of Western diet and lifestyles has led to the increased prevalence of ischemic heart disease in lower resource nations. As a result, the death rate for coronary artery disease in Africa, India, and Southeast Asia now exceeds that in the United States; Eastern European countries have rates three to five times higher than the United States and 7 to 12 times higher than Japan.

Risk Factors

The prevalence and severity of atherosclerosis and ischemic heart disease among individuals and groups are related to a number of risk factors identified through several prospective analyses (e.g., the Framingham Heart Study); some of these are constitutional (and therefore less controllable), but others are acquired or related to specific behaviors and potentially amenable to intervention (Table 11.3). These risk factors typically have greater than additive effects, but treatment (even less than optimal) can mitigate some of the risk (Fig. 11.6).

Constitutional Risk Factors

- *Genetics.* Family history is the most important independent risk factor for atherosclerosis. Certain Mendelian disorders are strongly associated with atherosclerosis (e.g., familial hypercholesterolemia; see Chapter 5), but they account for only a small percentage of cases. The well-established familial predisposition to atherosclerosis and ischemic heart disease is usually polygenic, relating to small effects of many shared alleles common to a family or population.
- *Age is a dominant influence.* The development of atherosclerotic plaque is a progressive process that usually becomes clinically

TABLE 11.3 Major Risk Factors for Atherosclerosis

Nonmodifiable (Constitutional)
Genetic variation
Family history
Increasing age
Male sex
Modifiable
Hyperlipidemia
Hypertension
Cigarette smoking
Diabetes
Inflammation

manifest in middle age or later (see later). Thus, between ages 40 and 60, the incidence of myocardial infarction increases fivefold. Death rates from ischemic heart disease rise with each decade even into advanced age. It is now clear, however, that this association with aging is related to acquired alterations in the innate immune system as well as the cumulative effects of vascular injury. One feature of aging is the outgrowth of cells derived from hematopoietic stem cells that bear mutations providing them with a selective advantage relative to normal stem cells, a phenomenon called *clonal hematopoiesis of indeterminate potential (CHIP)* (Chapter 13). In addition to increasing the risk of hematopoietic malignancies, CHIP is strongly associated with an increased risk of dying of cardiovascular disease. Work in animal models suggests this is because immune cells derived from the mutated clone, particularly monocytes, have an abnormal pro-inflammatory phenotype that exacerbates the development and severity of atherosclerosis (see later).

- *Sex.* Premenopausal females are relatively “protected” against atherosclerosis and its consequences compared with age-matched males. Thus, myocardial infarction and other complications of atherosclerosis are uncommon in premenopausal females unless they are otherwise predisposed by diabetes, hyperlipidemia, or severe hypertension. After menopause, however, the incidence of atherosclerosis-related diseases increases in females, eventually exceeding that in males. Although a favorable influence of estrogen has been proposed to explain this effect, clinical trials of estrogen replacement did not show any benefit; indeed, postmenopausal estrogen therapy actually increased cardiovascular risk in some older females (Chapter 9).

Modifiable Major Risk Factors

The modifiable risk factors include the level and composition of body lipids, as well as smoking, and control of diseases that predispose to atherosclerosis, notably hypertension and diabetes. The most important are described below.

- *Hyperlipidemia*—and more specifically hypercholesterolemia—is a major risk factor for atherosclerosis; even in the absence of other risk factors, hypercholesterolemia is sufficient to initiate lesion development. The major component of serum cholesterol associated with increased risk is LDL cholesterol (“bad cholesterol”). LDL is the lipid-cholesterol-protein complex that delivers cholesterol to peripheral tissues; in contrast, high-density lipoprotein (HDL) is the complex that mobilizes cholesterol from the periphery (including atheromas) and transports it to the liver for catabolism and biliary excretion. Higher levels of HDL (“good cholesterol”) correlate with reduced risk.

Understandably, dietary and pharmacologic interventions that lower LDL or total serum cholesterol are of considerable interest. Interestingly, approaches that exclusively raise HDL are not effective.

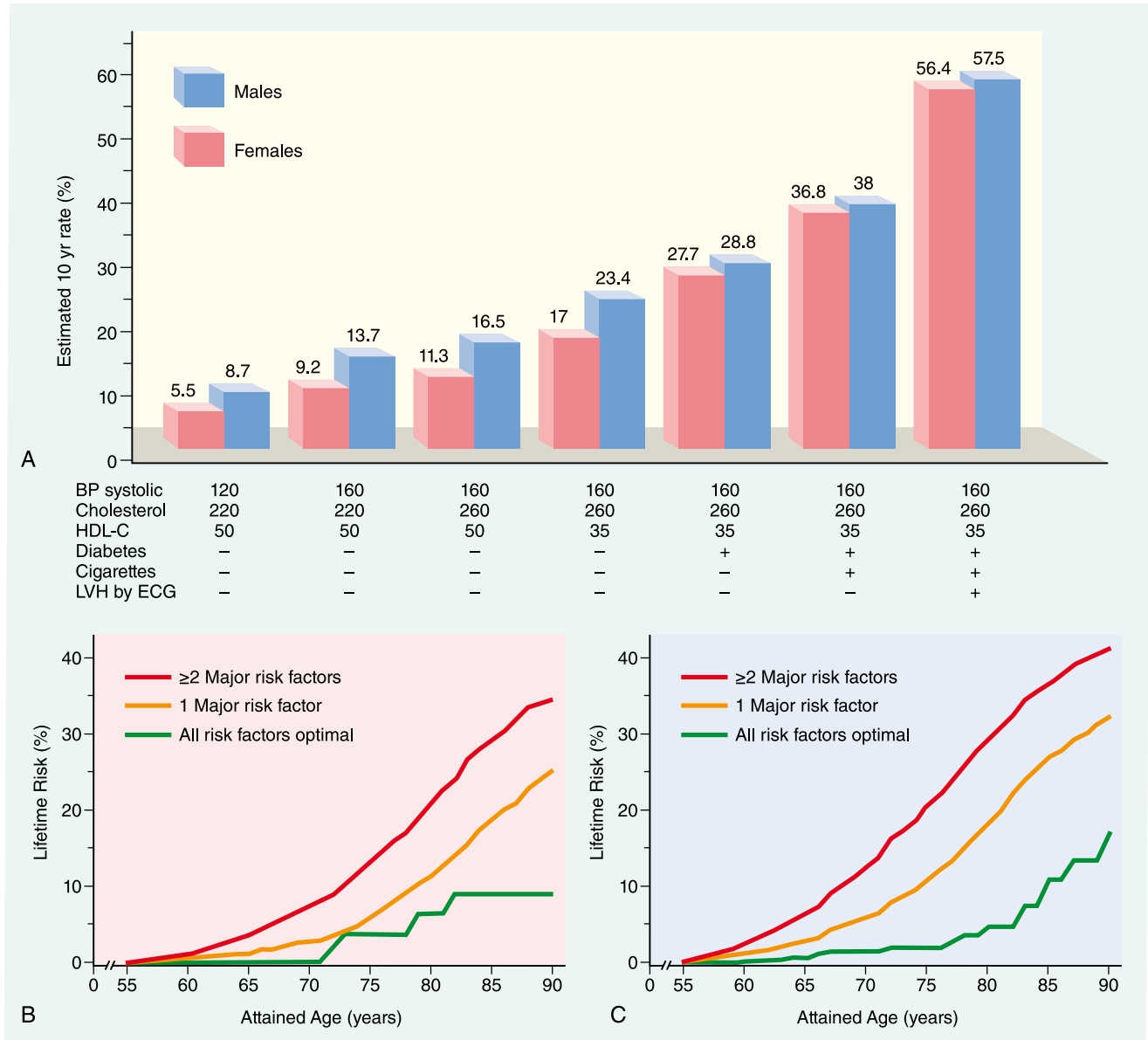


Fig. 11.6 Lifetime risk of death from cardiovascular disease. (A) Estimated 10-year risk of coronary artery disease in hypothetical 55-year-old males and females as a function of traditional risk factors (hyperlipidemia, hypertension, smoking, and diabetes). *BP*, Blood pressure; *ECG*, electrocardiogram; *HDL-C*, high-density lipoprotein cholesterol; *LVH*, left ventricular hypertrophy. In females (B) and males (C), one or more risk factors of blood pressure, cholesterol, diabetes, and cigarettes significantly increases the lifetime risk of a cardiovascular event. (A, From O'Donnell CJ, Kannel WB. Cardiovascular risks of hypertension: lessons from observational studies. *J Hypertens Suppl.* 1998;16(6):S3-S7. B–C, Modified from Berry JD, Dyer A, Cai X, et al. Lifetime risks of cardiovascular disease. *N Eng J Med.* 2012;366:321-329.)

Although previously considered important, the contribution of most dietary fats to atherosclerosis is now viewed as minimal. Nevertheless, *omega-3 fatty acids* (abundant in fish oils) are considered beneficial, whereas trans-unsaturated fats produced by artificial hydrogenation of polyunsaturated oils (used in baked goods and margarine) can adversely affect cholesterol profiles.

Statins are a class of drugs that lower circulating cholesterol levels by inhibiting hydroxymethylglutaryl coenzyme A (HMG CoA) reductase, the rate-limiting enzyme in hepatic cholesterol biosynthesis (Chapter 5). Statins are widely used to lower serum cholesterol levels,

lowering rates of myocardial infarctions and arguably one of the most significant success stories of translational research. Interestingly, some of the benefit of the statins may be due to “off-target” effects that reduce inflammation (see later).

More recently, a new player in regulating LDL levels has been identified; the intracellular serine protease PCSK9 (*proprotein convertase subtilisin/kexin type 9*). PCSK9 binds to LDL receptors and accelerates their degradation, leading to elevated LDL cholesterol levels. Conversely, loss-of-function *PCSK9* mutations reduce levels of cholesterol, and a new class of PCSK9 inhibitors results in markedly reduced LDL cholesterol.

- *Hypertension* can increase the risk of ischemic heart disease by approximately 60% versus normotensive populations. Chronic hypertension is the most common cause of left ventricular hypertrophy, and hence the latter is also a surrogate marker for cardiovascular risk.
- *Cigarette smoking*, and in particular prolonged use (years), doubles the death rate from ischemic heart disease. Smoking cessation reduces that risk substantially.
- *Diabetes* induces hypercholesterolemia (Chapter 24) and markedly increases the risk of atherosclerosis. Other factors being equal, the incidence of myocardial infarction is twice as high in patients with diabetes relative to normoglycemic individuals. There is also an increased risk of strokes and a 100-fold increased risk of atherosclerosis-induced gangrene of the lower extremities.

Additional Risk Factors

As many as 20% of all cardiovascular events occur in the absence of major risk factors (e.g., hypertension, hyperlipidemia, smoking, or diabetes). Indeed, more than 75% of cardiovascular events in previously healthy females occur with LDL cholesterol levels below 160 mg/dL (levels generally considered to connote low risk). Clearly, other factors are contributory. Among those that are proven or suspected are:

- *Inflammation*. Inflammation is present during all stages of atherogenesis and is intimately linked with atherosclerotic plaque formation and rupture (see later). With the increasing recognition that inflammation plays a significant causal role in ischemic heart disease, assessment of systemic inflammation has become important in overall risk stratification. A number of circulating markers of inflammation correlate with ischemic heart disease, such as *C-reactive protein (CRP)*.
CRP is an acute phase reactant synthesized primarily by the liver. Although CRP does not appear to be causally related to the development of atherosclerosis or its sequelae, it is well established that plasma CRP is a strong, independent marker of risk for myocardial infarction, stroke, peripheral arterial disease, and sudden cardiac death, even among apparently healthy individuals (Fig. 11.7). Accordingly, CRP levels are now incorporated into risk stratification algorithms. Interestingly, CRP levels are also typically lowered in concert with other risk reduction measures including smoking cessation, weight loss, exercise, and statin administration.
- *Hyperhomocysteinemia*. Serum homocysteine levels correlate with coronary atherosclerosis, peripheral vascular disease, stroke, and venous thrombosis. *Homocystinuria*, caused by a rare inborn error of metabolism, results in elevated circulating homocysteine ($>100 \mu\text{mol/L}$) and is associated with premature vascular disease.
- *Metabolic syndrome*. Associated with central obesity (Chapter 9), this entity is characterized by insulin resistance, hypertension, dyslipidemia (elevated LDL and depressed HDL), hypercoagulability, and a proinflammatory state. Dyslipidemia, hyperglycemia, and hypertension are all cardiac risk factors, while the systemic hypercoagulable and proinflammatory state may contribute to endothelial dysfunction and/or thrombosis.
- *Lipoprotein a (Lp[a])* is an altered form of LDL that contains the apolipoprotein B-100 portion of LDL linked to apolipoprotein A (apo A). Lp(a) impacts fibrinolysis, inflammation, oxidative stress, and overall endothelial cell function; high levels are associated with coronary and cerebrovascular disease risk, independent of total cholesterol or LDL levels.
- *Factors affecting hemostasis*. Several markers of hemostatic and/or fibrinolytic function (e.g., elevated plasminogen activator inhibitor 1) are potent predictors of risk for major atherosclerotic events, including myocardial infarction and stroke. Platelet-derived factors as well as thrombin—through both procoagulant and proinflammatory

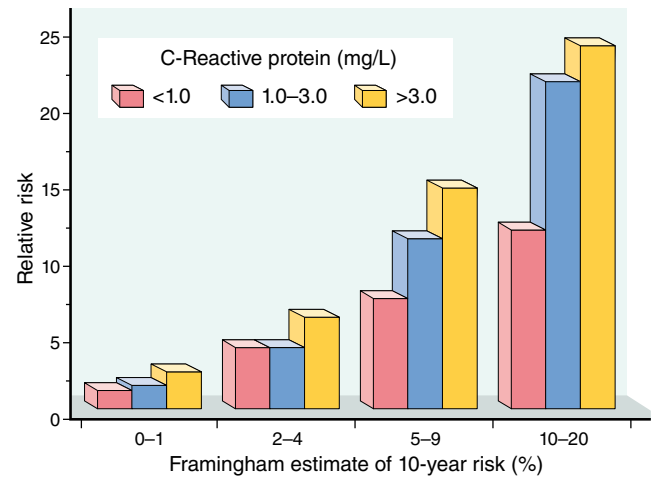


Fig. 11.7 C-reactive protein adds prognostic information at all levels of traditional risk identified from the Framingham Heart Study. Relative risk (y-axis) refers to the risk of a cardiovascular event (e.g., myocardial infarction). The x-axis is the 10-year risk of a cardiovascular event derived from the traditional risk factors identified in the Framingham study. In each risk group, C-reactive protein values further stratify the patients. (Data from Ridker PM, Rifai N, Rose L, Buring JE, Cook NR. Comparison of C-reactive protein and low-density lipoprotein cholesterol levels in the prediction of first cardiovascular events. *N Engl J Med*. 2002;347:1557-1565.)

effects—are increasingly recognized as major contributors to local vascular pathology.

- *Other factors*. Factors associated with a less pronounced and/or difficult to quantitate risk include lack of exercise; competitive, stressful lifestyle (type A personality); and obesity (the last-mentioned also being complicated by hypertension, diabetes, hypertriglyceridemia, and decreased HDL).

Pathogenesis

The clinical importance of atherosclerosis has stimulated enormous interest in understanding the mechanisms that underlie its evolution and complications. The contemporary view of atherogenesis, called the “response to injury” hypothesis, integrates the risk factors previously discussed. **This model views atherosclerosis as a chronic inflammatory and healing response of the arterial wall to endothelial injury. Lesion progression occurs through interaction of modified lipoproteins, macrophages, and T lymphocytes with ECs and SMCs of the arterial wall (Fig. 11.8).** According to this schema, atherosclerosis progresses in the following sequence:

- *Endothelial injury and dysfunction*, causing (among other things) increased vascular permeability, leukocyte adhesion, and thrombosis.
- *Accumulation of lipoproteins* (mainly LDL and its oxidized forms) in the vessel wall.
- *Monocyte adhesion to the endothelium*, followed by migration into the intima and transformation into *macrophages* and *foam cells*.
- *Platelet adhesion*.
- *Factor release* from activated platelets, macrophages, and vascular wall cells, inducing SMC recruitment, either from the media or from circulating precursors.
- *SMC proliferation, ECM production, and recruitment of T cells*.
- *Lipid accumulation* both extracellularly and within cells (macrophages and SMCs).
- *Calcification* of ECM and necrotic debris late in the pathogenesis.

We will now discuss the role of these factors in the pathogenesis of atherosclerosis in some detail, starting with endothelial injury.

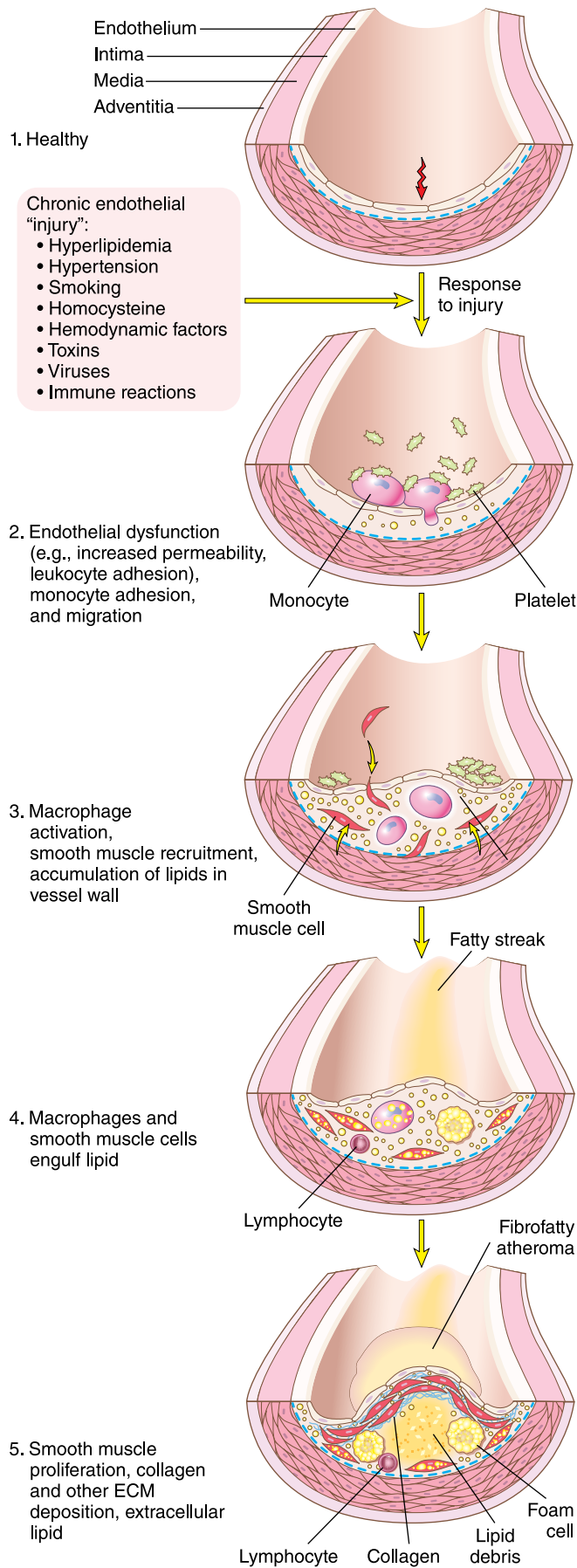


Fig. 11.8 Evolution of arterial wall changes in the response to injury hypothesis. 1, Healthy. 2, Endothelial injury with monocyte and platelet adhesion. 3, Monocyte and smooth muscle cell migration into the intima, with macrophage activation. 4, Macrophage and smooth muscle cell uptake of modified lipids, with further activation. 5, Intimal smooth muscle cell proliferation with extracellular matrix (ECM) production, forming a well-developed plaque.

Endothelial Injury. EC injury is the cornerstone of the response-to-injury hypothesis. Early atherosclerosis lesions begin at sites of morphologically *intact* endothelium that exhibit features of endothelial dysfunction—increased permeability, enhanced leukocyte adhesion, and altered gene expression. Causes of EC dysfunction include toxins from cigarette smoke, hypertriglyceridemia, and the local production of inflammatory cytokines. However, the **three most important causes of endothelial dysfunction are hemodynamic disturbances, hypercholesterolemia, and inflammation.**

Hemodynamic Disturbances. The importance of hemodynamic turbulence in atherogenesis is illustrated by the observation that plaques do not occur randomly, but rather tend to locate at the ostia of exiting vessels, branch points, and along the posterior abdominal aorta where flow patterns are disturbed and nonlaminar. This is explained by the fact that laminar nonturbulent flow increases the production of transcription factors and, in particular, Krüppel-like factor-2 (KLF2) that turn on “atheroprotective” genes and turn off inflammatory gene transcription. Conversely, turbulent, nonlaminar flow drives a repertoire of genetic transcription that makes those sites “atheroprone.” It is noteworthy that some of the atheroprotective effects of statins also occur through KLF2 upregulation.

Hypercholesterolemia. Lipids are transported in the bloodstream bound to specific apoproteins (forming lipoprotein complexes). *Dyslipoproteinemias* are lipoprotein abnormalities that may be present in the general population (and, indeed, are found in many myocardial infarction survivors). These include (1) increased LDL cholesterol levels, (2) decreased HDL cholesterol levels, and (3) increased levels of the abnormal Lp(a). These may result from mutations in apoproteins or lipoprotein receptors or arise from other underlying disorders that affect circulating lipid levels such as nephrotic syndrome, alcohol use disorder, hypothyroidism, or diabetes mellitus. All these abnormalities are associated with an increased risk of atherosclerosis.

The evidence implicating hypercholesterolemia in atherogenesis includes:

- *The dominant lipids in atheromatous plaques are cholesterol and cholesterol esters.*
- *Genetic defects in lipoprotein uptake and metabolism* that cause hyperlipoproteinemia are associated with accelerated atherosclerosis. Thus, familial hypercholesterolemia, caused by loss-of-function mutations in LDL receptors that result in inadequate hepatic LDL uptake and catabolism (Chapter 5), can precipitate myocardial infarctions before age 20 years in those with homozygous mutations. Similarly, accelerated atherosclerosis occurs in animal models with engineered deficiencies in apolipoproteins or LDL receptors.
- *Epidemiologic analyses* demonstrate a significant correlation between the severity of atherosclerosis and the levels of total plasma cholesterol or LDL.
- *Lowering serum cholesterol* by diet or drugs slows the rate of progression of atherosclerosis, causes regression of some plaques, and reduces the risk of cardiovascular events.

The mechanisms by which hyperlipidemia contributes to atherogenesis include:

- **Impaired EC function.** Chronic hyperlipidemia, particularly hypercholesterolemia, can directly impair EC function by increasing local reactive oxygen species production. Besides causing membrane and mitochondrial damage, oxygen free radicals accelerate NO decay, dampening its vasodilator activity.
- **Modified LDL.** With chronic hyperlipidemia, lipoproteins accumulate within the intima, where they may aggregate or become oxidized by free radicals produced by inflammatory cells. Such modified LDL is taken up by macrophages through a variety of scavenger receptors (distinct from the LDL receptor). Because the modified lipoproteins cannot be completely degraded, chronic ingestion leads to the formation of lipid-filled macrophages called *foam cells*. SMCs can similarly transform into lipid-laden foam cells by ingesting modified lipids through LDL receptor-related proteins. Not only are the modified lipoproteins toxic to ECs, SMCs, and macrophages, but their uptake also stimulates the release of growth factors, cytokines, and chemokines that create a vicious inflammatory cycle of monocyte recruitment and activation. The early lesions containing lipid-filled macrophages are called *fatty streaks*.

Inflammation. Chronic inflammation contributes to the initiation and progression of atherosclerotic lesions. It is believed that

inflammation is triggered by the accumulation of cholesterol crystals and free fatty acids in macrophages and other cells (Fig. 11.9). These cells sense the presence of abnormal materials via cytosolic innate immune receptors that are components of the inflammasome (Chapter 6). The resulting inflammasome activation leads to the production of the proinflammatory cytokine IL-1, which activates ECs and recruits and activates mononuclear cells including macrophages and T lymphocytes. Mononuclear cell activation, in turn, leads to local production of cytokines and chemokines that recruit and activate more inflammatory cells. Activated macrophages produce reactive oxygen species that enhance LDL oxidation and elaborate growth factors that drive SMC proliferation. Activated T lymphocytes in the growing intimal lesions elaborate inflammatory cytokines, (e.g., interferon- γ), which, in turn, can activate macrophages as well as ECs and SMCs. Thus, many of the lesions of atherosclerosis are attributable to the chronic inflammatory reaction in the vessel wall.

Infection. Although circumstantial evidence has been presented linking atherosclerosis to herpesvirus, cytomegalovirus, and *Chlamydia pneumoniae*, there is no established causal role for infection.

Smooth Muscle Proliferation and Matrix Synthesis. Vascular SMCs from the media can be recruited into the intima, where they can proliferate and synthesize a variety of ECM proteins. This process converts a fatty streak into a mature atheroma and contributes to the

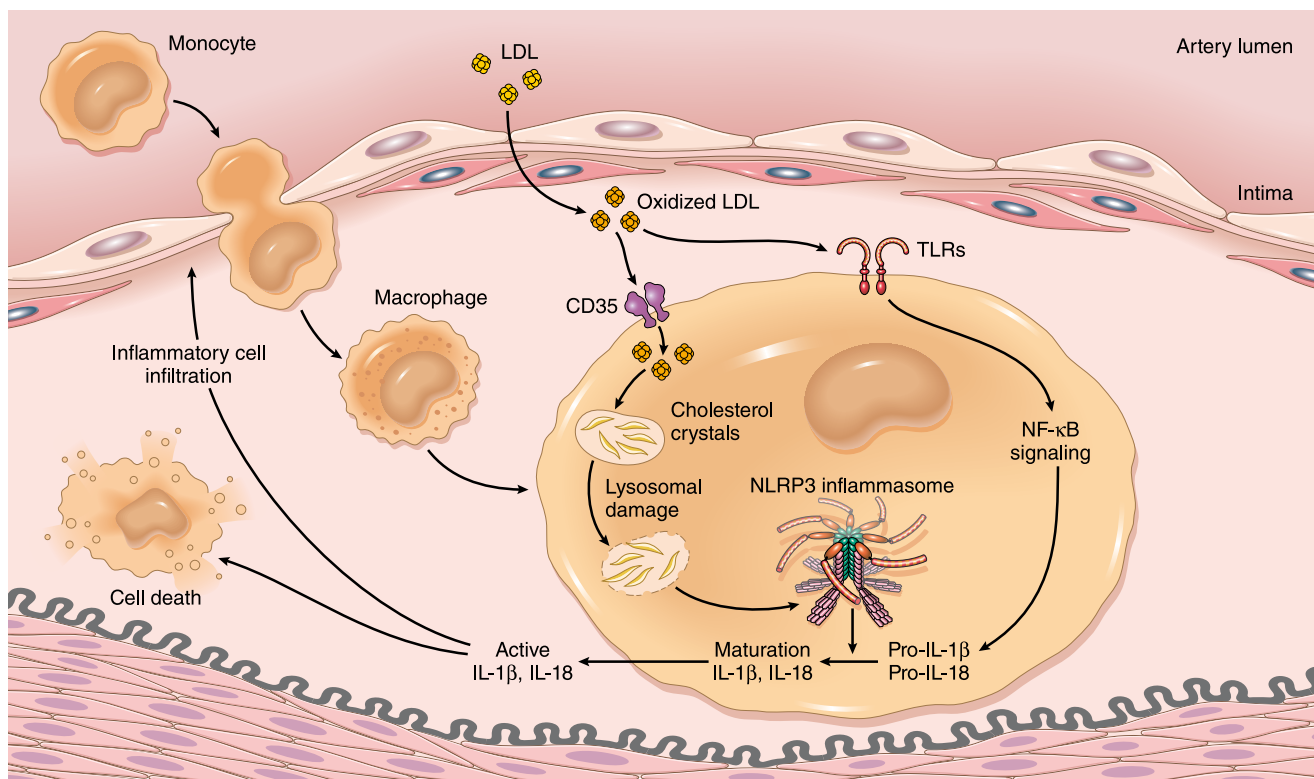


Fig. 11.9 The role of cholesterol crystals in inflammasome activation and cytokine production in atherosclerotic plaque. Oxidized low-density lipoprotein (LDL) leads to the activation of subendothelial macrophages through toll-like receptor (TLR) mediated initiation of downstream nuclear factor (NF)- κ B transcription factor activation. The NF- κ B complex then induces the production of proinflammatory mediators, including interleukin precursors. Uptake of oxidized LDL and subsequent cholesterol crystallization are recognized by a sensor (NLRP3), which results in assembly of the active inflammasome complex that proteolytically cleaves pro-interleukin-1 β (IL-1 β) and interleukin-18 (IL-18) molecules to generate the biologically active forms of these cytokines. Inflammasome activation can also trigger a pathway of cell death called pyroptosis (Chapter 2). (Modified from Li X, Deroide N, Mallat Z. The role of the inflammasome in cardiovascular disease. *J Mol Med (Berl)*. 2014;92:307-319.)

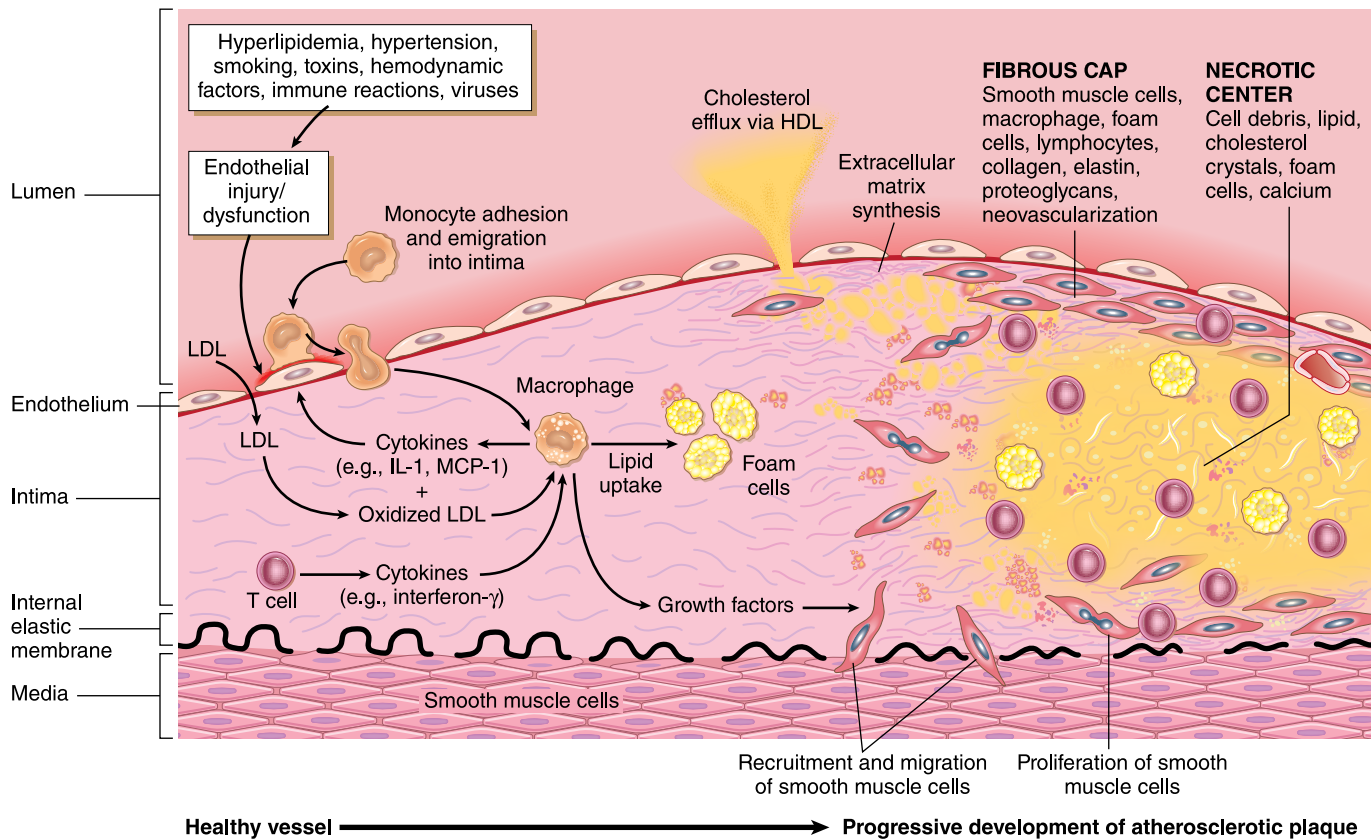


Fig. 11.10 Cells and inflammatory molecules contributing to atheroma formation and progression. Hyperlipidemia, hyperglycemia, hypertension, and other influences cause endothelial dysfunction. This results in platelet and monocyte adhesion, with subsequent cytokine and growth factor release leading to smooth muscle cell migration and proliferation. Foam cells in atheromatous plaques derive from macrophages and smooth muscle cells that have accumulated modified lipids (e.g., oxidized and aggregated low-density lipoprotein [LDL]) via scavenger and LDL-receptor-related proteins. Extracellular lipid is derived from insudation from the vessel lumen, particularly in the presence of hypercholesterolemia, as well as from degenerating foam cells. Cholesterol accumulation in the plaque reflects an imbalance between influx and efflux; high-density lipoprotein (HDL) likely helps clear cholesterol from these accumulations. In response to the elaborated cytokines and chemokines, smooth muscle cells migrate to the intima, proliferate, and produce extracellular matrix including collagen and proteoglycans. This matrix comprises the fibrous cap, which can overlay a necrotic center. A thin fibrous cap can be referred to as a vulnerable plaque, while a thick fibrous cap is considered stable. Atherosclerosis is an intimal-based process with a complex interplay of cells and extracellular materials. Plaques can have secondary effects on the underlying media including a reduction in smooth muscle cells. *IL-1*, Interleukin-1; *MCP-1*, monocyte chemoattractant protein-1.

progressive growth of atherosclerotic lesions (see Fig. 11.10). Several growth factors are implicated in SMC proliferation and ECM synthesis, including platelet-derived growth factor (PDGF) released by locally adherent platelets, as well as macrophages, ECs and SMCs; fibroblast growth factor; and transforming growth factor- α (Chapter 1). These factors also stimulate SMCs to synthesize ECM (notably collagen), which stabilizes atherosclerotic plaques. In contrast, activated inflammatory cells in atheromas may increase the breakdown of ECM components, destabilizing the plaques and making them prone to rupture (see later).

Overview. Figures 11.8 to 11.10 summarize the major pathogenic pathways in atherogenesis, emphasizing the multifactorial nature of the disease. These schematics highlight the concept of atherosclerosis as a chronic inflammatory response—and ultimately an attempt at vascular “healing”—driven by a variety of insults including EC injury, lipid oxidation, lipid accumulation, and inflammation. Atheromas are

dynamic lesions consisting of dysfunctional ECs, proliferating SMCs, and admixed T lymphocytes and macrophages. All four cell types are capable of liberating mediators that can influence atherogenesis. Thus, at early stages, intimal plaques are little more than aggregates of SMCs, macrophages, and foam cells; death of these cells releases lipids and necrotic debris. With progression, the atheroma is modified by ECM synthesized by SMCs; connective tissue is particularly prominent on the intimal aspect, forming a fibrous cap. Lesions typically retain a central core of lipid-laden cells and fatty debris that can become calcified. Although the intimal plaque can initially cause medial remodeling and outward expansion, the growing atherosclerotic plaque—with varying degrees of calcification depending on the nature of the surrounding matrix—eventually encroaches on the vessel lumen and compromises blood flow. The plaque can also compress the underlying media, leading to its degeneration, or can erode or rupture to expose thrombogenic factors, resulting in thrombus formation and acute vascular occlusion.

MORPHOLOGY

Fatty streaks. Fatty streaks are composed primarily of lipid-filled foamy macrophages. Beginning as small flat yellow macules, these can eventually coalesce into elongated streaks 1 cm long or longer. The lesions are slightly raised and do not cause any significant flow disturbance (Fig. 11.11). Although fatty streaks can evolve into plaques, not all are destined to become advanced lesions. Aortas of infants can exhibit fatty streaks, and such lesions are present in virtually all adolescents, even those without known risk factors. That coronary fatty streaks begin to form in adolescence at the same anatomic sites that later tend to develop plaques suggests a precursor-product relationship between these lesions and atheromas.

Atherosclerotic plaque. Atheromatous plaques are yellow-tan and are raised above the surrounding vessel wall; any superimposed thrombus over ulcerated plaques will be red-brown. Plaques vary in size but can coalesce to form larger masses (Fig. 11.12). Atherosclerotic lesions are patchy and rarely circumferential, usually involving only a portion of any given arterial wall; on cross-section, the lesions therefore appear eccentric (Fig. 11.13A). The focality of atherosclerotic lesions—despite the uniform exposure of vessel walls to such factors as cigarette smoke toxins, elevated LDL, hyperglycemia, and so forth—is attributable to the vagaries of vascular hemodynamics. Local flow disturbances, such as non-laminar flow (turbulence) at branch points, make certain portions of a vessel wall more susceptible to plaque formation. Although initially focal and sparsely distributed, atherosclerotic lesions tend to enlarge over time and become more numerous and broadly distributed. Moreover, in any given vessel, lesions at various stages often coexist.

In descending order, **the most extensively involved vessels are the lower abdominal aorta and iliac arteries, the coronary arteries, the popliteal arteries, the internal carotid arteries, and the vessels of the circle of Willis.** Vessels of the upper extremities are usually relatively spared, as are the mesenteric and renal arteries except at their ostia. Although most individuals tend to have a consistent degree of atherosclerotic burden in the affected vasculature, the severity of disease in one arterial distribution does not necessarily predict severity in another.

Atherosclerotic plaques have four principal components: (1) cells, including variable numbers of SMCs, macrophages, and T lymphocytes; (2) ECM, including collagen, elastic fibers, and proteoglycans; (3) intracellular and extracellular lipids; and (4) calcifications in late stage

plaques (Fig. 11.13B–C). These components occur in varying proportions and configurations in different lesions. Typically, there is a superficial fibrous cap composed of SMCs and relatively dense collagen. Beneath and to the side of the cap (the “shoulder”) is a more cellular area containing macrophages, T lymphocytes, and SMCs. Deep to the fibrous cap is a necrotic core containing lipid (primarily cholesterol and cholesterol esters), debris from dead cells, foam cells (lipid-laden macrophages and lipid-laden SMCs), fibrin, thrombus in varying degrees of organization, and other plasma proteins. The cholesterol content is frequently present as crystalline aggregates that are washed out during routine tissue processing and leave behind only empty “clefts.” The periphery of the lesions demonstrates **neovascularization** (proliferating small blood vessels) (see Fig. 11.13C). Most atheromas contain abundant lipid, but some plaques (“**fibrous plaques**”) are composed almost exclusively of SMCs and fibrous tissue, which can be heavily calcified.

Plaques generally continue to change and progressively enlarge as a consequence of cell death and degeneration, synthesis and degradation (remodeling) of ECM, organization of any superimposed thrombus, and secondary calcification of phospholipids or necrotic debris (see Fig. 11.13C).

Atherosclerotic plaques are susceptible to clinically important pathologic changes. As discussed further later, the thickness and ECM content of the overlying fibrous cap will impact the stability or fragility of the plaque and its tendency to undergo secondary changes.

- **Rupture, ulceration, or erosion** of the surface of atheromatous plaques exposes the blood stream to highly thrombogenic substances and leads to **thrombosis**, which can partially or completely occlude the vessel lumen (Fig. 11.14). If the patient survives, the thrombus may organize and become incorporated into the growing plaque.
- **Hemorrhage into a plaque.** Rupture of the overlying fibrous cap or of the thin-walled vessels in the areas of neovascularization can cause intraplaque hemorrhage; a contained hematoma may expand the plaque or induce plaque rupture.
- **Atheroembolism.** Plaque rupture can discharge atherosclerotic debris into the blood stream, producing microemboli.
- **Aneurysm formation.** Atherosclerosis-induced pressure or ischemic atrophy of the underlying media, with loss of elastic tissue, causes weakness and aneurysmal dilation, possibly leading to eventual rupture.

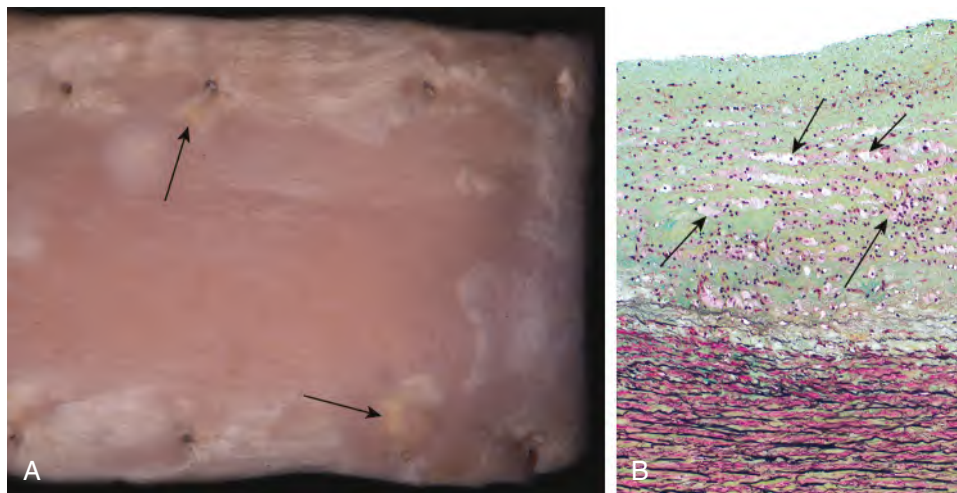


Fig. 11.11 Fatty streak. (A) Aorta with fatty streaks (arrows), associated largely with the ostia of branch vessels. (B) Photomicrograph of a fatty streak in the aorta demonstrating intimal, macrophage-derived foam cells (arrows in A). (A, Courtesy E. Rene Rodriguez, MD, and Carmela Tan, MD, Pathology and Laboratory Medicine Institute, Cleveland Clinic Foundation.)

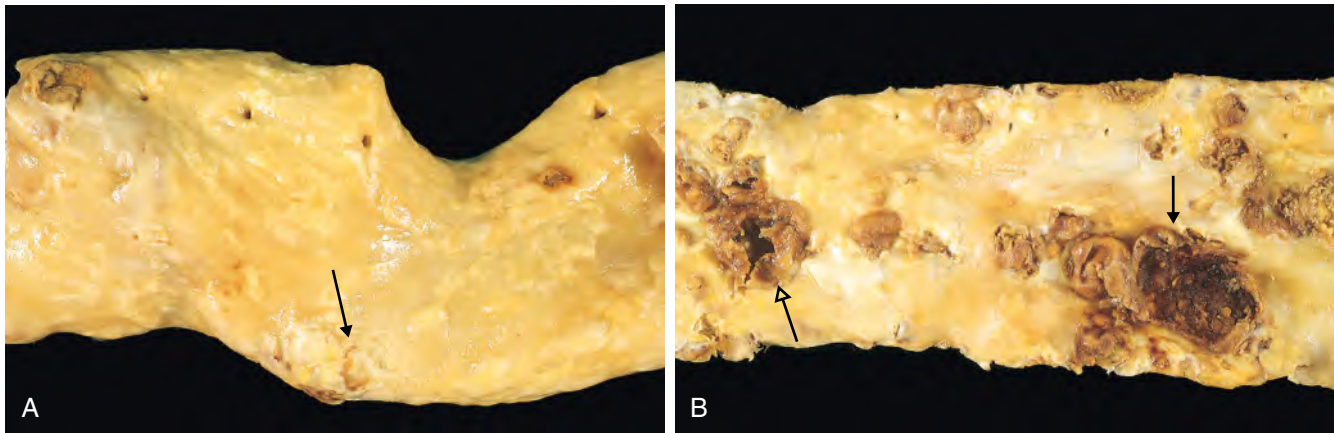


Fig. 11.12 Gross views of atherosclerosis in the aorta. (A) Mild atherosclerosis composed of fibrous plaques, one of which is denoted by the *arrow*. (B) Severe disease with diffuse, complicated lesions including an ulcerated plaque (*open arrow*) and a lesion with overlying thrombus (*closed arrow*).

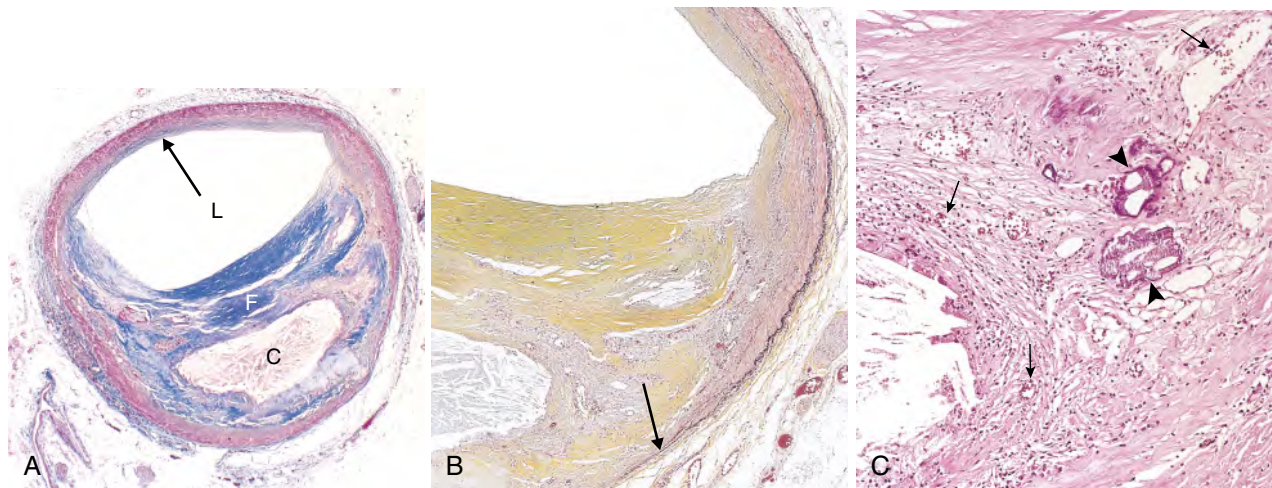


Fig. 11.13 Histologic features of an atheromatous plaque in the coronary artery. (A) This plaque consists of a fibrous cap (*F*) and a central necrotic core (*C*) containing cholesterol and other lipids. The lumen (*L*) has been moderately compromised. Note that a segment of the wall is plaque-free (*arrow*); the lesion is therefore “eccentric.” In this section, collagen has been stained blue (Masson trichrome stain). (B) Higher magnification photograph of a section of the plaque shown in (A), stained for elastin (black), demonstrating that the internal and external elastic membranes are attenuated and the media of the artery is thinned under the most advanced plaque (*arrow*). (C) Higher magnification photomicrograph at the junction of the fibrous cap and core showing scattered inflammatory cells, calcification (*arrowhead*), and neovascularization (*small arrows*).

Consequences of Atherosclerotic Disease

Myocardial infarction (heart attack), cerebral infarction (stroke), aortic aneurysm, and peripheral vascular disease (gangrene of the legs) are the major consequences of atherosclerosis. Large elastic arteries (e.g., aorta, carotid, and iliac arteries) and large- and medium-sized muscular arteries (e.g., coronary and popliteal arteries) are the major sites of atherosclerosis. Symptomatic atherosclerotic disease most often involves the arteries supplying the heart, brain, kidneys, and lower extremities. The natural history, principal morphologic features, and main pathogenic events are schematized in Fig. 11.15).

We next describe the features of atherosclerotic lesions that are typically responsible for the clinicopathologic manifestations.

Atherosclerotic Stenosis

In small arteries, atherosclerotic plaques can gradually occlude vessel lumens, compromising blood flow and causing ischemic injury. At early stages of stenosis, outward remodeling of the vessel media (so-called *Glagov phenomenon*) tends to preserve the size of the lumen to maintain a constant blood flow in the face of an increased atherosclerotic lesion. However, there are limits on the extent of remodeling, and eventually the expanding atheroma impinges on the lumen to such a degree that blood flow is compromised (see Fig. 11.15). *Critical stenosis* is the stage at which the occlusion is sufficiently severe to cause tissue ischemia. In the coronary (and other) circulations, this typically occurs when the plaque produces a 70% to 75% decrease in luminal cross-sectional area; with this

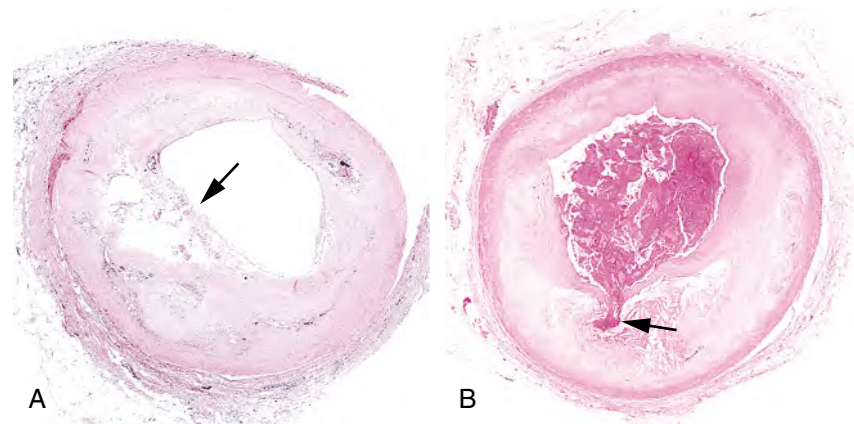


Fig. 11.14 Atherosclerotic plaque rupture. (A) Plaque rupture without superimposed thrombus in a patient who died suddenly. (B) Acute coronary thrombosis superimposed on an atherosclerotic plaque with focal disruption of the fibrous cap, triggering fatal myocardial infarction. In both A and B, the arrow points to the site of plaque rupture. (B, From Schoen FJ. *Interventional and Surgical Cardiovascular Pathology: Clinical Correlations and Basic Principles*. WB Saunders; 1989:61.)

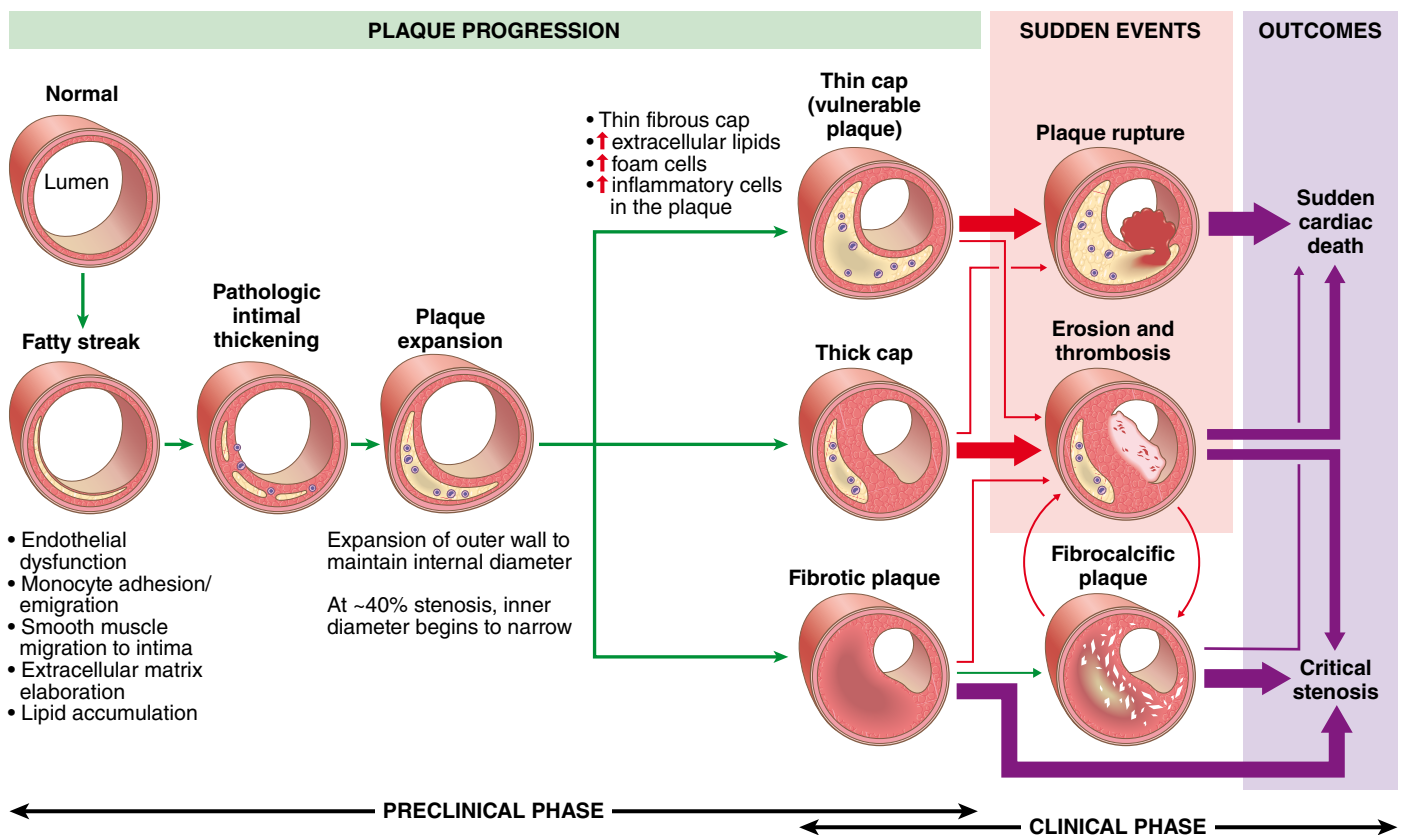


Fig. 11.15 Atherosclerotic plaque formation, activities, and outcomes. In a lesion-prone area along with other risk factors, a reversible fatty streak is formed that can progress to form an atheroma. Compensatory enlargement through remodeling initially prevents a reduction of blood flow through the vessel. A plaque can become a thin cap plaque (unstable, vulnerable), a thick cap plaque (stable), or a fibrotic plaque (stable). Thin cap plaques are the most prone to plaque rupture, often leading to sudden cardiac death. Stable plaques can undergo surface erosion and thrombosis, rapidly expanding the plaque size and leading to more prominent calcifications. This event also can lead to sudden cardiac death. Marked narrowing of the luminal diameter from large plaques results in critical stenosis, reducing blood supply to the heart sufficiently to produce angina.

degree of stenosis, chest pain may develop with exertion (so-called *stable angina*; see Chapter 12). Although acute plaque rupture (see later) is the most dangerous consequence, atherosclerosis also takes a toll through chronically diminished arterial perfusion. Mesenteric occlusion and bowel ischemia, sudden cardiac death, chronic ischemic heart disease, ischemic encephalopathy, and intermittent claudication (diminished perfusion of the extremities) are all consequences of flow-limiting stenoses. The effects of vascular occlusion ultimately depend on arterial supply and collateralization, as well as the metabolic demand of the affected tissue. If stenosis occurs slowly, smaller adjacent vessels can partially compensate by enlarging and creating collateral circulation to perfuse the organ.

Acute Plaque Change

Plaque erosion or rupture is typically promptly followed by partial or complete vascular thrombosis (see Figs. 11.11 and 11.15), resulting in acute tissue infarction (e.g., myocardial or cerebral infarction). Plaque changes fall into three general categories:

- *Rupture/fissuring*, exposing highly thrombogenic plaque constituents that activate coagulation and induce thrombosis that is often completely occlusive
- *Erosion/ulceration*, exposing the thrombogenic subendothelial basement membrane to blood, less frequently inducing fully occlusive thrombosis
- *Hemorrhage into the atheroma*, expanding its volume

It is now recognized that the plaques that are responsible for myocardial infarction and other acute coronary syndromes are often asymptomatic before undergoing a sudden, typically unpredictable change. Pathologic and clinical studies show that most plaques that undergo abrupt disruption and coronary occlusion previously showed only mild to moderate noncritical luminal stenosis. Thus, somewhat distressingly, a large number of asymptomatic adults may be at risk for a catastrophic coronary event. Although imaging modalities are being developed that could preemptively identify such lesions, it is clear that standard angiographic studies are woefully inadequate to visualize them until after the fact.

Plaques rupture when they are unable to withstand mechanical stresses generated by vascular shear forces. The events that trigger abrupt changes in plaques and subsequent thrombosis are complex and include both intrinsic factors (e.g., plaque structure and composition) and extrinsic elements (e.g., blood pressure, platelet reactivity, and vessel spasm).

The fibrous cap undergoes continuous remodeling that can stabilize the plaque or, conversely, render it more susceptible to rupture. Collagen is the major structural component of the fibrous cap and accounts for its mechanical strength and stability; the balance of collagen synthesis versus degradation affects cap integrity. Thus, plaques with thin fibrous caps and active inflammatory cells over a necrotic core are more likely to rupture; these are referred to as *vulnerable plaques* (see Figs. 11.10 and 11.15).

Collagen in atherosclerotic plaque is produced primarily by SMCs so that loss of these cells results in a more fragile cap. Moreover, collagen turnover is controlled by metalloproteinases (MMPs), enzymes elaborated largely by macrophages and SMCs within the atheromatous plaque. Conversely, tissue inhibitors of metalloproteinases (TIMPs) produced by ECs, SMCs, and macrophages modulate MMP activity. In general, plaque inflammation results in a net increase in collagen degradation and reduced collagen synthesis, thereby destabilizing the mechanical integrity of the fibrous cap (see later). The inflammation induced by cholesterol deposits themselves may contribute to plaque destabilization. Conversely, statins may have a beneficial therapeutic effect not only by reducing circulating cholesterol levels but also by stabilizing plaques through a reduction in plaque inflammation.

Influences extrinsic to plaques also contribute to acute plaque changes. Adrenergic stimulation can increase systemic blood pressure or induce local vasoconstriction, thereby increasing the physical stresses on a given plaque. Indeed, the adrenergic stimulation associated with awakening and rising can cause blood pressure spikes (followed by heightened platelet reactivity) that have been causally linked to the pronounced circadian periodicity of acute myocardial infarction (peaking between 6 a.m. and 12 noon). Intense emotional stress can also contribute to plaque disruption; this is most dramatically illustrated by the uptick in the incidence of sudden death associated with natural or other disasters, such as earthquakes and the September 11, 2001, attack on the World Trade Center.

It is also important to note that not all plaque ruptures result in occlusive thromboses with catastrophic consequences. Indeed, plaque disruption and an ensuing superficial platelet aggregation and thrombosis are probably common, repetitive, and often clinically silent complications of atheroma. Healing of these subclinical plaque disruptions—and organization of their overlying thrombi—is an important mechanism in the growth of atherosclerotic lesions.

Thrombosis, partial or total, associated with a disrupted plaque is a central factor in acute coronary syndromes. In its most serious form, the thrombus totally occludes the affected vessel. In contrast, in other coronary syndromes (Chapter 12), luminal obstruction by the thrombus is incomplete and may wax and wane with time.

Mural thrombi in a coronary artery can also embolize. Small embolic fragments of thrombus can often be found in the distal intramyocardial circulation or in association with microinfarcts in patients with atherosclerosis who die suddenly. Thrombin and other factors associated with thrombosis are potent activators of SMCs through protease activate receptors (Chapters 3 and 4), and can thereby contribute to the growth of atherosclerotic lesions.

Vasoconstriction compromises lumen size and, by increasing the local mechanical forces, can potentiate plaque disruption. Vasoconstriction at sites of atheroma may be stimulated by (1) circulating adrenergic agonists, (2) locally released platelet contents, (3) dysfunctional EC resulting in an imbalance between endothelial-derived relaxing factors (NO) relative to contracting factors (endothelin), and (4) mediators released from perivascular inflammatory cells.

KEY CONCEPTS

Atherosclerosis

- Atherosclerosis is an intimal-based lesion composed of a fibrous cap and an atheromatous core; the constituents of the plaque include SMCs, ECM, inflammatory cells, calcifications, lipids, and necrotic debris.
- Atherogenesis is driven by an interplay of vessel wall injury and inflammation. The multiple risk factors for atherosclerosis all cause EC dysfunction and influence inflammatory cell and SMC recruitment and stimulation.
- Atherosclerotic plaques develop and generally grow slowly over decades. Stable plaques can produce symptoms related to chronic ischemia by narrowing vessel lumens, whereas unstable plaques can cause dramatic and potentially fatal ischemic complications related to acute plaque rupture, thrombosis, or embolization.
- Stable plaques tend to have a dense fibrous cap, minimal lipid accumulation and little inflammation, whereas “vulnerable” unstable plaques have thin caps, large lipid cores, and relatively dense inflammatory infiltrates.
- Major consequences of atherosclerotic diseases include myocardial infarction, cerebral infarction, aortic aneurysms, and peripheral vascular diseases.

ANEURYSMS AND DISSECTION

An aneurysm is a localized abnormal dilation of a blood vessel or the heart that may be congenital or acquired (Fig. 11.16). When an

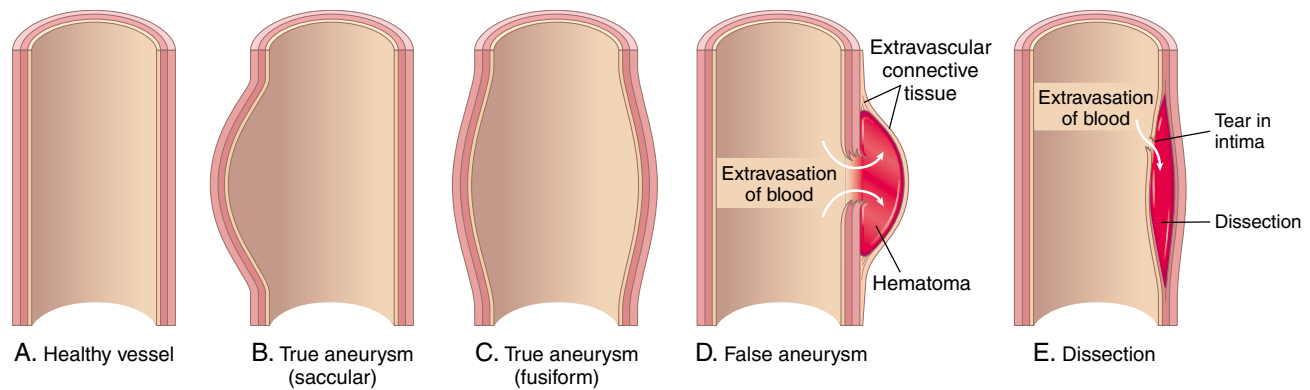


Fig. 11.16 Aneurysms. (A) Healthy vessel. (B) True aneurysm, saccular type. The wall focally bulges outward. (C) True aneurysm, fusiform type. (D) False aneurysm. The wall is ruptured, and there is a collection of blood (hematoma) that is bounded externally by adherent extravascular tissues. (E) Dissection. Blood has entered (*dissected*) the wall of the vessel and separated the layers. Although this is shown as occurring through a tear in the lumen, dissections can also occur by rupture of the vessels of the vasa vasorum within the media.

aneurysm involves all the layers of an intact (but attenuated) arterial wall or the thinned ventricular wall of the heart, it is called a true aneurysm. Atherosclerotic and congenital vascular aneurysms and ventricular aneurysms that follow transmural myocardial infarctions are of this type. In contrast, a *false aneurysm* (also called a *pseudoaneurysm*) is a defect in the vascular wall leading to an extravascular hematoma that freely communicates with the intravascular space (“pulsating hematoma”). Examples include a ventricular rupture after myocardial infarction that is contained by a pericardial adhesion or a leak at the sutured junction of a vascular graft with a natural artery. An arterial dissection arises when blood enters a defect in the arterial wall and tunnels through medial or medial-adventitial planes. Dissections mostly, but not always, occur at an aneurysmal site (see later). Both true and false aneurysms as well as dissections can rupture, often with catastrophic consequences.

Descriptively, aneurysms are classified by macroscopic shape and size (see Fig. 11.16). *Saccular aneurysms* are essentially spherical outpouchings (involving only a portion of the vessel wall); in intracranial vessels they generally measure 2 to 20 mm; however, in the aorta they range from 5 to 10 cm in diameter and often contain thrombus. *Fusiform aneurysms* involve diffuse, circumferential dilation of a long vascular segment; they vary in diameter (in the aorta generally from 5 to 10 cm) and can involve extensive portions of the aortic arch, abdominal aorta, or even the iliacs. These types are not specific for any disease or clinical manifestations.

Pathogenesis

To maintain their structural and functional integrity, arterial walls constantly remodel by synthesizing, degrading, and repairing damage to their ECM constituents (Fig. 11.17). Aneurysms occur when the structure or function of the connective tissue within the vascular wall is compromised. Defects in synthesis or breakdown of connective tissue contribute to the pathobiology of both heritable aneurysmal diseases, as well as the common, sporadic forms of aneurysms.

The two most important causes of aortic aneurysms are atherosclerosis and hypertension; atherosclerosis is a greater factor in abdominal aortic aneurysms (AAAs), while hypertension is the most common etiology associated with ascending aortic aneurysms (see Fig. 11.17B). Other pathologies and risk factors that weaken vessel walls and lead to aneurysms include advanced age, smoking, trauma, vasculitis (see later), congenital defects (e.g., fibromuscular dysplasia and berry aneurysms typically in the circle of Willis; see Chapter 28), and infections (mycotic aneurysms). Mycotic aneurysms can originate from (1) embolization of a septic embolus, usually as a complication of

infective endocarditis; (2) extension of an adjacent suppurative process; or (3) circulating organisms directly infecting the arterial wall.

Defects in the vessel wall by genetic disorders or inflammatory processes that are implicated in the pathogenesis of aneurysm are as follows:

- *The balance of collagen degradation and synthesis is altered by inflammation and associated proteases.* Inflammatory cells in the setting of an aortitis or associated with atherosclerosis can be found throughout the aortic wall (see Fig. 11.17C). Increased matrix metalloproteinase (MMP) production, especially by macrophages, can contribute to aneurysm development by degrading ECM (elastin, collagens, proteoglycans, laminin, fibronectin) in all layers of the wall; decreased expression of tissue inhibitors of metalloproteinases (TIMPs) can also impact ECM degradation. Both processes result in a loss of elastic fibers necessary for recoil in diastole.
- Genetic predisposition to aneurysm formation in the setting of inflammatory lesions (e.g., atherosclerosis) may be related to MMP and/or TIMP polymorphisms or to the nature of the local inflammatory response. Indeed, AAAs (see later) are associated with shifts in the local cytokine environments toward the production of T helper 2 (Th2) cytokines (e.g., IL-4 and IL-13) that can drive macrophages to produce increased amounts of elastolytic MMP.
- *The intrinsic quality of the vascular wall connective tissue is poor.* Weak vessel walls due to defective type III collagen synthesis are a hallmark of the vascular form of *Ehlers-Danlos syndrome* (Chapter 5).
- *Abnormal transforming growth factor- β (TGF- β) signaling.* Excessive TGF- β activity alters vascular wall remodeling, primarily of the ascending aorta, leading ultimately to diminished ECM content and integrity with aneurysmal dilation (see Fig. 11.17D). In *Marfan syndrome* (Chapter 5), defective synthesis of the scaffolding protein fibrillin leads to the inability to appropriately sequester endogenously produced TGF- β . In *Loeys-Dietz syndrome*, increased activity in the TGF- β signaling pathway results from mutations in TGF- β receptors, an intracellular downstream signaling molecule in the TGF- β pathway (SMAD3), or TGF- β itself. Aneurysms in patients with Loeys-Dietz syndrome can rupture at small sizes.
- *The vascular wall is weakened through loss of SMCs or the inappropriate synthesis of noncollagenous or nonelastic ECM.* Ischemia of the inner media occurs when there is atherosclerotic thickening of the intima, increasing the distance that oxygen and nutrients must diffuse (see Fig. 11.17B). Systemic hypertension can also cause significant narrowing of arterioles of the vasa vasorum (e.g., in the

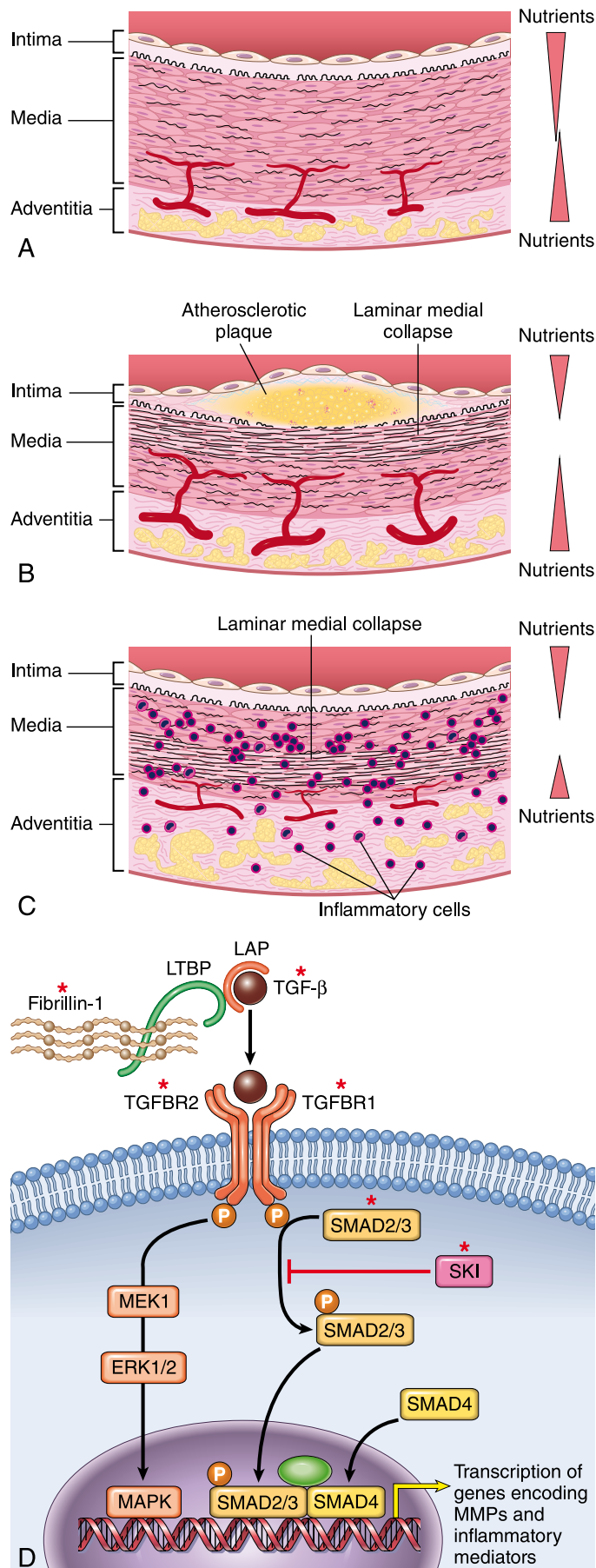


Fig. 11.17 Patterns of elastic artery injury. (A) Healthy aortic wall with three layers: intima, media, and adventitia. Vasa vasorum is present along the adventitia and outer one-third of the media. (B) Beneath an atherosclerotic plaque, laminar medial collapse can occur with a loss of smooth muscle cells (SMCs). The vasa vasorum may extend further up the media and the adventitia can be thickened. (C) With an inflammatory large vessel vasculitis, damage can occur within the media and adventitia with the presence of immune cells (lymphocytes, plasma cells, giant cells), laminar medial collapse with a loss of SMC. (D) A simplified schema of transforming growth factor β ($TGF\beta$) signaling pathway: $TGF\beta$ generally binds to LTBP that is further linked to extracellular matrix (e.g., fibrillin). Following tissue injury or mutational alteration, $TGF\beta$ is released from the ECM and binds to its receptors activating a signaling pathway that ultimately leads to transcription of various $TGF\beta$ -dependent genes. A number of syndromic aortic aneurysm diseases share a common dysregulation of the $TGF\beta$ pathway. *SKI* oncogene inhibits $TGF\beta$ signaling. It is mutated in Shprintzen-Goldberg syndrome which has considerable phenotypic overlap with Marfan syndrome, including aortic aneurysm. Asterisk indicates a gene known to be mutated in syndromic disease. *LAP*, Latency associated peptide; *LTBP*, latent $TGF\beta$ binding protein; *MMP*, matrix metalloproteinase; *TGFBR*, $TGF\beta$ receptor.

aorta), which causes outer medial ischemia. Tertiary syphilis is another rare cause of aortic aneurysms in which obliterative endarteritis occurs in the vasa vasorum of the thoracic aorta; this engenders medial ischemia that leads to SMC loss, elastic fiber loss, and inadequate or inappropriate ECM synthesis.

All of the above-described processes lead to histopathologic changes in the arterial wall, referred to as *medial degeneration* (Fig. 11.18). Certain findings may point to a cause, such as adventitial lymphoplasmacytic inflammation suggests IgG4-RD, and clinical features (e.g., skeletal changes in Marfan syndrome) may suggest a genetic basis, but generally, medial degeneration is a nonspecific finding common to all forms of aortic disease.

Abdominal Aortic Aneurysm

Aneurysms occurring because of atherosclerosis form most commonly in the abdominal aorta and common iliac arteries. A variety of factors discussed earlier collaborate to weaken the media and predispose to aneurysm formation.

AAAs occur more frequently in males and in smokers, rarely developing before age 50 years. Atherosclerosis is a major cause of AAAs, but other factors clearly contribute, since the incidence is <5% in males older than age 60 years, despite almost universal abdominal aortic atherosclerosis in that population.

MORPHOLOGY

Usually located between the renal arteries and the bifurcation of the aorta, AAAs can be saccular or fusiform: greater than 3 cm, but often >5.5 cm, in diameter, and up to 25 cm in length (Fig. 11.19). There is severe complicated atherosclerosis with destruction and thinning of the underlying aortic media; the aneurysm frequently contains a bland, poorly organized mural thrombus. AAAs can occasionally affect the renal and superior or inferior mesenteric arteries, either by direct extension or by occluding vessel ostia with mural thrombi. Not infrequently, AAAs will be accompanied by smaller aneurysms of the iliac arteries.

Three AAA variants merit special mention:

- **Inflammatory AAAs** account for 5% to 10% of all AAAs; these typically occur in younger patients who often present with back pain and elevated inflammatory markers (e.g., erythrocyte sedimentation rate). Inflammatory aneurysms are characterized by abundant lymphoplasmacytic inflammation with many

macrophages (and even giant cells) associated with dense periaortic scarring that can extend into the anterior retroperitoneum. The cause is a presumed localized immune response to the abdominal aortic wall; remarkably, most cases are not associated with inflammation of other arteries.

- A subset of inflammatory AAAs may be a vascular manifestation of **immunoglobulin G4-related disease (IgG4-RD)**. This is a disorder marked by (in most cases) high circulating levels of IgG4, with storiform fibrosis and IgG4-positive infiltrating plasma cells in affected tissues (Chapter 6). IgG4-RD can affect a host of organs including the pancreas and thyroid, as well as other vascular sites such as the ascending aorta and heart vessels. There may be retroperitoneal fibrosis with bilateral hydronephrosis. Recognition of this entity is important since it responds well to steroids and anti-B-cell therapies.
- **Mycotic AAAs** are lesions that have become infected by seeding of circulating microorganisms in the wall. In such cases, suppurative further destroys the media, potentiating rapid dilation and rupture.

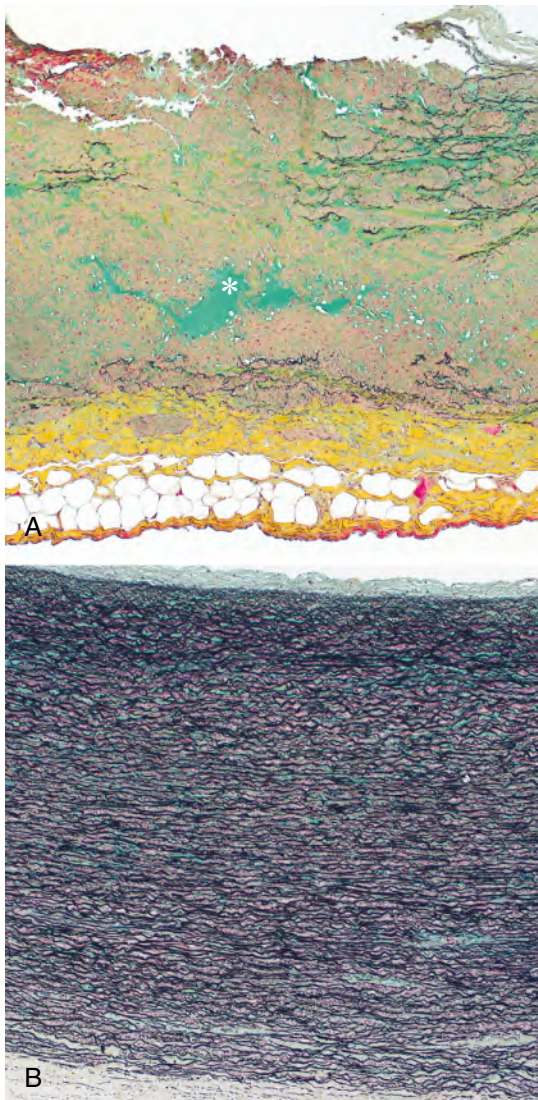


Fig. 11.18 Medial degeneration. (A) Cross-section of aortic media from a patient with Marfan syndrome showing marked elastic fiber fragmentation and loss (black fibers) and collections of (blue/green) mucoid extracellular matrix accumulation (cystic medial degeneration) (asterisks). (B) Healthy aortic media for comparison showing the regular layered lamellar units of elastic tissue.

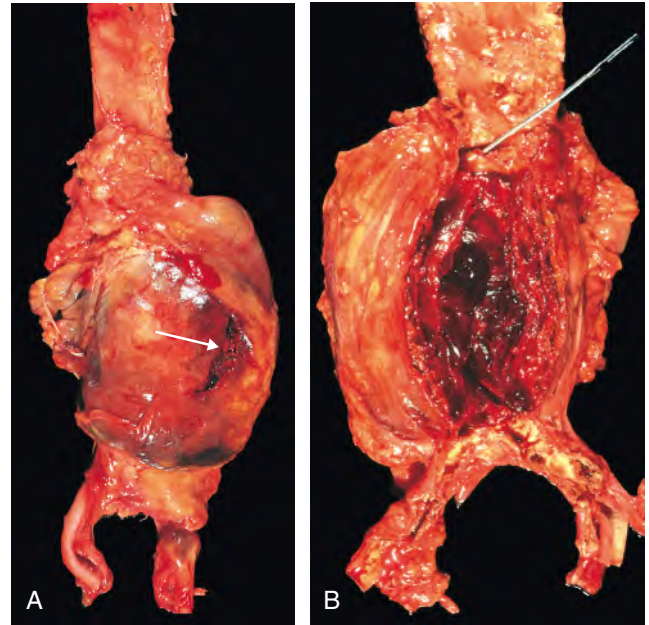


Fig. 11.19 Abdominal aortic aneurysm. (A) External view, gross photograph of a large aortic aneurysm that ruptured (rupture site is indicated by the arrow). (B) Opened view, with the location of the rupture tract indicated by a probe. The wall of the aneurysm is exceedingly thin, and the lumen is filled by a large quantity of layered but largely unorganized thrombus. Severe atherosclerosis is noted above and below the aneurysm, suggesting the cause of the wall weakness.

Clinical Features

Most cases of AAAs are asymptomatic and are discovered incidentally on physical examination as an abdominal mass (often palpably pulsating) that simulates a tumor. The other clinical manifestations of AAAs include:

- **Rupture** into the peritoneal cavity or retroperitoneal tissues with massive, potentially fatal hemorrhage
- **Obstruction** of a branch vessel resulting in downstream tissue ischemic injury; for example, iliac (leg), renal (kidney), mesenteric (gastrointestinal tract), or vertebral arteries (spinal cord)
- **Embolism** from atheroma or mural thrombus
- **Impingement** on an adjacent structure, for example, compression of a ureter or erosion of vertebrae

The risk of rupture is directly related to the size of the aneurysm, varying from nil for AAAs ≤ 4 cm in diameter to 1% per year for AAAs between 4 and 5 cm, 11% per year for AAAs between 5 and 6 cm, and 25% per year for aneurysms > 6 cm. Most aneurysms expand at a rate of 0.2 to 0.3 cm/year, but 20% expand more rapidly. In general, aneurysms ≥ 5 cm are managed aggressively, either through open surgeries involving prosthetic grafts or endovascular approaches through the femoral artery deploying a stent graft (expandable wire frame covered by a cloth sleeve). Timely surgery is critical; operative mortality for unruptured aneurysms is approximately 5%, whereas fewer than half of patients survive after rupture even with emergency surgery. It is worth reiterating that because atherosclerosis is a systemic disease, a patient with an AAA is also very likely to have atherosclerosis in other vascular beds and is at a significantly increased risk of ischemic heart disease and stroke.

Thoracic Aortic Aneurysm

Thoracic aortic aneurysms are most commonly associated with hypertension, although other causes such as Marfan syndrome, Loeys-Dietz

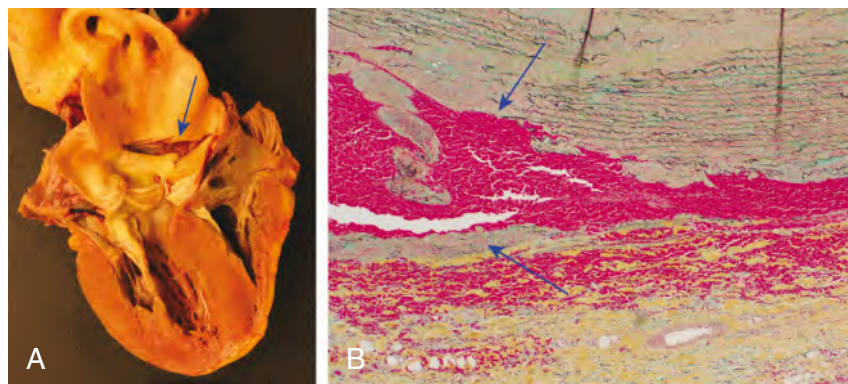


Fig. 11.20 Aortic dissection (A) Gross photograph of an opened aorta with proximal dissection of the ascending aorta approximately 2 cm above the aortic valve (*blue arrow*). This tear allows blood to enter the media; the dissection can then propagate either anterograde (away from the heart) and/or retrograde (back toward the heart). Acutely fatal complications include dissection through the adventitia leading to hemopericardium (and tamponade) or hemothorax. Note the dilated appearance of the ascending aorta which may be due to underlying lack of medial integrity. (B) Histologic view of a dissection demonstrating blood tearing through the media (*blue arrows*). There is a marked reduction in elastic fibers in the media, a form of medial degeneration. Fresh blood is in the adventitia, suggesting this dissection was not contained in the media (Movat stain).

syndrome, bicuspid aortic valve with aneurysm, and inflammatory conditions (aortitis) are increasingly recognized. Although now rare, syphilitic aortic aneurysms commonly affect the ascending and thoracic aorta. Before dissection or rupture, symptoms can include (1) chest pain from encroachment or erosion into bone, (2) myocardial ischemia from compression of a coronary artery, (3) difficulty swallowing due to compression of the esophagus, (4) hoarseness from irritation of or pressure on the recurrent laryngeal nerves, or (5) respiratory complications from compression of the bronchi; the vast majority cause no symptoms at all until a catastrophic event occurs (see later).

Aortic Dissection

Aortic dissection occurs when blood separates the laminar planes of the media and forms a blood-filled channel within the aortic wall (Fig. 11.20). Aortic dissection is generally associated with aortic dilation; it can be disastrous if the dissection ruptures through the adventitia and hemorrhages into adjacent spaces.

Aortic dissection occurs principally in two groups of patients: (1) males aged 40 to 60 years, with antecedent hypertension (more than 90% of cases) and (2) younger patients with syndromic diseases affecting the aorta (e.g., Marfan syndrome). Dissections can also be iatrogenic (e.g., following arterial cannulations during diagnostic catheterization or cardiopulmonary bypass). Pregnancy is rarely associated with aortic (or other vessel) dissection (roughly 10 to 20 cases per 1 million births), typically occurring during or after the third trimester. It may be related to hormonally induced vascular remodeling and the hemodynamic stresses of the perinatal period. Dissection is unusual in the presence of substantial atherosclerosis or other causes of medial scarring, such as syphilis, presumably because the medial fibrosis inhibits propagation of the dissecting hematoma. Once the tear has occurred, blood flow under systemic pressure dissects through the media, leading to progression of the hematoma.

Pathogenesis

Hypertension is the major risk factor for aortic dissection. Aortas of hypertensive patients have medial degenerative changes with SMC loss and altered ECM content. Other dissections occur in the setting of inherited or acquired connective tissue disorders with defective TGF- β signaling or defective ECM synthesis or degradation. However, specific patterns of medial damage are neither a prerequisite for dissection nor a

reliable predictor of its occurrence. Regardless of the underlying etiology, the ultimate trigger for the intimal tear and initial intramural aortic hemorrhage is not known in most cases. Nevertheless, once the tear has occurred, blood flow under systemic pressure dissects through the media, leading to progression of the hematoma. Accordingly, aggressive pressure-reducing therapy may be effective in limiting an evolving dissection. In some cases, disruption of penetrating vessels of the vasa vasorum can give rise to an intramural hematoma without an intimal tear.

MORPHOLOGY

In most cases of dissection, medial degeneration is present at the site of the tear but may not be seen in the area of propagation. Typical features include elastic fiber fragmentation and loss, mucoid ECM accumulation, and SMC attrition (see Fig. 11.18); inflammation is characteristically absent. Dissections can occur in the setting of rather trivial medial degeneration; conversely, marked degenerative changes are frequently seen at autopsy in patients who are completely free from dissection. Thus, the relationship of the structural changes to the pathogenesis of dissection is uncertain.

An aortic dissection usually initiates with an intimal tear. In most spontaneous dissections, the tear occurs in the ascending aorta, usually within 10 cm of the aortic valve (see Fig. 11.20A). Such tears are typically transverse with sharp, jagged edges up to 1 to 5 cm in length. The dissection can extend retrograde toward the heart as well as distally, sometimes as far as the iliac and femoral arteries. The dissecting hematoma spreads characteristically between lamellar units of the outer third of the media or between media and adventitial layers (see Fig. 11.20B). It can rupture through the adventitia causing massive hemorrhage (e.g., into the thoracic or abdominal cavities) or cardiac tamponade (hemorrhage into the pericardial sac). In some (fortuitous) instances, the dissecting hematoma reenters the lumen of the aorta through a second distal intimal tear, creating a new false vascular channel ("**double-barreled aorta**"). This averts a fatal extraaortic hemorrhage, and over time, such false channels can be endothelialized to become **chronic dissections**.

Traumatic chest injury (e.g., motor vehicle accident, or even extremely vigorous cardiopulmonary resuscitation) can cause intimal tears that originate after the origin of arch vessels at the *ligamentum arteriosum*. In such cases the connective tissue that tethers the aorta and pulmonary vessels in that location can lead to aortic tearing when there is sudden extreme movement of the heart toward the anterior chest wall.

Clinical Features

The morbidity and mortality associated with dissections depends on which part of the aorta is involved; the most serious complications occur with dissections between the aortic valve and the distal arch. Accordingly, aortic dissections are generally classified into two types (Fig. 11.21):

- **Type A dissections.** These are the more common (and dangerous) proximal lesions involving either the ascending aorta and the descending aorta or only the ascending aorta only (types I and II of the DeBakey classification, respectively).
- **Type B dissections.** These are distal lesions not involving the ascending aorta and usually beginning distal to the subclavian artery (called DeBakey type III).

The classic clinical symptoms of aortic dissection are the sudden onset of excruciating pain, usually beginning in the anterior chest, radiating to the back between the scapulae, and moving downward as the dissection progresses; the pain can be confused with that of myocardial infarction. The most common cause of death is rupture of the dissection and bleeding into the pericardial, pleural, or peritoneal cavities. Retrograde dissection into the aortic root can also disrupt the aortic valve annulus. Common clinical manifestations include cardiac tamponade and aortic insufficiency. Dissections can also extend into the great arteries of the neck or into the coronary, renal, mesenteric, or iliac arteries, causing vascular obstruction and ischemic consequences such as myocardial infarction; involvement of spinal arteries can cause transverse myelitis.

In type A dissections, rapid diagnosis and institution of intensive antihypertensive therapy coupled with surgical plication of the aortic tear can save 65% to 85% of patients. However, mortality

approaches 70% in those who present with hemorrhage or symptoms related to distal ischemia, and the overall 10-year survival is 40% to 88% depending on whether a patient is a surgical candidate and other prognostic factors. Most type B dissections can be managed conservatively; such patients have a short-term 75% survival rate whether they are treated with surgery or antihypertensive medication only.

KEY CONCEPTS

Aneurysms and Dissections

- Vascular aneurysms are congenital or acquired dilations of the blood vessels that involve the entire thickness of the wall. Complications are related to rupture, thrombosis, and embolization.
- Dissections occur when blood enters the wall of a vessel and dissects through the layers of the wall, generally between lamellar units. Complications arise due to rupture or obstruction of vessels branching off the aorta.
- Aneurysms and dissections result from structural weakness of the vessel wall caused by aberrant TGF- β signaling, loss of SMCs, and changes in the ECM, which can result from ischemia, genetic defects, or defective matrix remodeling.
- Atherosclerosis and hypertension are common etiological factors for the aneurysms of the abdominal aorta and ascending aorta, respectively.
- Abdominal aortic aneurysm (AAA) can be inflammatory (5%–10% of all AAAs, a subset of which may be associated with IgG4-RD) or mycotic (lodging of circulating microorganisms in the vessel media).
- Unlike Type B (distal) dissections, type A dissections (proximal) are associated with serious complications and greatest mortality.

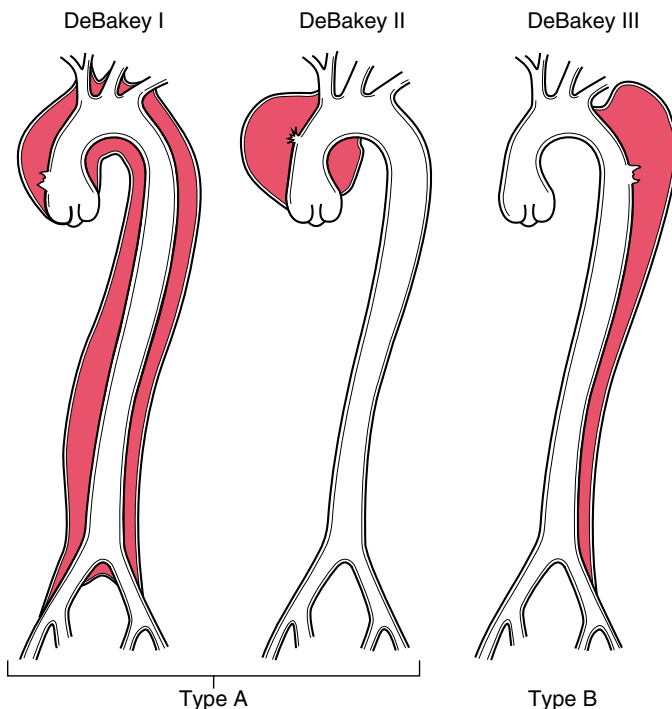


Fig. 11.21 Classification of dissections. Type A (proximal) involves the ascending aorta, either as part of a more extensive dissection (DeBakey I) or in isolation (DeBakey II). Type B (distal or DeBakey III) dissections arise after the take-off of the great vessels. Type A dissections typically have the most serious complications and greatest mortality.

VASCULITIS

Vasculitis is a general term for vessel wall inflammation; it can have protean manifestations depending on the vascular bed affected. Vasculitis can be systemic and affect multiple vascular beds, or can be limited (at least initially) to vessels in a single organ or tissue. Besides the findings referable to the specific tissues involved, the clinical manifestations typically include constitutional signs and symptoms associated with systemic inflammation, such as fever, myalgias, arthralgias, and malaise.

The two pathogenic mechanisms of vasculitis are **direct invasion of vascular walls by infectious pathogens and noninfectious immune-mediated inflammation.** Infections can also indirectly induce a noninfectious vasculitis, for example, by generating immune complexes or triggering vascular cross-reactivity (see later). In any given patient, it is obviously important to distinguish between infectious and immunologic mechanisms—immunosuppressive therapy is appropriate for immune-mediated vasculitis but would likely be harmful in those with an infectious vasculitis. Physical and chemical injury (e.g., from irradiation, mechanical trauma, and toxins) can also cause vasculitis.

Infectious Vasculitis

Arteritis can be caused by the direct invasion of infectious agents, usually bacteria, such as *Pseudomonas*, or fungi, such as *Aspergillus* and *Mucor* species. Vascular invasion can be part of a localized tissue infection (e.g., bacterial pneumonia or adjacent to abscesses) or, less commonly, can arise from hematogenous spread of microorganisms during septicemia or embolization from infective endocarditis.

Vascular infections can weaken arterial walls and culminate in mycotic aneurysms (see earlier), or can induce thrombosis and downstream infarction. Thus inflammation-induced thrombosis of meningeal vessels in bacterial meningitis can eventually give rise to infarction of the underlying brain.

Noninfectious Vasculitis

Noninfectious vasculitides are characterized based on the caliber of vessels that are typically affected (large, medium, or small), on the tissues typically affected, and—in small vessel vasculitis—by the presence or relative absence of *immune complex deposition*. Over 25 primary forms of noninfectious vasculitis are recognized (we will only discuss the major ones); besides vessel size, location, and immune complexes, the most widely utilized classification scheme also considers the presence of specific autoantibodies, granulomas, and even population demographics (Table 11.4; Fig. 11.22). Depending on the specific entity (see later), the main immunologic mechanisms underlying noninfectious vasculitis are immune complex deposition, antineutrophil cytoplasmic antibodies (ANCA), antiendothelial cell antibodies, and/or autoreactive T cells.

As we will see, there is considerable clinical and pathologic overlap among many of the various vasculitides, and despite the

attempts to strictly fit them into unique pigeonholes, there are frequent exceptions.

Large Vessel Vasculitis

Large vessel vasculitis involves the aorta, great arch vessels, and the larger bore vessels in the extremities more commonly than other sized vessels; the two main forms are *giant cell arteritis*, and *Takayasu arteritis* (Table 11.4 and Fig. 11.22).

Giant Cell (Temporal) Arteritis

Giant cell (temporal) arteritis is a chronic, classically granulomatous, inflammation affecting large- to medium-sized arteries, especially those in the head; it is the most common form of vasculitis among elderly adults in the United States and Europe. Although it is characterized as a “large vessel vasculitis,” and indeed can involve the aorta (*giant cell aortitis*), medium and even small-bore vessels are often affected. Thus, temporal arteries are not uniquely vulnerable, but their name is nevertheless attached to the disorder (*temporal arteritis*) since they are the most readily biopsied to establish a diagnosis. Importantly, when giant cell arteritis is present in temporal arteries, it is likely that the vertebral and ophthalmic arteries are also involved. Because

TABLE 11.4 Classification and Characteristic Pathologic Features of Vasculitides^a

Name	Characteristic Pathologic Features
Large vessel vasculitis	Predominantly affects large arteries
Giant cell arteritis	Arteritis, often granulomatous; usually affecting the aorta and/or its major branches (carotid, vertebral, temporal); patients usually older than 50 years
Takayasu arteritis	Arteritis, often granulomatous; patients usually younger than 50 years
Medium vessel vasculitis	Predominantly affects medium arteries (main visceral arteries and their branches); inflammatory aneurysms and stenoses are common
Thromboangiitis obliterans (Buerger Disease)	Arteritis with associated thrombosis and distal ischemia, typically involving extremity arteries in young male smokers
Polyarteritis nodosa	Necrotizing arteritis of medium or small arteries (not affecting pulmonary circulation, glomeruli, arterioles, capillaries, or venules) not associated with antineutrophil cytoplasmic antibodies (ANCA)
Kawasaki disease	Arteritis associated with mucocutaneous lymph node syndrome; predominantly affects medium and small arteries (especially coronary arteries); most common in infants and young children
Small vessel vasculitis	Vasculitis predominantly affecting small vessels (small intraparenchymal arteries, arterioles, capillaries, and venules)
ANCA-associated vasculitis	Necrotizing vasculitis, with few or no immune deposits; associated with myeloperoxidase (MPO) ANCA or proteinase 3 (PR3) ANCA
Microscopic polyangiitis	Necrotizing vasculitis, with few or no immune deposits; necrotizing glomerulonephritis is very common and pulmonary capillaritis often occurs; associated with MPO-ANCA
Granulomatosis with polyangiitis	Necrotizing granulomatous inflammation of upper and lower respiratory tract, and necrotizing vasculitis affecting predominantly small to medium vessels; necrotizing glomerulonephritis is common. PR3-ANCA are present in 95% of cases
Eosinophilic granulomatosis with polyangiitis	Eosinophil-rich and necrotizing granulomatous inflammation of respiratory tract, and necrotizing vasculitis of small to medium vessels; associated with asthma and eosinophilia; MPO-ANCA in over 50% of cases; ANCA is more frequent when glomerulonephritis is present
Immune complex vasculitis Antiglomerular basement membrane disease Cryoglobulinemic vasculitis Immunoglobulin A (IgA) vasculitis Vasculitis associated with systemic disease (SLE, rheumatoid arthritis)	Moderate to marked vessel wall deposits of immunoglobulin and/or complement components predominantly affecting small vessels; glomerulonephritis is frequent.

ANCA, Antineutrophilic cytoplasmic antibodies; SLE, systemic lupus erythematosus.

^aModified from Jennette JC, Falk RJ, Bacon PA, et al. 2012 revised International Chapel Hill Consensus Conference Nomenclature of Vasculitides. *Arthritis Rheum.* 2013;65:1-11.

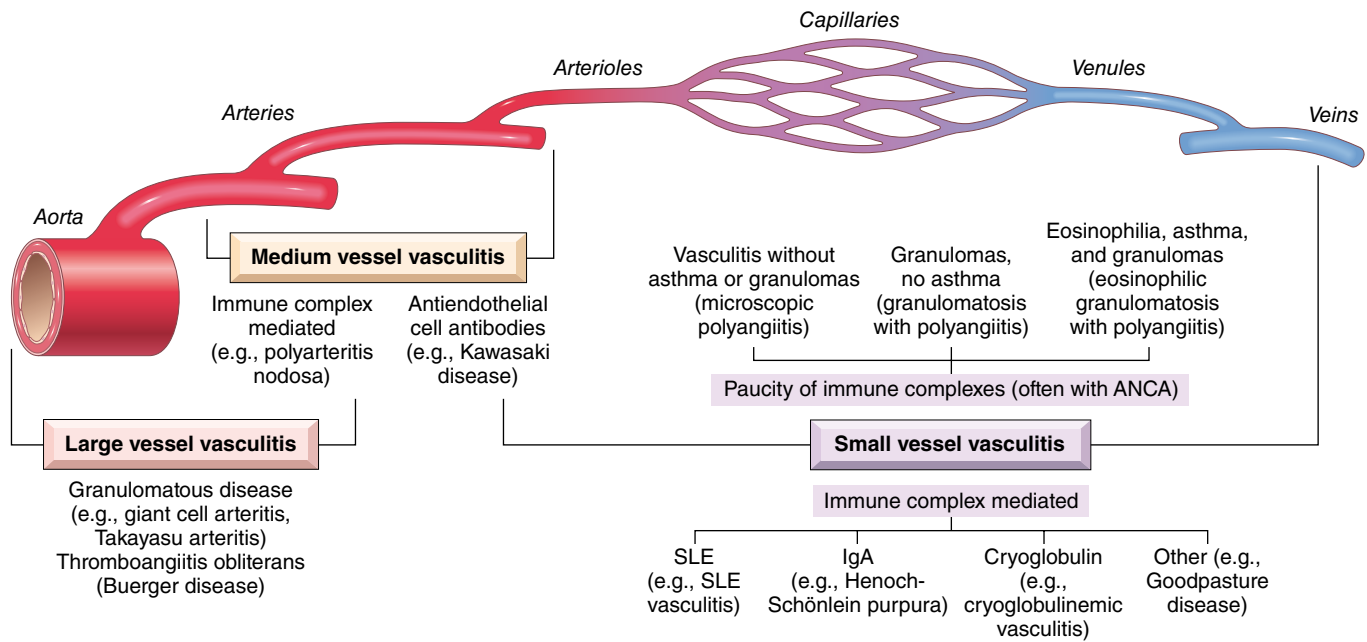


Fig. 11.22 Vascular sites typically involved with the more common forms of vasculitis as well as their presumptive etiologies. Note that there is a substantial overlap in distributions. ANCA, Antineutrophil cytoplasmic antibody; SLE, systemic lupus erythematosus.

ophthalmic artery involvement can lead to sudden and permanent blindness, affected persons must be diagnosed and treated promptly.

Pathogenesis

Giant cell arteritis likely occurs as a result of a T-cell-mediated immune response to an as yet uncharacterized vessel wall antigen. Proinflammatory cytokines (especially IFN- γ and IL-17) and anti-EC antibodies also contribute. The characteristic granulomatous inflammation, an association with certain major histocompatibility complex (MHC) class II haplotypes, and the excellent therapeutic response to steroids all strongly support an immune etiology. The predilection for vessels of the head remains unexplained, although one hypothesis is that vessels in various parts of the body develop from distinct anlagen and may therefore express unique antigens.

MORPHOLOGY

The pathologic changes of giant cell arteritis are notoriously patchy along the length of any affected vessel; accurate diagnosis therefore requires a sufficiently long (>1 cm) biopsy. Involved arterial segments develop **intimal thickening** (with occasional thromboses) that reduces the luminal diameter. Classic lesions exhibit medial **granulomatous inflammation** centered on the internal elastic membrane with **elastic lamina fragmentation**; there is an infiltrate of T cells (CD4+ > CD8+) and macrophages. Although multinucleated giant cells are seen in approximately 75% of adequately biopsied specimens (Fig. 11.23), granulomas and giant cells can be rare or absent. Healed lesions exhibit medial attenuation and scarring with intimal thickening, fragmentation of greater than 30% of the circumference of the internal elastic lamina, and adventitial fibrosis.

Clinical Features

Giant cell (temporal) arteritis is rare before age 50 years. Symptoms may be only vague and constitutional—fever, fatigue, weight loss—or may take the form of facial pain or headache, most intense along the

course of the superficial temporal artery, which can be painful to palpation. Ocular symptoms (associated with involvement of the ophthalmic artery) appear abruptly in about 50% of patients; these range from diplopia to complete vision loss. Diagnosis depends on biopsy and histologic confirmation; even with adequate sampling, the patchy distribution of lesions means that a negative biopsy result does not exclude the diagnosis. Corticosteroids or anti-TNF therapies are usually effective.

Takayasu Arteritis

Takayasu arteritis is a granulomatous vasculitis of medium and larger arteries characterized principally by ocular disturbances and marked weakening of the pulses in the upper extremities (hence its alternate name, *pulseless disease*). Takayasu arteritis manifests with transmural fibrous thickening of the aorta—particularly the aortic arch and great vessels—with severe luminal narrowing of the major branch vessels (Fig. 11.24). Takayasu arteritis shares many attributes with giant cell aortitis, including clinical features and histology; indeed, the distinction is typically made based on the age of the patient. Those over 50 years of age are designated as *giant cell arteritis*, while those under 50 are designated as *Takayasu arteritis*. Although traditionally associated with the Japanese population and a subset of HLA haplotypes, Takayasu arteritis has a global distribution. A T-cell-mediated autoimmune etiology is likely.

MORPHOLOGY

Takayasu arteritis classically involves the aortic arch. In a third of patients, it also affects the remainder of the aorta and its branches, with **pulmonary artery** involvement in half of cases; **coronary and renal arteries** may be similarly affected. There is irregular thickening of the vessel wall with intimal hyperplasia; when the aortic arch is involved, the great vessel lumens can be markedly narrowed or even obliterated (see Fig. 11.24A–B). Such narrowing explains the weakness of the peripheral pulses. Histologically the changes range from adventitial mononuclear infiltrates with perivascular

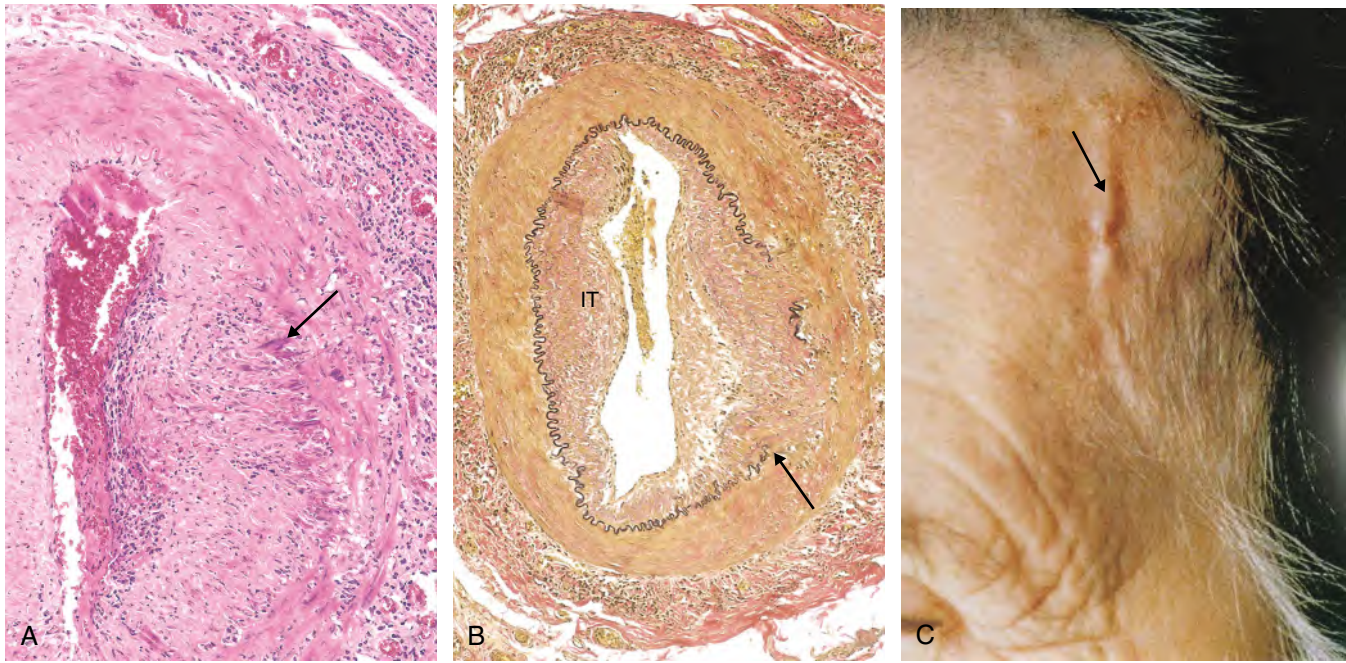


Fig. 11.23 Temporal (giant cell) arteritis. (A) Hematoxylin and eosin stain of temporal artery showing giant cells and degenerated internal elastic membrane in active arteritis (arrow). (B) Elastic tissue stain demonstrating focal destruction of internal elastic membrane (arrow) and intimal thickening (IT) characteristic of long-standing or healed arteritis. (C) The temporal artery of a patient with classic giant cell arteritis shows a thickened, nodular, and tender segment of a vessel on the surface of head (arrow). (C, From Salvarani C, Cantini F, Boiardi L, Hunder GG. Polymyalgia rheumatica and giant-cell arteritis. *N Engl J Med*. 2002;347:261-271.)

cuffing of the vasa vasorum to intense mononuclear inflammation in the media. Granulomatous inflammation, replete with giant cells and patchy medial necrosis, is also seen. The histology (see Fig. 11.24C) is essentially indistinguishable from giant cell (temporal) arteritis. As the disease progresses, collagenous scarring, with admixed chronic inflammatory infiltrates, occurs in all three layers of the vessel wall. Occasionally, aortic root involvement causes dilation and aortic valve insufficiency.

Clinical Features

Initial symptoms are usually nonspecific, including fatigue, weight loss, and fever. With progression, vascular symptoms appear and dominate the clinical picture, including reduced blood pressure and weaker pulses in the upper extremities; ocular disturbances, including visual defects, retinal hemorrhages, and total blindness; and neurologic deficits. Involvement of the more distal aorta may lead to claudication of the legs; pulmonary artery involvement can cause pulmonary hypertension. Narrowing of the coronary ostia may lead to myocardial infarction, and involvement of the renal arteries leads to systemic hypertension in roughly one-half of patients. The course of the disease is variable. In some patients there is rapid progression, while others enter a quiescent stage at 1 to 2 years, permitting long-term survival, albeit with visual or neurologic deficits.

Medium Vessel Vasculitis

Although any size artery may be affected, medium vessel vasculitis most commonly involves medium-sized arteries and their branches; the three main forms are *thromboangiitis obliterans*, *polyarteritis nodosa*, and *Kawasaki disease* (Table 11.4).

Thromboangiitis Obliterans (Buerger Disease). Thromboangiitis obliterans (as precisely described by Leo Buerger) is characterized by

segmental, thrombosing, acute and chronic inflammation of medium-sized arteries, especially the tibial and radial arteries, that often leads to vascular insufficiency, typically of the extremities. It usually occurs in young males (<35 years of age) who are heavy cigarette smokers.

Pathogenesis

The strong relationship with cigarette use involves either direct idiosyncratic EC toxicity caused by some tobacco component, or an immune response to the same agents that have modified host vascular wall proteins. Most patients with Buerger disease have hypersensitivity to intradermally injected tobacco extracts, and their vessels exhibit impaired endothelium-dependent vasodilation when challenged with acetylcholine. There is an increased prevalence in certain geographic regions (Israel, India, Japan) and an association with certain MHC haplotypes.

MORPHOLOGY

Affected vessels of thromboangiitis obliterans show acute and chronic inflammation, accompanied by luminal thrombosis. The thrombus can contain small **microabscesses** composed of neutrophils surrounded by granulomatous inflammation (Fig. 11.25); the thrombus may eventually organize and recanalize. The inflammatory process extends into contiguous veins and nerves (rare with other forms of vasculitis), and with time all three structures can be encased in fibrous tissue.

Clinical Features

Early manifestations include Raynaud phenomenon (see later), instep foot pain induced by exercise (instep claudication), and a superficial nodular phlebitis (venous inflammation). The vascular insufficiency of Buerger disease tends to be accompanied by severe pain—even at

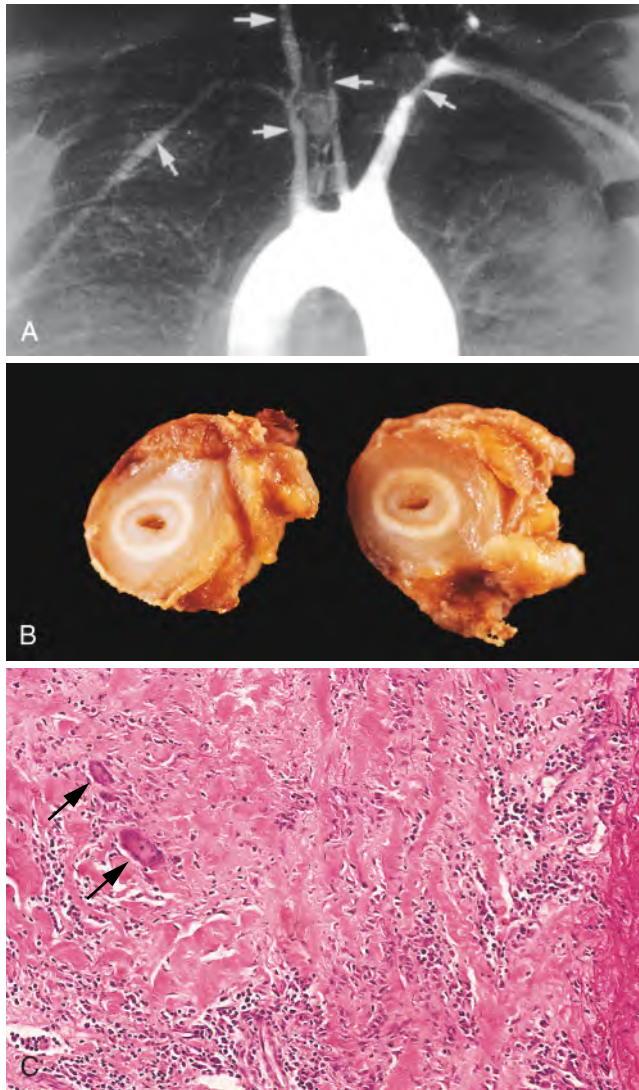


Fig. 11.24 Takayasu arteritis. (A) Aortic arch angiogram showing narrowing of brachiocephalic, carotid, and subclavian arteries (*arrows*). (B) Gross photograph of two cross-sections of the right carotid artery taken at autopsy of the patient shown in (A) demonstrating marked intimal thickening and adventitial fibrosis a narrowed lumen. (C) Histologic appearance in active Takayasu aortitis illustrating destruction and fibrosis of the arterial media associated with mononuclear infiltrates and inflammatory giant cells (*arrows*).

rest—undoubtedly due to the neural involvement. Chronic extremity ulcerations may develop, progressing over time (occasionally precipitously) to frank gangrene. Smoking abstinence in the early stages of the disease can often ameliorate further attacks; however, once established, the vascular lesions do not respond to smoking abstinence.

Polyarteritis Nodosa. Polyarteritis nodosa (PAN) is a systemic vasculitis of small- or medium-sized muscular arteries that typically affects renal and visceral vessels and spares the pulmonary circulation. There is no association with ANCA. One-third of patients with PAN have chronic hepatitis B (HB), which leads to the formation of HB surface antigen-antibody complexes that deposit in affected vessels; the cause is unknown in the remaining patients. Clinical manifestations of PAN result from ischemia and infarction of affected tissues and organs.

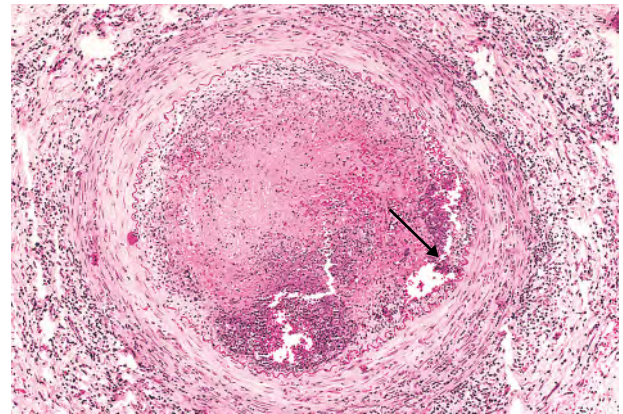


Fig. 11.25 Thromboangiitis obliterans (Buerger disease). The lumen is occluded by a thrombus containing abscesses (*arrow*), and the vessel wall is infiltrated with leukocytes.

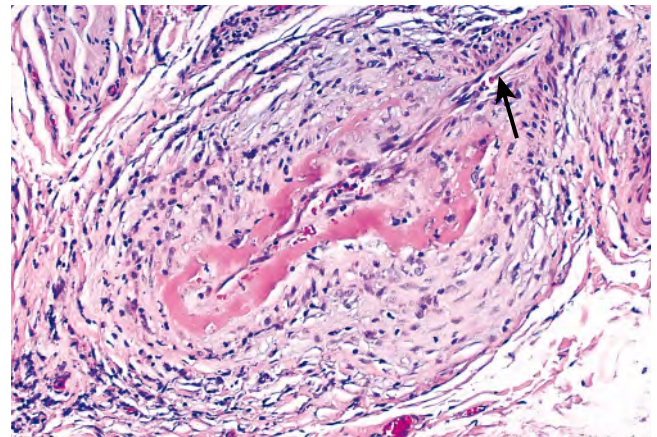


Fig. 11.26 Polyarteritis nodosa. There is segmental fibrinoid necrosis and thrombotic occlusion of the lumen of this small artery. Note that part of the vessel wall at the upper right (*arrow*) is uninvolved. (Courtesy Sidney Murphree, MD, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

MORPHOLOGY

Classic PAN is a segmental transmural necrotizing inflammation of small- to medium-sized arteries, often with superimposed aneurysm and/or thrombosis. Kidney, heart, liver, and gastrointestinal tract vessels are involved in descending order of frequency. Lesions usually involve only part of the vessel circumference and show a predilection for branch points. Impaired perfusion of parenchyma distal to the lesions results in ulcerations, infarcts, ischemic atrophy, or hemorrhage.

During the acute phase, there is transmural inflammation of the arterial wall with a mixed infiltrate of neutrophils, eosinophils, and mononuclear cells, frequently accompanied by **fibrinoid necrosis and luminal thrombosis** (see Fig. 11.26). Older lesions show fibrous thickening of the vessel wall extending into the adventitia. Characteristically, all stages of activity (from early to late) coexist in different vessels or even within the same vessel, suggesting ongoing injury.

Clinical Features

PAN is primarily a disease of young adults but can occur in all age groups. The course is frequently remitting and episodic, with long

symptom-free intervals. Because the vascular involvement is widely scattered, the clinical signs and symptoms of PAN are quite varied. Typical presentations involve some combination of rapidly accelerating hypertension due to renal artery involvement; abdominal pain and bloody stools caused by vascular gastrointestinal lesions; diffuse myalgias; and peripheral neuritis, predominantly affecting motor nerves. Renal involvement is often a major cause of mortality. Untreated, PAN is typically fatal; however, immunosuppression provides up to 80% 5-year survival.

Kawasaki Disease. Kawasaki disease is an acute, febrile, usually self-limited illness of infancy and childhood associated with large- to medium-sized vessel arteritis; 80% of the patients are younger than 5 years of age. Its clinical significance stems from the involvement of coronary arteries. Coronary arteritis can result in aneurysms that rupture or thrombose, causing myocardial infarction. Originally described in Japan, the disease is now recognized in the United States and elsewhere.

In genetically susceptible persons, a variety of infectious agents (mostly viral) have been posited to trigger the disease; interestingly, a Kawasaki-like disease (*multisystem inflammatory syndrome in children* [MIS-C]) has been documented in children infected with SARS-CoV-2. The vasculitis may result from a delayed-type hypersensitivity response directed against cross-reactive or newly uncovered vascular antigens. Subsequent cytokine production and polyclonal B-cell activation result in autoantibodies to ECs and SMCs that precipitate the vasculitis.

MORPHOLOGY

Kawasaki disease vasculitis resembles that seen in PAN. There is a dense transmural inflammatory infiltrate, although the fibrinoid necrosis is usually less prominent than in PAN. The acute vasculitis typically subsides spontaneously or in response to treatment, but aneurysm formation due to wall damage can appear. As with other arteritides, healed lesions can also exhibit obstructive intimal thickening. Pathologic changes outside the cardiovascular system are rarely significant.

Clinical Features

Kawasaki disease typically develops following a presumed viral illness associated with conjunctival and oral erythema and blistering, edema of the hands and feet, erythema of the palms and soles, a desquamative rash, and cervical lymph node enlargement—*mucocutaneous lymph node syndrome*. Approximately 20% of untreated patients develop cardiovascular sequelae, ranging from asymptomatic coronary arteritis, to coronary artery ectasia, to giant coronary artery aneurysms (7 to 8 mm); the latter may lead to rupture, thrombosis, myocardial infarction, or sudden death. With intravenous immunoglobulin therapy and aspirin, the rate of symptomatic coronary artery disease is markedly diminished.

Small Vessel Vasculitis

Small vessel vasculitis predominantly affects small intraparenchymal arteries, arterioles, capillaries, and venules (although medium-sized vessels can also be involved). This category is subdivided depending on whether there is a paucity (ANCA-associated vasculitis) or plethora (immune complex vasculitis) of immunoglobulin deposition in lesional vessels (Table 11.4 and Fig. 11.27).

Immune complex vasculitis is associated with moderate to extensive immunoglobulin and/or complement deposits in the affected vessel walls; immune complex glomerulonephritis is a common accompaniment. The vascular lesions resemble those found in experimental immune complex-mediated disorders (Arthus phenomenon and serum sickness) and have readily identifiable antibody and complement deposition. However, the specific antigen responsible for immune complex-associated vasculitis is only rarely identified, and it is not clear whether the pathogenic antigen-antibody complexes are deposited from the circulation or form in situ. Once immune complexes are deposited, vessel damage occurs due to complement activation and/or the recruitment and activation of Fc-receptor-bearing cells (e.g., neutrophils and monocyte/macrophages; Chapter 6).

Immune complex deposition is implicated in the following vasculitides:

- **Drug hypersensitivity vasculitis.** In some cases (e.g., penicillin), drugs act as haptens by binding to serum proteins; other agents are themselves foreign proteins (e.g., streptokinase). Regardless, antibodies directed against the drug-modified proteins or foreign molecules result in immune complex formation. The clinical manifestations can

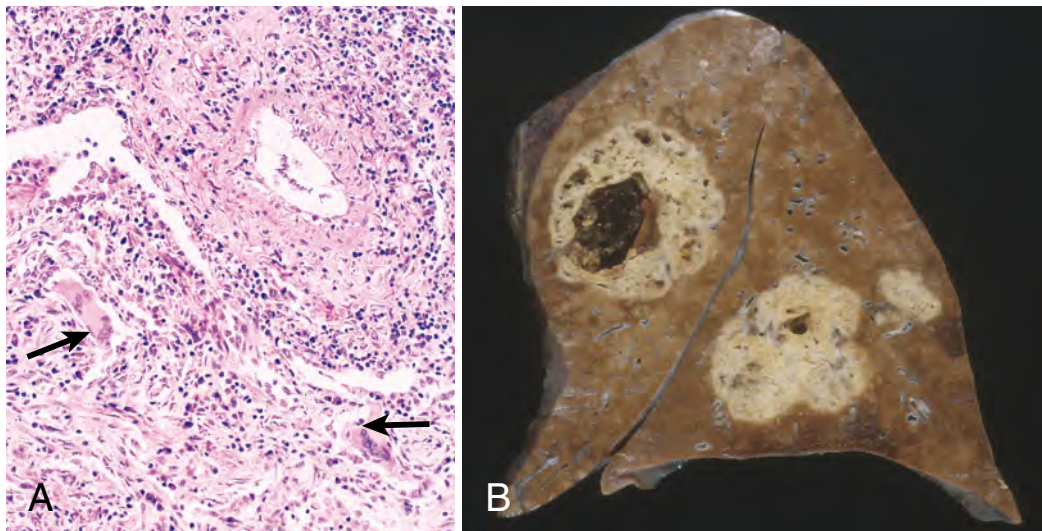


Fig. 11.27 Granulomatosis with polyangiitis. (A) Vasculitis of a small artery with adjacent granulomatous inflammation including epithelioid cells and giant cells (arrows). (B) Gross photo from the lung of a patient with fatal granulomatosis with polyangiitis demonstrating large, nodular, centrally cavating lesions. (B, Courtesy Sidney Murphree, MD, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

be mild and self-limiting or severe and even fatal; skin lesions are most common. It is always important to consider drug hypersensitivity as a cause of vasculitis since discontinuation of the offending agent usually leads to resolution.

- *Vasculitis secondary to infections.* Antibodies to microbial constituents can form immune complexes that circulate and deposit in vascular lesions. In up to 30% of patients with polyarteritis nodosa (see earlier), the vasculitis is attributable to immune complexes composed of hepatitis B surface antigen (HBsAg) and anti-HBsAg antibody.
- *Associated with other immunologic disorders* where there is autoantibody production (e.g., *systemic lupus erythematosus [SLE]*, *antiphospholipid antibody syndrome*, and *rheumatoid arthritis*) (Chapter 6). In SLE, DNA/anti-DNA immune complexes are common precipitants in small- and medium-sized vessel vasculitis, but SLE is also associated with numerous other autoantibodies that can be causal. Rheumatoid vasculitis of small and medium vessels occurs predominantly in the setting of long-standing, severe rheumatoid arthritis; it may also cause a clinically significant aortitis, and EC-specific antibodies and cell-mediated immunity may also contribute to the pathogenesis. Other systemic diseases that can cause immune complex vasculitis include *IgA vasculitis* (previously called *Henoch-Schönlein purpura*) caused by IgG forming immune complexes with abnormally glycosylated IgA, and *cryoglobulinemic vasculitis* due to cryoglobulin immune deposits (cryoglobulins are immunoglobulins that precipitate at temperatures $<37^{\circ}\text{C}$).
- *Antiglomerular basement membrane (anti-GBM) vasculitis (Goodpasture disease).* Rather than antigen-antibody immune complex deposition, this entity is caused by autoantibodies that directly bind to elements of the glomerular and pulmonary capillary basement membranes; this leads to a similar immunoglobulin deposition in vessels, and the same complement- and leukocyte-mediated injury seen in immune complex vasculitis (Chapter 20).

Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis is caused by circulating autoantibodies that react with neutrophil cytoplasmic antigens; these bind to various constituents (mainly enzymes) of neutrophil primary granules, monocyte lysosomes, and ECs. Two of several different ANCAs are clinically most important. These were previously grouped according to the intracellular distribution of the target antigens (cytoplasmic [c-ANCA] or perinuclear [p-ANCA]), but are now classified according to their antigen specificity.

- *Antiproteinase-3 (PR3-ANCA, previously c-ANCA).* Proteinase-3 (PR3) is a neutrophil azurophilic granule constituent that shares homology with numerous microbial peptides, possibly explaining the generation of PR3-ANCAs. PR3-ANCAs are associated with granulomatosis with polyangiitis (see later).
- *Anti-myeloperoxidase (MPO-ANCA, previously p-ANCA).* Myeloperoxidase (MPO) is a lysosomal granule constituent involved in oxygen free radical generation (Chapter 3). MPO-ANCAs can be induced by therapeutic agents (e.g., propylthiouracil used to treat hyperthyroidism). They are associated with microscopic polyangiitis and eosinophilic granulomatosis and polyangiitis (Churg-Strauss syndrome; see later).

While the antigenic targets of ANCA are primarily intracellular, it is now clear that ANCA antigens are either expressed constitutively at low levels on the plasma membrane or are translocated to the cell surfaces of activated and apoptotic leukocytes. Because of this, ANCAs can directly activate neutrophils, stimulating the release of reactive oxygen species and proteolytic enzymes; in vascular beds, such activation also leads to destructive interactions between inflammatory cells and ECs.

Since the ANCA autoantibodies are directed against cellular constituents and do not form circulating immune complexes, the vascular lesions do not typically contain demonstrable antibody and complement. Thus, ANCA-associated vasculitides are often described as *pauci-immune*. **ANCAs are not only pathogenic, but they are also useful diagnostic markers; their circulating titers generally mirror clinical severity, and spikes predict disease recurrence.** We describe next three ANCA-associated vasculitis that affect small vessels.

Microscopic Polyangiitis. Microscopic polyangiitis (MPA) is a small vessel necrotizing vasculitis that generally affects capillaries as well as small arterioles and venules. Unlike PAN, all lesions of microscopic polyangiitis tend to be of the same age in any given patient. Skin, mucous membranes, lungs, brain, heart, gastrointestinal tract, kidneys, and muscle all can be involved; necrotizing glomerulonephritis (90% of patients) and pulmonary capillaritis are particularly common. Most cases of MPA are associated with MPO-ANCA, and immune complexes are typically absent; drugs (e.g., hydralazine) and certain infections (e.g., *Staphylococcus aureus*) can trigger the disease.

MORPHOLOGY

Microscopic polyangiitis is characterized by **segmental fibrinoid necrosis of the media** with focal transmural necrotizing lesions; granulomatous inflammation is absent. These lesions morphologically resemble PAN but typically spare medium-sized and larger arteries; consequently, macroscopic infarcts are uncommon. Although immunoglobulins and complement components can be demonstrated in early skin lesions of MPA, most lesions are pauci-immune. In some areas (typically postcapillary venules), only infiltrating and fragmenting neutrophils are seen, giving rise to the term *leukocytoclastic vasculitis*. Note that similar neutrophil fragmentation can also be seen in immune complex-associated vasculitis, so that the pattern is not ANCA-specific.

Clinical Features

Depending on the vascular bed involved, major clinical features include hemoptysis, hematuria and proteinuria, bowel pain or bleeding, muscle pain or weakness, and palpable cutaneous purpura. With the exception of those who develop widespread renal or brain involvement, immunosuppression (and removal of potential inciting agents) induces durable remission.

Granulomatosis With Polyangiitis. Previously called Wegener granulomatosis, granulomatosis with polyangiitis (GPA) is an ANCA-positive necrotizing vasculitis characterized by a triad of:

- *Acute necrotizing granulomas* of the upper respiratory tract (ear, nose, sinuses, throat) or the lower respiratory tract (lung) or both
- *Necrotizing or granulomatous vasculitis* affecting small- to medium-sized vessels (e.g., capillaries, venules, arterioles, and arteries), most prominent in the lungs and upper airways but affecting other sites as well
- *Focal necrotizing, often crescentic, glomerulonephritis*

“Limited” forms of GPA may be restricted to the respiratory tract. Conversely, a widespread form of the disease can affect eyes, skin, and other organs, notably the heart; clinically, widespread GPA resembles PAN except that there is also respiratory involvement.

Pathogenesis

GPA likely represents a form of T cell-mediated hypersensitivity response to normally “innocuous” inhaled microbial or other environmental agents. Such a pathogenesis is supported by the presence of granulomas and a dramatic response to immunosuppressive therapy. PR3-ANCAs are

present in up to 95% of cases. In addition to likely being integral to GPA's pathogenesis, PR3-ANCA's are useful markers of response to therapy and early disease relapse.

MORPHOLOGY

Upper respiratory tract lesions include **granulomatous sinusitis** and ulcerative lesions of the nose, palate, or pharynx; lung findings vary from diffuse parenchymal infiltrates to granulomatous nodules. There is typically multifocal granulomatous vasculitis with a surrounding zone of proliferating fibroblasts (see Fig. 11.27B). Multiple granulomata can coalesce to produce radiographically visible nodules that can also cavitate; late-stage disease may be marked by extensive necrotizing granulomatous involvement of the parenchyma (see Fig. 11.27C), and alveolar hemorrhage may be prominent. Lesions may ultimately undergo progressive fibrosis and organization.

Renal lesions range from mild, focal glomerular necrosis with isolated capillary loop thrombosis (**focal and segmental necrotizing glomerulonephritis**) to more advanced glomerular lesions with diffuse necrosis and parietal cell proliferation resulting in crescent formation (**crescentic glomerulonephritis**).

Clinical Features

Males are affected more often than females, at an average age of about 40 years. Classic features include bilateral pneumonitis with nodules and cavitary lesions (95%), chronic sinusitis (90%), mucosal ulcerations of the nasopharynx (75%), and renal disease (80%). Rashes, myalgias, articular involvement, neuritis, and fever can also occur. Left untreated, the disease is fatal in 80% of cases within 1 year. Treatment with steroids, cyclophosphamide, TNF antagonists, and anti-B-cell antibodies have turned GPA into a chronic relapsing and remitting disease.

Eosinophilic Granulomatosis With Polyangiitis (Churg-Strauss Syndrome). Eosinophilic granulomatosis with polyangiitis (EGPA) is a small vessel necrotizing vasculitis classically associated with asthma, allergic rhinitis, lung infiltrates, peripheral eosinophilia, extravascular necrotizing granulomas, and eosinophilic infiltrates in vessels and other tissues.

Cutaneous involvement (with palpable purpura), gastrointestinal bleeding, and renal disease (primarily as focal and segmental glomerulosclerosis) are major associations. Cytotoxicity produced by myocardial eosinophilic infiltrates can cause cardiomyopathy, which is seen in 60% of patients and is a major cause of morbidity and death.

EGPA is likely a consequence of hyperresponsiveness to some normally innocuous allergic stimulus. MPO-ANCA's are present in 40% to 60% of cases overall and in 75% to 100% of cases with concurrent renal involvement. The vascular lesions differ from those of PAN or microscopic polyangiitis by virtue of the presence of both granulomas and eosinophils.

Behçet Syndrome. Behçet syndrome is a systemic inflammatory disorder associated with a vasculitis of veins and arteries. Virtually all cases are marked by transient, self-healing ulcerative lesions of the oral mucosa, and larger and more persistent ulcers may also appear on the genitals. Other commonly affected tissues include the skin, joints, central nervous system (CNS) (particularly the eyes), gastrointestinal tract, and cardiovascular system.

The cause of Behçet syndrome is unknown. Higher risk is associated with variants in genes that regulate Th17 T cell responses. It is hypothesized that exposure of genetically "at-risk" individuals to unknown

environmental agents triggers an abnormal Th17 T cell mediated immune response, resulting in tissue injury.

The average age of onset of Behçet syndrome is 30 years and the disease is more common and more severe in males. None of its features is specific and diagnosis depends on the constellation of clinical findings and abnormalities identified radiologically, such as focal inflammatory CNS lesions or arterial aneurysms. Skin lesions in Behçet syndrome may resemble acne or erythema nodosum. Biopsy of such lesions reveals a leukocytoclastic vasculitis of dermal vessels associated with an infiltrate composed of neutrophils and admixed T cells and macrophages. Cardiovascular involvement may take the form of retinal artery vasculitis, arterial aneurysm, or venous thrombosis; the latter may lead to complications such as pulmonary embolism or Budd-Chiari syndrome. Eye involvement is seen in approximately 50% of cases and most commonly manifests as uveitis or hypopyon (the accumulation of neutrophils in the anterior chamber). Brain lesions most commonly involve the brainstem and may lead to permanent injury or death.

KEY CONCEPTS

Vasculitis

- Vasculitis is defined as inflammation of vessel walls; it is frequently associated with systemic manifestations (including fever, malaise, myalgias, and arthralgias) and tissue pathology and organ dysfunction that depends on the distribution of vascular involvement.
- Vasculitis most commonly has an immunologic basis resulting from immune complex deposition, ANCA's, or anti-EC or anti-GBM antibodies. Less commonly it is caused by infections.
- Different forms of vasculitis tend to specifically affect vessels of a particular caliber and location (see Fig. 11.22 and Table 11.4).

DISORDERS OF BLOOD VESSEL HYPERREACTIVITY

Several disorders are characterized by inappropriate or excessive blood vessel vasoconstriction.

Raynaud Phenomenon

Raynaud phenomenon results from exaggerated vasoconstriction of arteries and arterioles in responses to cold or emotion that most commonly affects the extremities, particularly the fingers and toes, but also occasionally the nose, earlobes, or lips. The restricted blood flow induces paroxysmal pallor and even cyanosis in severe cases; involved digits classically show "red, white, and blue" color changes from most proximal to most distal, correlating with proximal vasodilation, central vasoconstriction, and more distal cyanosis (Fig. 11.28). Raynaud phenomenon can be a primary entity or secondary to other disorders.

- **Primary Raynaud phenomenon** affects 3% to 5% of the general population and occurs most often in young females. It tends to symmetrically affect the extremities, and the severity and extent of involvement typically does not progress. It results from intrinsic hyperreactivity of medial SMCs. Structural changes in the arterial walls are absent except in long-standing disease when intimal thickening can develop. The course is usually benign, but chronicity can lead to atrophy of the skin, subcutaneous tissues, and muscles. Ulceration and ischemic gangrene are rare.
- **Secondary Raynaud phenomenon** refers to vascular insufficiency due to arterial disease caused by other entities, including SLE, scleroderma, Buerger disease, or even atherosclerosis. Clinically, secondary Raynaud

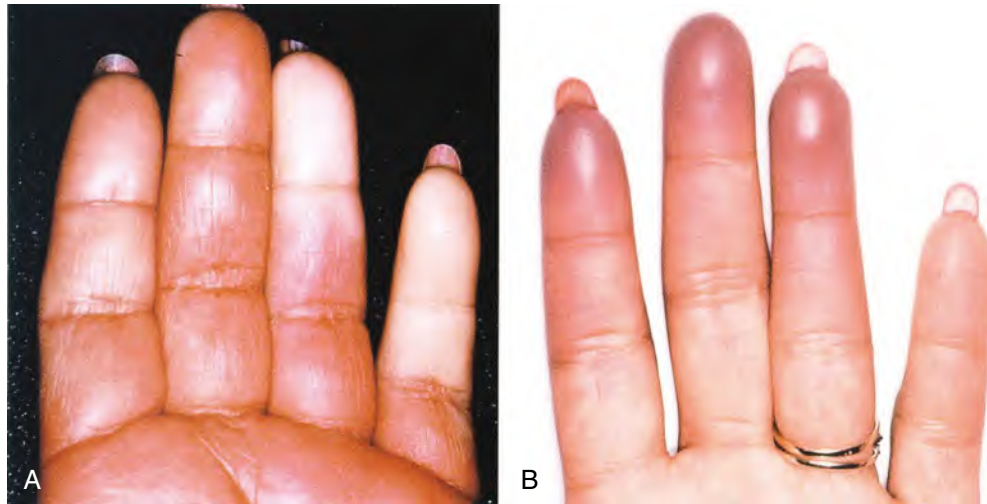


Fig. 11.28 Raynaud phenomenon. (A) Sharply demarcated pallor of the distal fingers resulting from the closure of digital arteries. (B) Cyanosis of the fingertips. (From Salvarani C, Cantini F, Boiardi L, Hunder GG. Polymyalgia rheumatica and giant-cell arteritis. *N Engl J Med*. 2002;347:261-271.)

phenomenon tends to have asymmetric involvement of the extremities and progressively worsens (extent and severity) over time.

Since Raynaud phenomenon may be the first manifestation of immune-mediated vasculitides, any patient with new symptoms should be evaluated. Of these, some 10% will eventually manifest an underlying disorder.

Myocardial Vessel Vasospasm

Excessive constriction of arteries or arterioles may cause ischemia, and persistent vasospasm can even cause myocardial infarction. In addition to intrinsic hyperreactivity of medial SMCs, as described for primary Raynaud disease (see earlier), high levels of vasoactive mediators can precipitate prolonged vascular contraction. Such agents can be endogenous (e.g., epinephrine released by pheochromocytomas) or exogenous (cocaine or phenylephrine). Elevated thyroid hormone causes a similar effect by increasing the sensitivity of vessels to circulating catecholamines, while autoantibodies and T cells in scleroderma (Chapter 6) can cause vascular instability and vasospasm. Extreme psychological stress and the attendant release of catecholamines can lead to pathologic vasospasm.

When vasospasm of cardiac arterial or arteriolar beds (so-called *cardiac Raynaud*) is of sufficient duration (20 to 30 minutes), myocardial infarction or sudden cardiac death can occur.

VEINS AND LYMPHATICS

Varicose veins and phlebothrombosis/thrombophlebitis together account for at least 90% of clinical venous disease.

Varicose Veins

Varicose veins are abnormally dilated, tortuous veins produced by prolonged, increased intraluminal pressure resulting in vessel dilation and subsequent incompetence of the venous valves. The superficial veins of the upper and lower leg are commonly involved because venous pressures in these sites can be markedly elevated (up to 10 times normal) by prolonged dependent posture. Up to 20% of males and one-third of females develop lower extremity varicose veins. Obesity increases the

risk, and the higher incidence in females probably reflects the prolonged elevation in venous pressure caused by compression of the inferior vena cava by the gravid uterus during pregnancy. A familial predilection to varicose veins reflects faulty venous wall development.

Clinical Features

Incompetence of the venous valves leads to stasis, congestion, edema, pain, and thrombosis. Secondary tissue ischemia results from chronic venous congestion and poor vessel drainage and leads to stasis dermatitis (also called “*brawny induration*”; the brawny color comes from the hemolysis of extravasated red cells) and ulcerations; Poor wound healing and superimposed infections are common additional complications. Notably, embolism from these superficial veins is very rare, as opposed to the relatively frequent thromboembolism that arises from thrombosed deep veins (see later and Chapter 4).

Varicosities in two other sites also deserve mention:

- **Esophageal varices.** Liver cirrhosis (less frequently, portal vein obstruction or hepatic vein thrombosis) causes portal vein hypertension (Chapter 18). Portal hypertension leads to the opening of portosystemic shunts that increase blood flow into veins at the gastroesophageal junction (forming esophageal varices), rectum (forming hemorrhoids), and periumbilical veins of the abdominal wall (forming a *caput medusa*). Esophageal varices are the most important since their rupture can lead to massive (even fatal) upper gastrointestinal hemorrhage.
- **Hemorrhoids** can result from primary varicose dilation of the venous plexus at the anorectal junction (e.g., through prolonged pelvic vascular congestion due to pregnancy or straining to defecate). Hemorrhoids may be a source of bleeding; they can also thrombose and are prone to painful ulceration.

Thrombophlebitis and Phlebothrombosis

Thrombophlebitis and *phlebothrombosis* are interchangeable designations for venous thrombosis and inflammation; deep leg veins account for more than 90% of cases. The periprosthetic venous plexus in males and the pelvic venous plexus in females are additional sites of involvement, along with the large veins in the skull and the dural sinuses (especially in the setting of infection or inflammation). Portal vein

thrombosis can occur in association with peritoneal infections (peritonitis, appendicitis, salpingitis, and pelvic abscesses), as well as certain thrombophilic conditions associated with platelet hyperactivity (e.g., polycythemia vera) (Chapter 14).

Systemic hypercoagulability, including genetic hypercoagulability syndromes (Chapter 4), can also contribute to thrombophlebitis. In patients with cancer, particularly adenocarcinoma, hypercoagulability often occurs as a paraneoplastic syndrome caused by procoagulant factors released from the tumor cells (Chapter 7). In this setting, venous thromboses classically appear in one location, disappear, and then occur in another site—so-called migratory thrombophlebitis (*Trousseau syndrome*).

Decreased blood flow in the setting of prolonged immobilization is the most common cause of lower extremity *deep venous thrombosis* (DVT). This can occur with extended bed rest or sitting during long airplane or automobile trips; the postoperative state can also precipitate DVT, and congestive heart failure, pregnancy, oral contraceptive use, malignancy, and obesity all constitute additional risk factors.

Thrombi in the legs tend to produce few, if any, reliable signs or symptoms. Indeed, typical local manifestations of symptomatic DVTs, such as vein dilation, edema, cyanosis, heat, erythema, or pain, may be entirely absent, especially in bedridden patients. In some cases, pain can be elicited by pressure over affected veins, squeezing the calf muscles, or forced dorsiflexion of the foot (*Homan sign*); absence of these findings does not exclude the presence of a DVT.

Pulmonary embolism is a serious clinical complication of DVT (Chapter 4), resulting from fragmentation or detachment of the venous thrombus. In many cases the first manifestation of thrombophlebitis is a pulmonary embolus. Chronic thromboembolic disease can also give rise to pulmonary hypertension. Depending on the size and number of emboli, the outcome can range from no symptoms to death.

Superior and Inferior Vena Caval Syndromes

The superior vena caval syndrome is usually caused by neoplasms that compress or invade the superior vena cava (e.g., bronchogenic carcinoma or mediastinal lymphoma). The resulting obstruction produces a characteristic clinical complex including marked dilation of the veins of the head, neck, and arms associated with cyanosis. Pulmonary vessels can also be compressed, inducing respiratory distress.

The inferior vena caval syndrome can be caused by neoplasms that compress or invade the inferior vena cava (IVC) or by thrombosis of the hepatic, renal, or lower extremity veins that propagates cephalad. Certain neoplasms—particularly hepatocellular carcinoma and renal cell carcinoma—show a striking tendency to grow within veins that drain the liver or kidney, respectively, and these can ultimately occlude the IVC. IVC obstruction induces marked lower extremity edema, distention of the superficial collateral veins of the lower abdomen, and—with renal vein involvement—massive proteinuria.

Lymphangitis and Lymphedema

Although primary disorders of lymphatic vessels are extremely uncommon, secondary processes frequently develop in association with inflammation or malignancies.

Lymphangitis represents acute inflammation caused by the spread of bacterial infections into lymphatics; group A β -hemolytic streptococcus is the most common agent. Affected lymphatics are dilated and filled with an exudate of neutrophils and monocytes; the infiltrates can extend through the vessel wall and, in severe cases, can produce cellulitis or focal abscesses. *Lymphangitis* is manifested by red, painful subcutaneous streaks (the inflamed lymphatics) and painful enlargement of the draining lymph nodes (*lymphadenitis*). If bacteria are not successfully contained within the lymph nodes, subsequent drainage into the venous circulation can result in bacteremia and sepsis.

Primary lymphedema can occur as an isolated congenital defect (simple congenital lymphedema) or as the familial *Milroy disease* (*heredofamilial congenital lymphedema*), which results in lymphatic agenesis or hypoplasia. Secondary or obstructive lymphedema stems from blockage of previously normal lymphatics, including:

- *Tumors* obstructing lymphatic channels or the regional lymph nodes
- *Surgical procedures* that sever lymphatic connections (e.g., axillary lymph node resection in a radical mastectomy)
- *Postradiation fibrosis*
- *Filariasis*
- *Postinflammatory thrombosis and scarring*

Regardless of the cause, lymphedema increases the hydrostatic pressure in the lymphatics distal to the obstruction and causes edema. Chronic edema, in turn, can lead to ECM deposition and fibrosis, producing induration or a *peau d'orange* (orange peel) appearance of the overlying skin. Eventually, inadequate tissue perfusion can lead to skin ulceration. Rupture of dilated lymphatics (e.g., secondary to obstruction from a tumor or following surgical injury) leads to milky accumulations of lymph designated chylous ascites (abdomen), chylothorax, or chylopericardium, depending on the site of lymph accumulation.

VASCULAR TUMORS

Tumors of blood vessels and lymphatics constitute a spectrum from benign hemangiomas (extremely common) to locally aggressive neoplasms that rarely metastasize, to rare, highly malignant angiosarcomas (Table 11.5).

Vascular neoplasms arise either from endothelium (e.g., *hemangioma*, *lymphangioma*, *angiosarcoma*) or from cells that support or surround blood vessels (e.g., *glomus tumor*). Primary tumors of large vessels (aorta, pulmonary artery, and vena cava) are mostly sarcomas. Although a benign well-differentiated hemangioma is not usually confused with an anaplastic angiosarcoma, the distinction between benign and malignant can occasionally be difficult. Congenital or developmental malformations and nonneoplastic reactive vascular proliferations (e.g., *bacillary angiomatosis*) can also present as tumor-like

TABLE 11.5 Classification of Vascular Tumors and Tumor-Like Conditions

Benign Neoplasms, Developmental and Acquired Conditions
Hemangioma - Capillary hemangioma - Cavernous hemangioma - Pyogenic granuloma
Lymphangioma - Simple (capillary) lymphangioma - Cavernous lymphangioma (cystic hygroma)
Glomus tumor
Vascular ectasias - Nevus flammeus - Spider telangiectasia (arterial spider) - Hereditary hemorrhagic telangiectasia (Rendu-Osler-Weber disease)
Reactive vascular proliferations Bacillary angiomatosis
Intermediate-Grade Neoplasms
Kaposi sarcoma Hemangioendothelioma
Malignant Neoplasms
Angiosarcoma

lesions that can be diagnostically challenging. In general, benign and malignant vascular neoplasms can be distinguished by the following features.

- *Benign tumors* usually contain obvious vascular channels filled with blood cells (lymphatics will be filled with lymph), lined by a monolayer of normal-appearing ECs.
- *Malignant tumors* are more solidly cellular and proliferative, and exhibit cytologic atypia; they usually do not form well-organized vessels. The endothelial derivation of such neoplastic proliferations may require immunohistochemical detection of EC-specific markers such as CD31 or von Willebrand factor.

Benign Tumors and Tumor-Like Conditions

Vascular Ectasias

Ectasia is a generic term for any local dilation of a structure, while *telangiectasia* is used to describe a permanent dilation of preexisting small vessels (capillaries, venules, and arterioles, usually in the skin or mucous membranes) that forms a discrete red lesion. These can be congenital or acquired and are not true neoplasms.

- *Nevus flammeus* (a “birthmark”), the most common form of vascular ectasia, is a light pink to deep purple flat lesion on the head or neck composed of dilated vessels. Most ultimately regress spontaneously. The so-called *port wine stain* is a type of nevus flammeus that tends to grow during childhood, thickening the associated skin surface, and persisting over time. Such lesions in the distribution of the trigeminal nerve can be associated with *Sturge-Weber syndrome* (also called encephalotrigeminal angiomatosis). This uncommon congenital disorder is associated with facial port wine nevi, ipsilateral venous angiomas in the cortical leptomeninges, intellectual disability, seizures, hemiplegia, and skull radiopacities. A large facial telangiectasia in a child with intellectual deficiency may indicate the presence of additional vascular malformations. Most cases of port wine stains (including those associated with Sturge-Weber syndrome) are caused by a somatic gain-of-function mutation in GNAQ (the alpha subunit of heterotrimeric transmembrane signaling molecule).
- *Spider telangiectasias* are nonneoplastic vascular malformations grossly resembling a spider. These manifest as radial, often pulsatile lesions composed of dilated subcutaneous arteries or arterioles (resembling spider legs) about a central core (resembling a spider’s body) that blanch with pressure. These commonly occur on the face, neck, or upper chest and are most frequently associated with hyperestrogenic states (e.g., in pregnant females or patients with cirrhosis).
- *Hereditary hemorrhagic telangiectasia (Rendu-Osler-Weber disease)* is an autosomal dominant disorder caused by loss-of-function mutations in genes that encode components of the TGF- β signaling pathway in ECs. The telangiectasias are malformations composed of dilated capillaries and veins that are present at birth. They are widely distributed over the skin and oral mucous membranes, as well as in the respiratory, gastrointestinal, and urinary tracts. The lesions can spontaneously rupture, causing serious epistaxis (nosebleed), gastrointestinal bleeding, or hematuria.

Hemangiomas

Hemangiomas are very common tumors composed of blood-filled vessels (Fig. 11.29); they are the most common tumor of infancy and childhood, occurring in 1 in 10 individuals. Most hemangiomas that present early in life are present from birth and initially increase in size, but many eventually regress spontaneously (see Fig. 10.21). While hemangiomas are typically confined to the head, neck, and

thoracic skin, they can also arise internally and can occasionally be more extensive (*angiomatosis*). Nearly one-third of internal lesions occur in the liver. Malignant transformation is rare. Several histologic and clinical variants have been described.

- *Capillary hemangiomas* are the most common type; these occur in the skin, subcutaneous tissues, and mucous membranes of the oral cavities and lips, as well as in the liver, spleen, and kidneys (see Fig. 11.29A). Histologically, they are composed of thin-walled capillaries with scant stroma (see Fig. 11.29B).
- *Juvenile hemangiomas* (so-called strawberry-type hemangiomas) of the newborn skin are extremely common (1 in 200 births) and can be multiple. These grow rapidly for a few months, but then fade by 1 to 3 years of age and completely regress by age 7 years in most cases.
- *Cavernous hemangiomas* are composed of large, dilated vascular channels. In contradistinction to capillary hemangiomas, cavernous hemangiomas are more infiltrative, frequently involve deep structures, and do not spontaneously regress. Moreover, cavernous hemangiomas detected by imaging studies may be difficult to distinguish from their malignant counterparts. On histologic examination, the mass is sharply delineated but unencapsulated and composed of large, cavernous blood-filled vascular spaces, separated by connective tissue stroma (see Fig. 11.29C). Intravascular thrombosis with associated dystrophic calcification is common. They can be locally destructive, but more often are of little clinical significance outside of a cosmetic disfigurement or their vulnerability to traumatic ulceration and bleeding. Brain hemangiomas are problematic, as they can cause symptoms related to compression of adjacent tissue or rupture. Cavernous hemangiomas are one component of *von Hippel-Lindau disease* (Chapter 28), in which vascular lesions are commonly found in the cerebellum, brain stem, retina, pancreas, and liver.
- *Pyogenic granulomas* are capillary hemangiomas that present as rapidly growing red pedunculated lesions on the skin, gingiva, or oral mucosa. They bleed easily and are often ulcerated (see Fig. 11.29D). Roughly one-quarter of lesions develop after trauma, reaching a size of 1 to 2 cm within a few weeks. Curettage and cautery is usually curative. Pregnancy tumor (*granuloma gravidarum*) is a pyogenic granuloma that occurs infrequently (1% of patients) in the gingiva of pregnant females. These lesions may spontaneously regress (especially after pregnancy) or undergo fibrosis, but occasionally require surgical excision.

Lymphangiomas

Lymphangiomas are the benign lymphatic counterparts of hemangiomas.

- *Simple (capillary) lymphangiomas* are slightly elevated or sometimes pedunculated lesions up to 1 to 2 cm in diameter that occur predominantly in the head, neck, and axillary subcutaneous tissues. Histologically, lymphangiomas exhibit networks of endothelium-lined spaces that can be distinguished from capillary channels by lymphatic endothelial markers (e.g., VEGFR-3, LYVE-1, and others) or by the absence of red cells.
- *Cavernous lymphangiomas (cystic hygromas)* are typically found in the neck or axilla of children and more rarely in the retroperitoneum. Cavernous lymphangiomas can occasionally be enormous (up to 15 cm), filling the axilla or producing gross deformities about the neck; cavernous lymphangiomas of the neck are common in *Turner syndrome*. The lesions are composed of massively dilated lymphatic spaces lined by ECs and separated by intervening connective tissue stroma containing lymphoid aggregates. The tumor margins are indistinct and unencapsulated, making resection difficult.

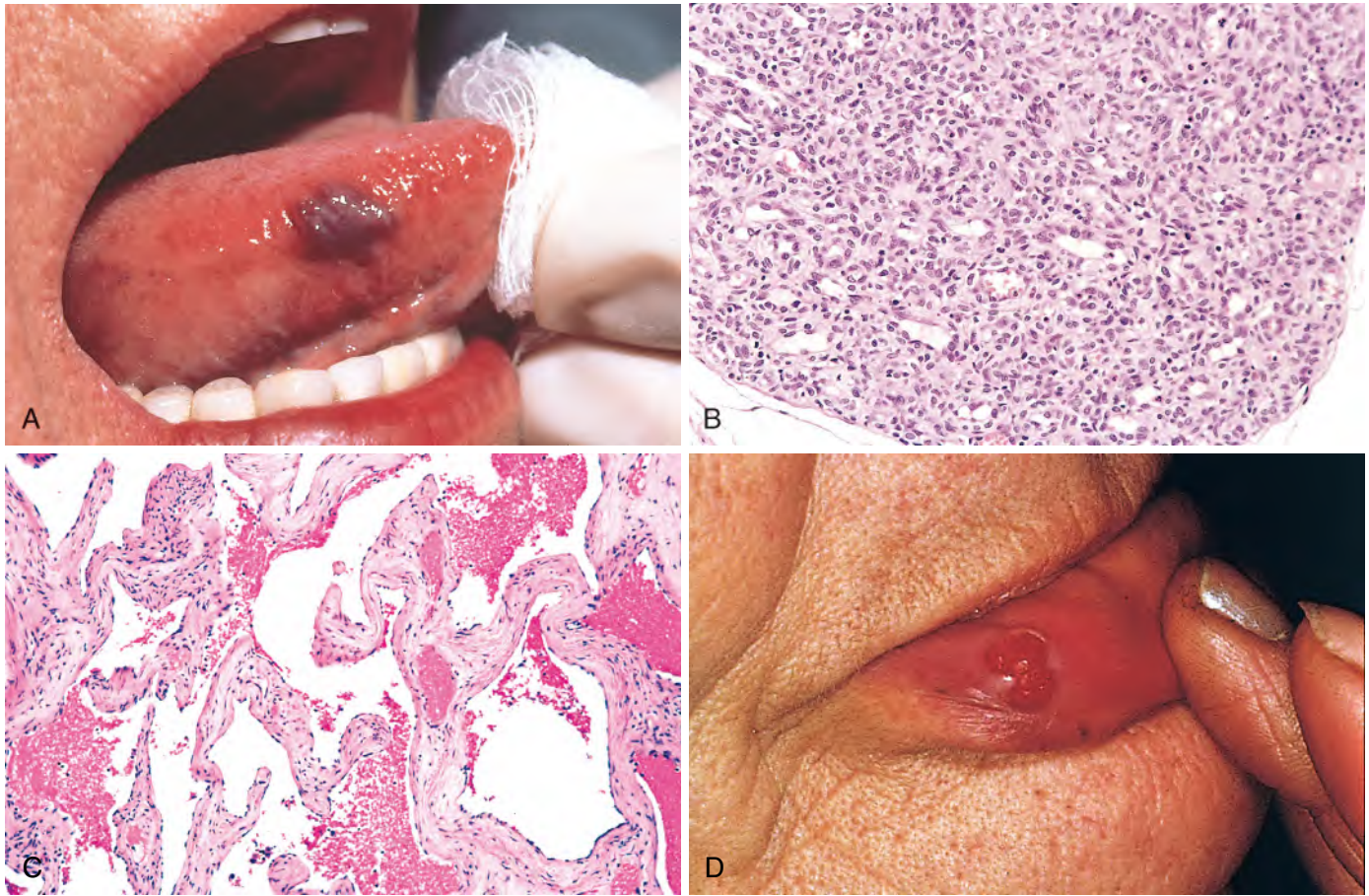


Fig. 11.29 Hemangiomas. (A) Hemangioma of the tongue. (B) Histology of juvenile capillary hemangioma. (C) Histology of cavernous hemangioma. (D) Pyogenic granuloma of the lip. (A and D, Courtesy John Sexton, MD, Beth Israel Hospital, Boston. B, Courtesy Christopher DM Fletcher, MD, Brigham and Women's Hospital, Boston. C, courtesy Thomas Rogers, MD, University of Texas Southwestern Medical School, Dallas.)

Glomus Tumor (Glomangioma)

Glomus tumors are benign but exquisitely painful tumors arising from modified SMCs of the glomus bodies, arteriovenous structures involved in thermoregulation. Although they may superficially resemble hemangiomas, glomangiomas arise from SMCs rather than ECs. They are most commonly found in the distal portion of the digits, especially under the fingernails. Excision is curative.

Bacillary Angiomatosis

Bacillary angiomatosis is a vascular proliferation in immunocompromised hosts (e.g., patients with acquired immunodeficiency syndrome [AIDS]) caused by opportunistic bacilli of the *Bartonella* family. They can involve the skin, bone, brain, and other organs. Two species are implicated.

- *Bartonella henselae*, whose principal reservoir is the domestic cat, causes *cat-scratch disease* (a necrotizing granulomatous disorder of lymph nodes) in immunocompetent hosts.
- *Bartonella quintana* is transmitted by human body lice; this microbe was the cause of trench fever in World War I.

The bacteria induce host tissues to produce hypoxia-inducible factor-1 α (HIF-1 α), which, in turn, drives vascular endothelial growth factor (VEGF) production and ultimately vascular proliferation. The infections (and lesions) are cleared by antibiotics.

MORPHOLOGY

Skin lesions in bacillary angiomatosis are red papules and nodules or rounded subcutaneous masses; histologically, there is capillary proliferation with prominent epithelioid ECs exhibiting nuclear atypia and mitoses, with scattered stromal neutrophils and causal bacteria (Fig. 11.30).

Intermediate-Grade (Borderline) Tumors Kaposi Sarcoma

Kaposi sarcoma (KS) is a vascular neoplasm caused by human herpesvirus 8 (HHV8, also known as Kaposi sarcoma herpesvirus [KSHV]). Although it occurs in a number of contexts, KS is currently seen most commonly in patients with AIDS; indeed, its presence is used as a criterion for diagnosis of AIDS. Based on population demographics and risk factors, KS is categorized into four forms:

- **Classic KS** is a disorder of older males of Mediterranean, Middle Eastern, or Eastern European descent (especially Ashkenazi Jewish people); it is uncommon in the United States. It can be associated with malignancy or altered immunity, but it is *not* associated with human immunodeficiency virus (HIV) infection. Classic KS manifests as multiple red-purple skin plaques or nodules, usually in the distal lower extremities; these progressively increase in size and number and spread proximally. Although persistent, the tumors are typically asymptomatic and remain localized to the skin and subcutaneous tissue.

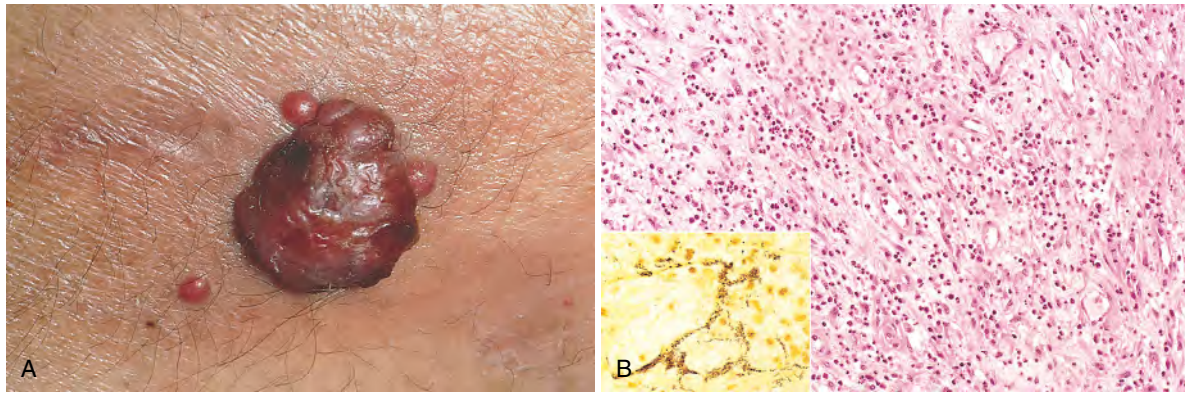


Fig. 11.30 Bacillary angiomatosis. (A) Characteristic cutaneous lesion. (B) Histologic appearance with acute neutrophilic inflammation and vascular (capillary) proliferation. *Inset*, Modified silver (Warthin-Starry) stain demonstrates clusters of tangled bacilli (black). (A, Courtesy Richard Johnson, MD, Beth Israel Deaconess Medical Center, Boston. B and *inset*, Courtesy Scott Granter, MD, Brigham and Women's Hospital, Boston.)

- **Endemic African KS** typically occurs in HIV-seronegative individuals younger than age 40 years and can follow an indolent or aggressive course; it involves lymph nodes much more frequently than the classic variant. In combination with AIDS-associated KS (see later), KS is one of the more common tumors in central Africa. The prevalence of endemic African KS is inversely related to the wearing of shoes, suggesting a cutaneous route of exposure to some infectious agent. A particularly severe form, with prominent lymph node and visceral involvement, occurs in prepubertal children; the prognosis is poor, with almost 100% mortality within 3 years.
- **Transplant-associated (iatrogenic) KS** occurs in solid-organ transplant recipients receiving T-cell immunosuppression; the risk can be 100-fold greater than for immunocompetent patients. Transplant-associated KS typically follows an aggressive course involving lymph nodes, mucosa, and viscera; cutaneous lesions may be absent. Lesions can regress if immunosuppression is reduced, but at the risk of organ rejection.
- **AIDS-associated (epidemic) KS** is an AIDS-defining illness; worldwide, it represents the most common HIV-related malignancy. Although the incidence of KS has fallen more than 80% with the advent of aggressive antiretroviral therapies, it still occurs in HIV-infected individuals 300-fold more often than in transplant recipients and 1000-fold more often than in the general population. AIDS-associated KS often involves lymph nodes and disseminates widely to viscera early in its course; most patients eventually die of opportunistic infections rather than KS.

Pathogenesis

Virtually all KS lesions are infected by *HHV8*. Like Epstein-Barr virus, *HHV8* is a γ -herpesvirus. It is transmitted sexually and potentially via oral secretions and cutaneous exposures. *HHV8* and altered T-cell immunity are likely required for KS development; in elderly adults, diminished T-cell immunity may be related to aging. Acquired somatic mutations in the cells of origin also contribute to tumor development and progression.

HHV8 causes lytic and latent infections in ECs, both of which contribute to KS pathogenesis. A virally encoded G protein induces VEGF production, stimulating endothelial growth; in addition, cytokines produced by inflammatory cells recruited to sites of lytic infection create a local proliferative milieu. In latently infected cells, *HHV8*-encoded proteins disrupt normal cellular proliferation controls (e.g., through synthesis of a viral homologue of cyclin D) and prevent apoptosis by inhibiting p53. Thus, the local inflammatory environment favors cellular proliferation, and latently infected cells have a growth advantage. In its early stages, only a few cells are *HHV8*-infected, but with time, virtually all of

the proliferating cells carry the virus. The molecular pathogenesis of *HHV8* infection and KS are discussed in more detail in Chapter 6.

MORPHOLOGY

In **classic KS** (and sometimes in other variants), the cutaneous lesions progress through three stages:

- **Patches** are red-purple macules (Fig. 11.31A). Histology shows dilated irregular EC-lined vascular spaces with interspersed lymphocytes, plasma cells, and macrophages (sometimes containing hemosiderin). The lesions can be difficult to distinguish from granulation tissue.
- With time, lesions become larger, violaceous, **raised plaques** (see Fig. 11.31A) composed of dermal accumulations of dilated, jagged vascular channels lined and surrounded by plump spindle cells. Scattered between the vascular channels are extravasated red cells, hemosiderin-laden macrophages, and other mononuclear inflammatory cells.
- Eventually, lesions become **nodular** and more distinctly neoplastic. These lesions are composed of sheets of plump, proliferating spindle cells, mostly in the dermis or subcutaneous tissues (Fig. 11.31B), containing small vessels and slit-like spaces with red cells. Marked hemorrhage, hemosiderin pigment, and mononuclear inflammation are present; mitotic figures are common, as are round, pink, cytoplasmic globules representing degenerating red cells within phagolysosomes. The nodular stage often heralds lymph node and visceral involvement.

Clinical Features

The course of KS varies widely and is significantly influenced by the clinical setting. Most primary *HHV8* infections are asymptomatic. Classic KS is—at least initially—largely restricted to the skin, and surgical resection is usually adequate for an excellent prognosis. Radiation can be used for multiple lesions in a restricted area, and chemotherapy yields satisfactory results for more disseminated disease including nodal involvement. In immunosuppression-associated KS, withdrawal of immunosuppression (perhaps with adjunct chemotherapy or radiotherapy) is often effective. For AIDS-associated KS, HIV antiretroviral drugs are generally beneficial, with or without additional therapy. In addition, interferon- γ and angiogenesis inhibitors have also proved variably effective, while newer strategies aimed at specific kinases downstream of VEGF receptors show promise.

Hemangioendothelioma

Hemangioendotheliomas comprise a spectrum of borderline vascular neoplasms with clinical behaviors intermediate between benign,

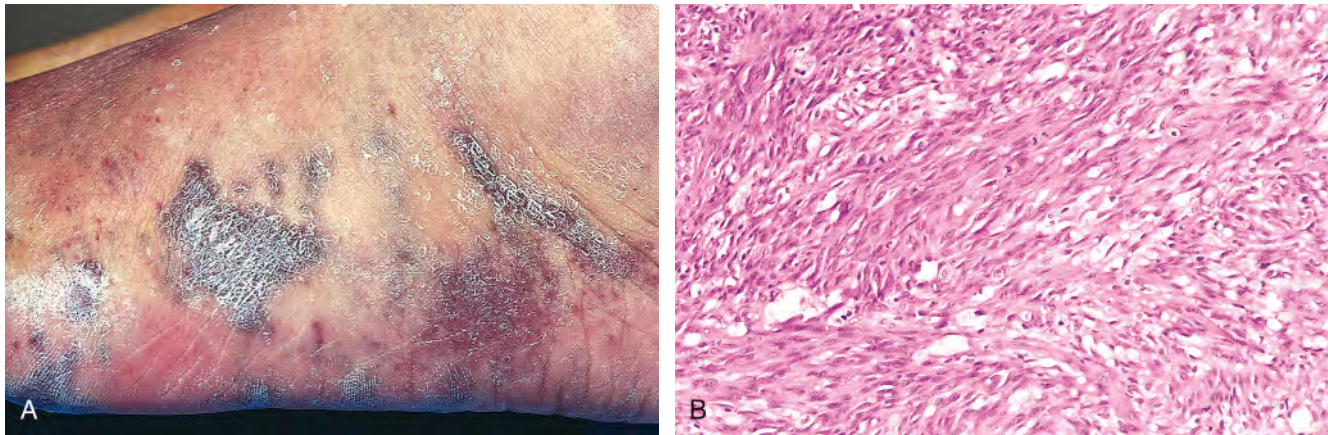


Fig. 11.31 Kaposi sarcoma. (A) Gross photograph illustrating coalescent red-purple macules and plaques of the skin. (B) Histologic appearance of the nodular stage of Kaposi sarcoma, demonstrating sheets of plump, proliferating spindle cells. (B, Courtesy Christopher DM Fletcher, MD, Brigham and Women's Hospital, Boston.)

well-differentiated hemangiomas and frankly anaplastic angiosarcomas.

Epithelioid hemangioendothelioma is a tumor of adults occurring around medium- and large-sized veins. Well-defined vascular channels are inconspicuous, and neoplastic cells are plump and often cuboidal (resembling epithelial cells). The clinical behavior is extremely variable; most are cured by excision, but the 5-year mortality rate is roughly 25%.

Malignant Tumors

Angiosarcoma

Angiosarcomas are malignant endothelial neoplasms (Fig. 11.32) with histology varying from highly differentiated tumors that resemble hemangiomas to profoundly anaplastic lesions. Older adults are more commonly affected; angiosarcomas occur at any site, but most often involve skin, soft tissue, breast, and liver.

Angiosarcomas can also arise in the setting of lymphedema, classically in the ipsilateral upper extremity several years after radical mastectomy for breast cancer i.e., after lymph node resection and/or axillary radiation); in such cases the tumor presumably arises from lymphatic vessels (*lymphangiosarcoma*). Angiosarcomas have also been induced by radiation and are rarely associated with prolonged insertion of foreign material (e.g., prosthetic devices).

Hepatic angiosarcomas are associated with a variety of carcinogenic exposures including arsenic (e.g., in pesticides), Thorotrast (a radioactive contrast agent formerly used for radiologic imaging), and polyvinyl chloride (one of the best-known examples of human chemical carcinogenesis). All of these agents have long latencies between initial exposure and eventual tumor development.

Angiosarcomas are locally invasive and can readily metastasize, with an overall poor prognosis; 5-year survival rates are only 10% to 30%.

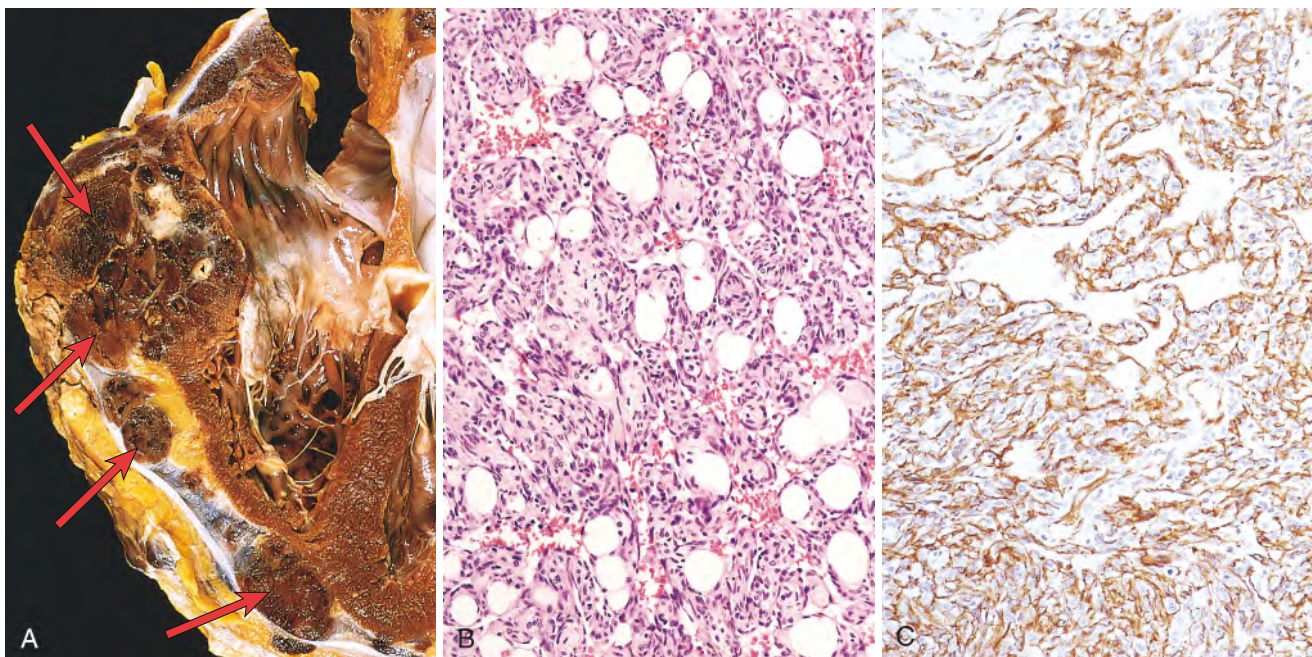


Fig. 11.32 Angiosarcoma. (A) Angiosarcoma involving the right ventricle. (B) Moderately differentiated angiosarcoma with dense clumps of atypical cells lining distinct vascular lumens. (C) Immunohistochemical staining for the endothelial cell marker CD31 demonstrating the endothelial nature of the tumor cells.

MORPHOLOGY

Cutaneous angiosarcomas can begin as multiple, deceptively small and asymptomatic red papules or nodules. More advanced lesions are large, fleshy masses of red-tan to gray-white tissue with margins blurring imperceptibly into surrounding structures (see Fig. 11.32A). Necrosis and hemorrhage are common.

Microscopically, **all degrees of differentiation can be seen**, from plump, atypical ECs forming vascular channels (see Fig. 11.32B) to wildly undifferentiated tumors with no discernible blood vessels. The endothelial origin of poorly differentiated tumors can be demonstrated by immunohistochemical staining for CD31 or von Willebrand factor (see Fig. 11.32C).

KEY CONCEPTS

Vascular Tumors

- Vascular ectasias are not neoplasms, but rather dilations of existing vessels.
- Vessel neoplasms can derive from either blood vessels or lymphatics and can be composed of ECs (hemangioma, lymphangioma, angiosarcoma) or other components of vascular wall cells (glomus tumor).
- Most vascular tumors are benign (e.g., hemangiomas); some have an intermediate, locally aggressive behavior (e.g., Kaposi sarcoma); and others are highly malignant (e.g., angiosarcoma).
- Benign tumors typically form obvious vascular channels lined by normal-appearing ECs. Malignant tumors are more often solid and cellular, exhibit cytologic atypia, and lack well-defined vessels.
- Hemangiomas, primarily tumors of infancy and childhood, are of two main types: capillary and cavernous hemangiomas.
- Lymphatic counterparts of hemangiomas are called lymphangiomas. Cavernous lymphangiomas (cystic hygromas) are often associated with Turner syndrome.
- Development of Kaposi sarcoma stems from the infections (both lytic and latent) of endothelial cells by HHV8 and altered T-cell immunity. Based on the geographical distribution and the types of population affected, KS is categorized into four forms: classic (older males of Mediterranean, Middle Eastern, and Eastern European descent), endemic (younger, equatorial Africa), transplant-associated (acquired, following organ or bone marrow transplantation), and AIDS-associated (epidemic, the most common HIV-related malignancy).

PATHOLOGY OF VASCULAR INTERVENTION

The morphologic changes that occur in vessels following therapeutic intervention (e.g., stenting or bypass surgery) largely recapitulate the changes that occur in the setting of any vascular insult. Local trauma or thrombosis (e.g., due to a stent) or abnormal mechanical forces (e.g., a saphenous vein inserted into the arterial circulation as a coronary artery bypass graft) all induce the same stereotypical healing responses—fibromuscular intimal hyperplastic lesions composed of SMCs and ECM. Thus, just as with several risk factors for atherosclerosis, interventions that injure the endothelium also tend to induce intimal thickening by recruiting SMCs and promoting ECM deposition.

Endovascular Stenting

Arterial stenoses (especially those in coronary arteries) can be dilated by transiently inflating a balloon catheter to pressures sufficient to rupture the occluding plaque (*balloon angioplasty*). Although most patients improve symptomatically following angioplasty alone, abrupt reclosure frequently occurs due to luminal compression caused by angioplasty-induced vascular dissection, by vessel wall spasm, or by thrombosis. Thus, more than 95% of endovascular coronary procedures now involve both angioplasty and concurrent *coronary stent* placement.

Coronary stents are expandable tubes of metallic mesh. They provide a larger and more regular lumen, “tack down” the intimal flaps and dissections that occur during angioplasty, and mechanically limit vascular spasm. Nevertheless, due to endothelial injury, thrombosis is an important immediate poststenting complication, and patients must receive potent antithrombotic agents (primarily platelet antagonists) to prevent acute catastrophic thrombotic occlusions. The long-term success of angioplasty is limited by the development of proliferative in-stent restenosis. This intimal thickening is due to SMC ingrowth, proliferation, and ECM synthesis, all driven by the initial vascular wall injury; it can cause clinically significant luminal occlusion in up to one-third of patients within 6 to 12 months of stenting (Fig. 11.33).

The newest generation of *drug-eluting stents* is designed to avoid this complication by leaching antiproliferative drugs (e.g., paclitaxel or sirolimus) into the adjacent vessel wall to block SMC activation. Although the duration of drug elution is short (on the order of days to weeks), these drug-eluting stents nevertheless reduce the incidence

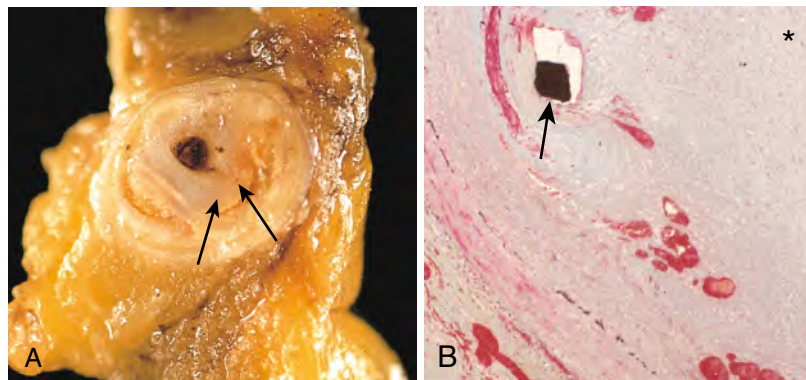


Fig. 11.33 Restenosis after angioplasty and stenting. (A) Gross view demonstrating residual yellow atherosclerotic plaque (*arrows*) and a new, tan-white concentric intimal lesion inside of that plaque. (B) Photomicrograph shows a thickened neointima separating and overlying the stent wires (black diamond indicated by the *arrow*), which encroaches on the lumen (*asterisk*). Movat stain with matrix staining gray-green. (B, From Schoen FJ, Edwards WD. Pathology of cardiovascular interventions, including endovascular therapies, revascularization, vascular replacement, cardiac assist/replacement, arrhythmia control, and repaired congenital heart disease. In: Silver MD, Gotlieb AI, Schoen FJ, eds. *Cardiovascular Pathology*. 3rd ed. Churchill Livingstone; 2001.)

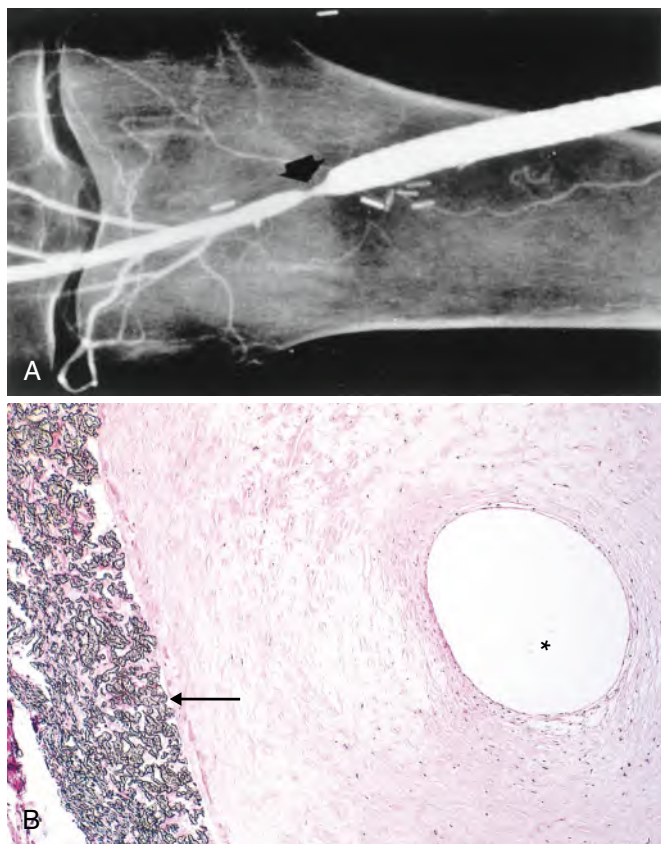


Fig. 11.34 Intimal hyperplasia at the distal anastomosis of a synthetic femoral-popliteal graft. (A) Angiogram demonstrating constriction (arrow). (B) Photomicrograph demonstrating Gore-Tex graft (arrow) with prominent intimal proliferation and very small residual lumen (asterisk). (A, Courtesy Anthony D. Whittemore, MD, Brigham and Women's Hospital, Boston.)

of restenosis at 1 year by 50% to 80%. However, because of the antiproliferative effect of the drug-eluting stents, the time to reendothelialization is prolonged, and patients require extended courses of anticoagulation to prevent stent thrombosis.

Vascular Replacement

Synthetic or autologous vascular grafts can be used to replace damaged vessels or bypass diseased arteries. Large-bore (12- to 18-mm) synthetic conduits function well in high-flow locations such as the aorta; unfortunately, small-diameter artificial grafts (≤ 8 mm in diameter) generally fail because of early thrombosis or late intimal hyperplasia, the latter at the junction of the graft with the native vasculature (Fig. 11.34).

Consequently, when small-bore vessel replacement is required (e.g., in the $>200,000$ coronary bypass surgeries performed annually in the United States), the grafts are fashioned from saphenous veins (taken from the patient's own leg) or left internal mammary arteries. The long-term patency of saphenous vein grafts averages only 50% at 10 years; grafts occlude due to thrombosis (typically early), intimal

thickening (months to years postoperatively), and vein graft atherosclerosis—sometimes with superimposed plaque rupture, thrombosis, or aneurysms. By contrast, more than 90% of internal mammary artery grafts, which can be typically used only to bypass the left anterior descending artery, are patent at 10 years.

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